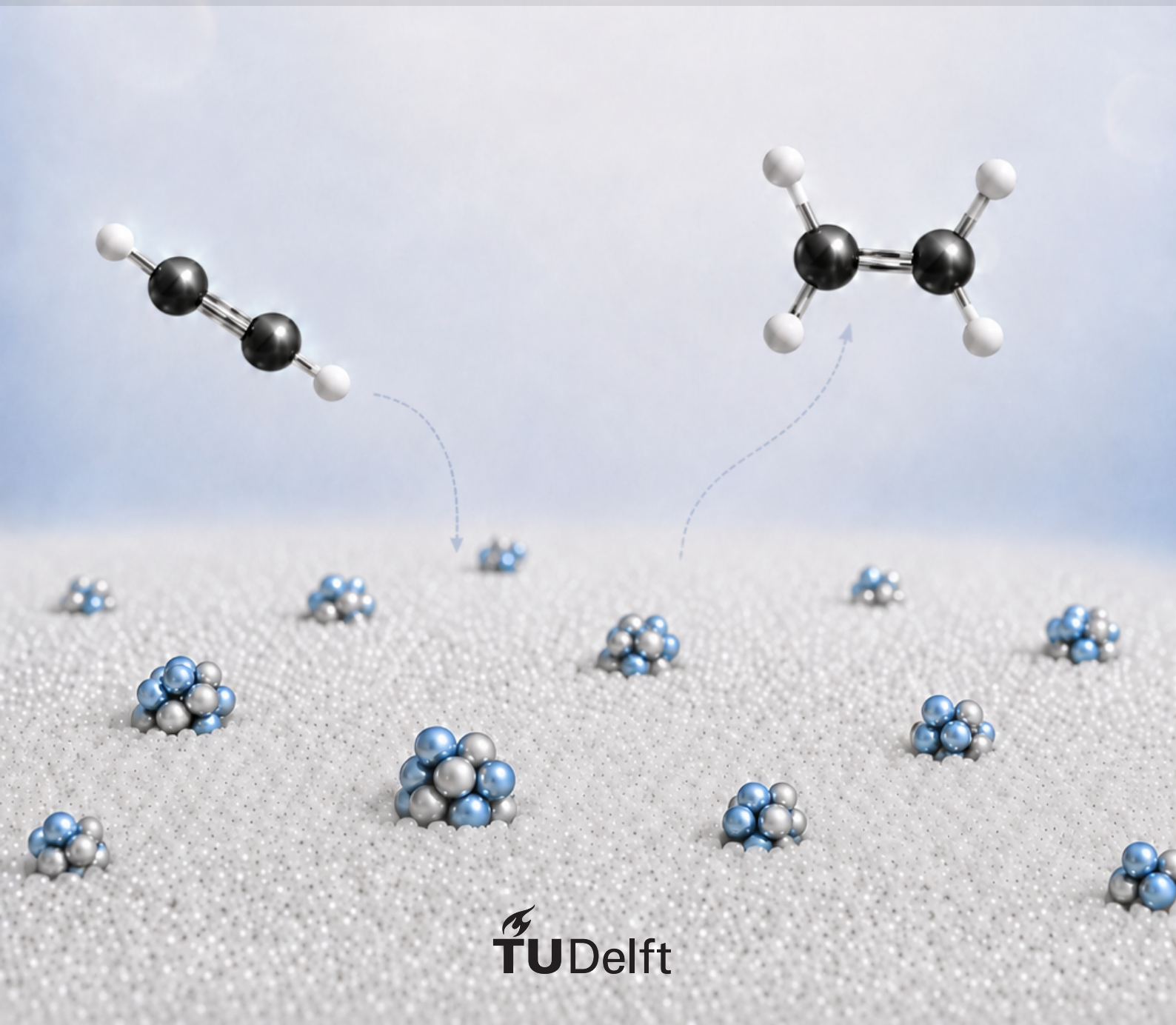
BEP

Semihydrogenation of acetylene to ethylene with a PdAg catalyst

Victor Larsen

Under supervision of:
Ir. Damian Vico van Berkel – Daily Supervisor
Prof. Atsushi Urakawa – Responsible Thesis Supervisor



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1

Abstract

The selective hydrogenation of acetylene to ethylene remains an industrially relevant process used to remove acetylene impurities from ethylene-rich product streams derived from stream cracking. Palladium based catalysts are widely used for this reaction, but their tendency to over-hydrogenate ethylene to ethane limits its selectivity. The addition of silver as diluent metal has been shown to improve the selectivity through both the geometric and electronic effects on the active palladium sites.

In this work, three different $PdAg/SiO_2$ catalysts with varying palladium loadings (Pd:Ag ratios of 1:200, 1:100, and 1:50) were prepared by incipient wetness co-impregnation and tested for selective acetylene hydrogenation at $100\,°C$ and $1000\,\mathrm{mbar}$. The effects of increasing flow rate, acetylene, ethylene and hydrogen concentrations were tested. Catalyst characterization was performed using powder XRD and H_2-TPR.

The experimental results have shown a trade-off between conversion and selectivity as a function of palladium loading. The catalysts with higher palladium loading have higher conversion, but also promoted over-hydrogenation, reducing selectivity. Catalysts with lower palladium loading have shown to maintain higher selectivity, but reduced activity.

Kinetic modeling using an extended LHHW model was applied to describe the observed catalytic behavior. The final model includes a maximum rate cap for acetylene hydrogenation and a flexible acetylene inhibition term for the over-hydrogenation reaction. This models has provided the best fit for all three catalysts with overall RMSE values of 5.10, 5.43, and 7.14 percentage points for Pd_0Ag, Pd_2Ag, and Pd_3Ag respectively. The results suggest that acetylene and ethylene hydrogenation are governed by different physical limitations, and that the Pd_2Ag offers the most favorable balance between conversion and selectivity.

Key Words: *Acetylene hydrogenation, palladium silver bimetallic catalysts, LHHW kinetic modeling*

2

Introduction

A world without ethylene (C_2H_4) is unthinkable, as it is one of the largest standard chemical building blocks. Ethylene is widely used as a feedstock for the production of polymers and polymer precursors, including polyethylene and polyvinyl chloride (PVC). Industrial ethylene is produced mainly through oil steam cracking, a process that is not free of hurdles. This is due to the resulting ethylene rich product stream can contain small amounts of acetylene (C_2H_2). Although acetylene can only be found in the stream in very small quantity, it still is known to poison to the Ziegler-Natta catalyst used in downstream ethylene conversion processes.[1] Therefore, acetylene must be removed from this product stream before further processing.

This is typically done through selective hydrogenation of acetylene to ethylene ($C_2H_2 + H_2 \longrightarrow C_2H_4$). This route is attractive because it removes acetylene, while possibly increasing the amount of ethylene in the stream. However, for this selective hydrogenation reaction a problem arises: ethylene can also hydrogenate to ethane ($C_2H_4 + H_2 \longrightarrow C_2H_6$). The formation of ethane lowers the ethylene yield and decreases the value of the purified ethylene stream. Therefore, for this process an effective catalyst must not only achieve high conversion, but also needs to suppress ethylene hydrogenation. This balance between acetylene conversion and ethylene selectivity has proven to be the main challenge in selective acetylene hydrogenation.

Palladium (Pd) is among the most widely studied and industrially relevant catalysts for acetylene hydrogenation because of its high hydrogenation activity. Palladium has proven to very efficiently activate hydrogen and adsorb unsaturated hydrocarbons. However, this efficiency also causes serious limitations. If acetylene concentration decreases, more active sites become available for ethylene to adsorb on. As a result, the catalysts selectivity can drop when high acetylene conversion is achieved.

In addition to over-hydrogenation, other unwanted side reactions can occur during hydrogenation. One important side reaction is C-C coupling, which can lead to the formation of so called ”green oil”. Green oil refers to heavier C_{4+} hydrocarbon species formed through coupling reactions between adsorbed acetylene- and ethylene-derived intermediates. These C_{4+} species can accumulate on the catalyst surface and ultimately lead to deactivation of the catalyst due to blockage of its active sites.[2,3]

Coke formation is another side reaction that can occur. Coke is formed through decomposition or further transformation of strongly adsorbed hydrocarbon species, which again leads to catalyst deactivation.[2,4–6] The main reactions and, side reactions, that can occur during this reaction are shown in Figure 2.1.

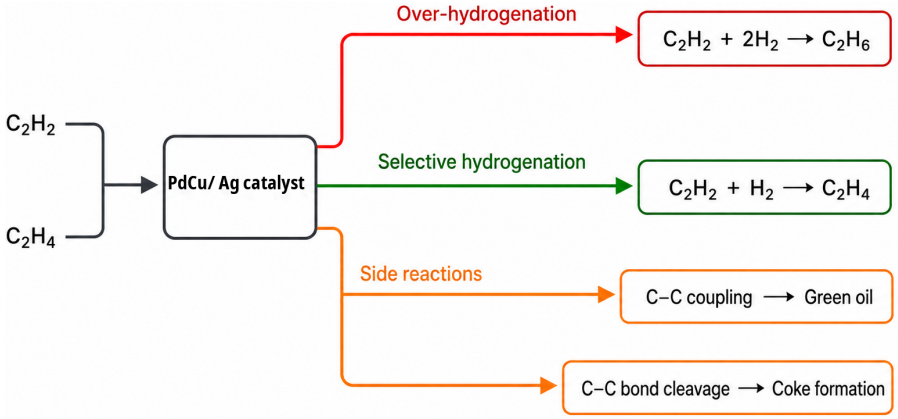


Figure 2.1: Overview of all possible reactions

A common strategy used to improve the selectivity of palladium based catalysts is through dilution with another metal. Metals such as copper (Cu) and silver (Ag) have proven to increase palladium based catalysts selectivity.[7–13] The addition of a second metal can influence the catalyst both geometrically and electronically. Geometrically, a second metal can dilute the catalyst surface and break up large palladium ensembles, reducing the number of sites that promote over-hydrogenation. Electronically, the second metal can modify the electronic structure of palladium. This can shift the d-band of palladium, resulting in a decrease of adsorption strength of the catalyst.[3,7,14] Through these effects, diluted palladium catalysts can improve the balance between acetylene conversion and ethylene selectivity.

Besides the addition of a second metal, the catalyst support also plays an important role. The support mainly influences the metal dispersion, but can also influence active site accessibility. Commonly investigated supports for acetylene hydrogenation are metal oxides such as Al_2O_3, TiO_2 and SiO_2.[2,4,7,10,15,16] In this research, SiO_2 was used as a support because it is generally considered to be more inert compared to other supports. This makes it suitable for studying the effect of different Pd/Ag compositions while minimizing additional support related effects.

Among bimetallic catalysts, PdAg remains among the most studied and industrially relevant catalysts for this reaction.[2,5,17] Previous studies have shown that PdAg catalysts can achieve high conversion, while maintaining relatively high selectivity. However, exact performance depends strongly on catalyst composition, particle structure, support, and reaction conditions.[2] This means that the palladium silver ratio is a crucial design parameter. A too low loading of silver may leave larger palladium ensembles to promote over-hydrogenation, while a too high loading can dilute the palladium too much thereby decreasing catalyst activity. Therefore, identifying an optimal palladium silver ratio requires a systematic comparison of catalytic performance.

Kinetic modeling is a useful tool for making this comparison more quantitative. Experimental conversion and selectivity show how a catalyst performs under tested condition, but kinetic modeling can help relate this performance to reaction rates, adsorption effects, inhibition terms, and reaction orders. Especially for this reaction, kinetic modeling is useful because it can describe both the desired and undesired hydrogenation reactions that can occur at the same time. With kinetic modeling these reaction can be described independently from each other. The kinetic parameters can provide insight into how different catalysts promote the desired reaction while suppressing other reactions. In addition, a sufficiently accurate kinetic model can be used to estimate catalyst performance under reaction conditions that were not tested experimentally.

The aim of this research is therefore to investigate different PdAg ratios in PdAg/SiO_2 catalysts on the selective semi hydrogenation of acetylene to ethylene. The catalysts are evaluated on their conversion and selectivity, and kinetic modeling is used to quantify differences in catalytic behavior between the different compositions. By combining experimental and simulated data, this work aims to

identify which PdAg ratio provides the most favorable ratio between conversion and selectivity.

3

Literature Background

3.1. Industrial relevance of the semi-hydrogenation reaction

After pumping crude oil, it still needs to undergo some changes before it can be used as fossil fuels or feedstock for plastics. The crude oil mainly exists of long hydrocarbon chains, which need to be broken apart to smaller hydrocarbon chains. Breaking these hydrocarbon chains is called the cracking of crude oil. This cracking happens in a cracking furnace, after which the cracked hydrocarbons go to a fractional distillation tower and get separated based on boiling points and chain length. However, this separation is not very precise, meaning that there can be some unwanted by products. One of these "over-cracked" by products is acetylene, which stays mixed with the desired ethylene, usually this stream contains only an acetylene concentration up to 3%.[18] Still, concentrations such as these poison the Ziegler-Natta catalyst used for the polymerization of ethylene, which is the reason it needs to be reduced to below 5 ppm in the feed stream to obtain higher yields.[2] One of the most effective ways to date remains the catalytic semi-hydrogenation of acetylene. The hydrogenation of acetylene creates more ethylene and thus improves yield and product quality.[19]

3.1.1. Reaction and possible side reactions

Generally speaking, the hydrogenation reaction has been found to be best represented by the Horuiti-Polanyi mechanism.[20] This mechanism describes the overall reaction by the sequential addition of a singular hydrogen atom, giving intermediates like vinyl (C_2H_3*). The full reaction following the Horuiti-Polanyi mechanism is described below.

$$C_2H_2 \rightleftharpoons C_2H_2* \overset{H*}{\rightleftharpoons} C_2H_3* \overset{H*}{\rightleftharpoons} C_2H_4* \rightleftharpoons C_2H_4 \qquad \Delta H = -172\ \mathrm{kJ\,mol^{-1}}$$

In addition to this desired hydrogenation reaction, ethylene can also hydrogenate again to ethane. This leads to a decrease in the selectivity of ethylene.

$$C_2H_2 \rightleftharpoons C_2H_2* \overset{H*}{\rightleftharpoons} C_2H_3* \overset{H*}{\rightleftharpoons} C_2H_4* \overset{H*}{\rightleftharpoons} C_2H_5* \overset{H*}{\longrightarrow} C_2H_6 \qquad \Delta H = -308\ \mathrm{kJ\,mol^{-1}}$$

Both reactions are exothermic, and increasing the temperatures can increase the reaction rate and therefore increase the conversion of acetylene. However, higher temperatures not only promote the wanted semi-hydrogenation but also promote the over-hydrogenation to ethane, leading to a decrease in selectivity and yield.

Next to the hydrogenation reaction there are also other reactions that can happen on the catalyst. One of these side reactions is so called C-C coupling, this is is the creation of longer carbon chains which happens when adsorbed acetylene and ethylene couple and together form C_{4+} products.[2] These chains keep on growing on the catalysts surface and after a while form polymeric deposits, which are called green oil. These oils form a sticky layer on the catalysts surface which blocks active sites, and ultimately leads to catalyst deactivation because acetylene can't reach the sites anymore.

$$2\,C_2H_2 + H_2 \longrightarrow CH_2CHCHCH_2* \longrightarrow C_{4+}\ (\mathrm{green\,oil})$$

Another side reaction that can occur is coke formation, this happens when C-C bond cleavage occurs. The cleavage modifies the active phase of the catalyst, like the formation of PdC_x on palladium based catalysts.

$$C_2H_2* \longrightarrow C_xH_y* \longrightarrow C(s)\ (coke)$$

Both of these side reactions are nearly impossible to avoid on industrial scale and are often compensated by increasing the temperature over time until the catalyst activity is below acceptable level.[2]

3.2. Current catalysts, and their challenges

For this hydrogenation reaction of acetylene, a lot of different metals have been found to be able to act as catalysts. The catalysts used for this reaction all work in the same principle, bringing the acetylene and hydrogen molecules together while activating hydrogen. The working of the catalyst starts with the adsorption of acetylene on its surface, because of the triple bond of acetylene it can relatively easy adsorb on the catalysts surface. In the meantime, the hydrogen also gets adsorbed and activated on the surface. This activation of hydrogen is one of the main reasons why a catalyst is needed for this reaction. Hydrogen on its own is a stable molecule and therefore is normally quite difficult to react with. After both these molecules are adsorbed the earlier described Horuiti-Polanyi mechanism can happen and the newly produced ethylene can desorb. However, this is where the challenge for these catalysts arise because if the acetylene concentration gets to low ethylene also can get adsorbed by the catalyst. When ethylene is adsorbed, the same mechanism can occur again leading to the formation of ethane and giving the low selectivity towards ethylene known for this reaction, and therefore the biggest challenge at this point in time.

3.2.1. Palladium catalysts

Throughout the years, transition metals like palladium (Pd), gold (Au), nickel (Ni) and copper (Cu) have been proven useful for this reaction.[2] Nickel was one of the first catalysts found to be able to promote hydrogenation. However, after some time, other transition metals have also been found to be good catalysts with great conversion and reasonable selectivity towards ethylene. Since the 1950s monometallic palladium has been the standard industrially, because of its ability to easily activate hydrogen and ease in adsorbing acetylene, while also desorbing ethylene relatively well.[5] Catalyst existing of palladium on a carbon support have been found to achieve conversion of 92% and selectivity of 50% at $75\,^{\circ}C$.[21] While other metals like gold for example achieve a conversion of 80%, while having a selectivity of 76% at $150\,^{\circ}C$ on a C-TiO_2-R250 support.[22] For this reason, palladium remains an important catalyst in industrial applications, despite its relatively high cost. This also helps explain why it continues to be extensively investigated in catalyst research.

However, even for palladium catalysts, improvements can be made. The main area for improvement is the catalysts low selectivity when high conversion is achieved. Achieving this higher selectivity can be achieved by different methods, but the method with the greatest potential has been catalyst modification by the addition of another metal. This is the area in which most research is done nowadays and is also showing significant improvements since research has started.

3.2.2. Catalyst dilution

As stated above, the palladium catalyst is continuously improving through catalyst modification, and the most researched is the addition of another metal. The beneficial effect, on semi-hydrogenation, of adding a second metal was first discovered in 1952 by Herbert Lindlar, and the catalyst found is called the Lindlar catalyst. Lindlar added palladium on calcium carbonate ($CaCO_3$) after which it was poisoned by lead (Pb) or sulfur (S).[23] Specifically for the selective hydrogenation of acetylene bimetallic catalysts were introduced in the 1980s.[5] All of these catalysts used palladium as the base metal and promoted it with different metals such as the previously mentioned silver (Ag) and gold (Au), but also metals such as nickel (Ni), zinc (Zn) or lead (Pb).[5] The purpose of diluting palladium with a second metal is to modify the active sites of palladium geometrically and electronically. This weakens the adsorption of ethylene on the catalyst surface, allowing ethylene to desorb more easily and thus suppressing its further hydrogenation to ethane, as can be seen in Figure 3.1.[17] Palladium diluted with silver is the catalyst in which most research has been done and is also the most widely used form of industrial catalyst for the hydrogenation of acetylene to ethylene.[2,5,17] This proven effectiveness industrially is the reason most research keeps concentrating on improving these palladium-silver catalysts.

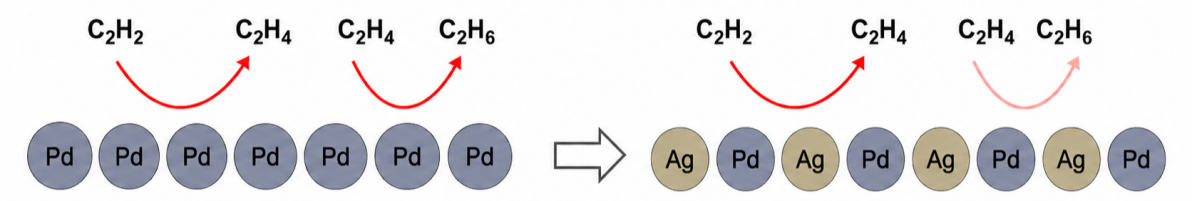


Figure 3.1: Schematic view of monometallic vs. bimetallic (Pd vs. PdAg) catalyst

Geometric effect

Bimetallic catalysts improve the selectivity of the palladium catalyst in two different ways, the introduction of the second metal tunes the geometric and electronic structure of the catalyst.[12] By changing the geometric structure, the selectivity of the catalyst is increased because it breaks up large ensembles of palladium atoms.

Large ensembles as described above are the main components that promote hydrogenation towards ethane. On these palladium rich ensembles acetylene can strongly bind to the catalyst through multi-coordinated π/di-σ type bonding.[10] Besides the ensemble effect affecting acetylene and ethylene adsorption, it also influences hydrogen activation. Between palladium atoms hydrogen can be moved or stored which increases the chance of hydrogen being close to an adsorbed acetylene or ethylene molecule.[13] But this is the same effect is where the problem towards selectivity originates from, the bonds these large ensembles form are so strong the newly formed ethylene can remain adsorbed, or can get readsorbed further in the reactor. If there is an activated hydrogen atom close to this adsorbed ethylene the hydrogenation will continue, ultimately forming ethane as an end product.

Through breaking these large ensembles up, with metals like silver, the possibility of forming such strong multi-coordinated π/di-σ type bond becomes harder. This leads to acetylene only being able to bond to the catalyst surface through a weaker π-bond.[10] However, the bond is still strong enough for the conversion of acetylene to ethylene, but the bond is not strong enough for ethylene to remain adsorbed or be readsorbed further in the reaction, preventing further hydrogenation to ethane. In addition to this effect, the dilution of palladium in the catalyst also ensures that the chance of a hydrogen atom being close enough to adsorbed ethylene for further hydrogenation gets smaller.[13] This shows that the ensemble effect has a large influence on the selective hydrogenation reaction by being able to significantly increase the selectivity of the reaction towards ethylene.

Possible catalyst configurations

This ensemble effect can also be influenced by the way these bimetallic catalysts active sites are configured. The most researched configurations are disordered alloy sites, ordered intermetallic compound (IMC) phases, and single-atom (SA) sites. These configuration possibilities are schematically shown in Figure 3.2.

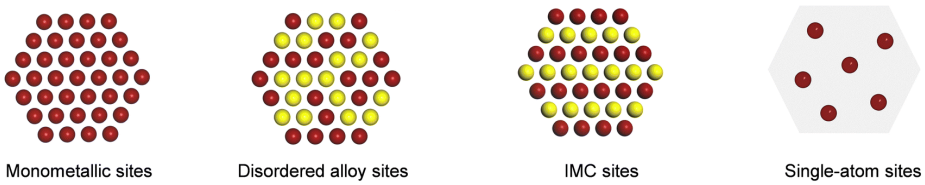


Figure 3.2: Schematic view of distinct types of active sites for acetylene hydrogenation. Adapted from Lan, Chen, and Wang [2]

In disordered alloy sites, the chance of large palladium ensembles is still significant. These catalysts are commonly prepared through methods such as co-impregnation or sequential impregnation. After which they are reduced through thermal treatment.[17] However, both of these preparation methods do not include any way of influencing the way both metallic atoms are ordered, leaving us with limited atomic-level control over the ordering of palladium-silver.[24] As a result, isolated palladium atoms, small palladium ensembles, and large palladium ensembles can co-exist depending on the total palladium loading.[25,26] This explains how with disordered alloy sites, the ensemble effect can still have an influence on the catalysts performance.

For intermetallic compounds this is different, the atomic arrangement is more ordered. These catalysts are prepared with controlled metal ratios, after which they undergo a high temperature reduction. This high temperature reduction allows the atoms to rearrange in an ordered structure.[27] The ordered structure into which these atoms rearrange often has a predefined stoichiometry and crystal structure.[28] However, whether the ensemble effect is still at play depends on the specific intermetallic phase and the surface structure.[29] When palladium atoms are still neighboring each other with this structure, the ensemble effect can still become active, once again having lower selectivity of the catalyst towards ethylene as consequence.

The final configuration on which a lot of research has been done is single-atom sites. In these types of catalysts, palladium atoms are isolated in a sliver rich matrix.[30] For this to be possible, very dilute amounts of palladium are introduced together with silver precursors onto a support by incipient wetness impregnation. This is then followed by drying and reduction in hydrogen, resulting in a palladium-silver alloy phase, and when the palladium loading is sufficiently low, palladium atoms can become atomically dispersed within the silver matrix, preventing the formation of larger palladium ensembles.[31] Without neighboring palladium atoms, the known palladium ensemble effect is severely suppressed, which therefore prevents the multi-coordinated bond with acetylene or ethylene.[8] The danger with SA-sites is that the palladium load is too low, which ultimately decreases the conversion of acetylene and thus the catalyst becomes inactive.[32]

Electronic effect

The other effect described is the change in electronic structure that the addition of a second metal brings with it. Whenever a metal is combined with another metal, they influence each other on an electronic level, this electronic level is the d-band of palladium.[17] Silver has been found to move an electron from silver to palladium filling the d-band of palladium.[8] The filling of the d-band improves the selectivity of the reaction because it lowers the d-states to lower energy. And since both acetylene and ethylene bind through their π-orbitals with the d-states of palladium, favorable back donation interactions by palladium are reduced.[33,34] This ultimately leads to a weaker interaction between acetylene and ethylene with palladium, which again reduces over-hydrogenation to ethane, thus improving selectivity.[11]

3.2.3. Effect of metal-oxide catalyst support

In addition to the composition of the bimetallic catalyst, the metal-oxide support on which it is placed also influences its selectivity towards hydrogenation of acetylene. The strongest influence of the support is to stabilize the metal atoms during the reaction. If palladium atoms become mobile, they can migrate and aggregate into larger ensembles, a phenomenon known as sintering, which especially has a great influence on the selectivity of a catalyst at higher temperatures.[16] Oxide supports prevent this phenomenon by separating the metal atoms and anchoring them to the support surface.[35] First of all, by adding a support to the active palladium atoms are more spread out than without support, making the chance of them finding each other during heating smaller.[36] Secondly, the anchoring through oxygen atoms and hydroxyl groups; palladium atoms can coordinate towards the surface oxygen atoms, forming palladium-oxygen bonds. This bond improves adhesion between the active catalyst and the support, making sintering less favorable, once again preventing the formation of larger palladium ensembles.[35] Next, the addition of a catalyst support also helps spread the heat evenly, therefore preventing local hot spots where sintering or side reactions can take place.[15] There are many metal-oxides that can be used as a support for this specific reaction, among the most common are Al_2O_3, SiO_2, and TiO_2.[2,5,15,16,18,20] However, industrially the most used support is Al_2O_3, specifically α-Al_2O_3 supports are very common to find.[2,5,16,20] However, SiO_2 has proven its worth in research towards single-atom palladium-silver catalysts because SiO_2 is a relatively inert support, therefore making it easier to observe the metal-alloy effect without support effects.[37]

3.3. Competitive Adsorption acetylene, ethylene and hydrogen

As mentioned above, the feed stream for this reaction contains only a small amount of acetylene, whereas the ethylene concentration is much higher. This creates an ethylene-rich feed stream in which acetylene, ethylene, and hydrogen compete for adsorption on the catalyst surface. Therefore, competitive adsorption plays an important role in determining both acetylene conversion and ethylene selectivity. The surface coverage of the catalyst can be calculated through the equation in 3.1, where K_i is the adsorption constant and p_i is the partial pressure.

$$\theta_i \propto K_i p_i \tag{3.1}$$

Acetylene itself forms a stronger bond with palladium than ethylene; acetylene (CH≡CH) has a triple bond through which both π-bonds can strongly interact with the d-orbitals of palladium, compared to the double bond of ethylene ($CH_2=CH_2$) which only has one π-bond that can interact with palladium. The formation of this bond between the d-orbitals of palladium and the π-bonds of acetylene and ethylene can be described through the Dewar–Chatt–Duncanson model, which can be seen in Figure 3.3. This model describes the donation of an electron from the π-system of acetylene or ethylene to the d-bands of palladium, while the palladium return an electron back through back-donation to the π^* (anti bonding) orbitals of acetylene and ethylene, which weakens the triple and double bond.[38] Since acetylene has two π-bonds, the adsorption onto palladium is stronger than that between palladium and ethylene, meaning that the adsorption constant of acetylene is significantly higher than that of ethylene. This is beneficial because that means that even though the feed stream is ethylene rich, acetylene will still adsorb stronger than ethylene. However, if the bond is too strong acetylene derived products can remain on the catalysts surface which can block sites for hydrogen adsorption, or help the coke and green oil formation.[2,39]

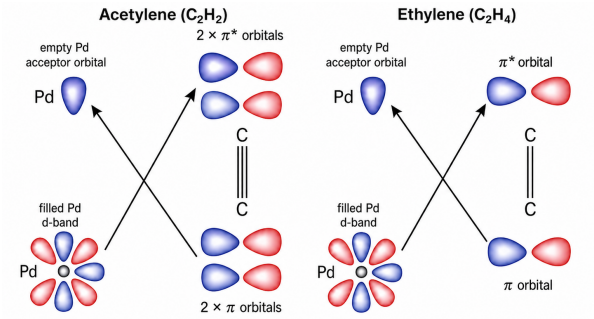


Figure 3.3: Schematic representation of the Dewar–Chatt–Duncanson model for the bond between palladium and acetylene/ethylene

In addition to the binding strength, the amount of molecule available in the gas phase also influences the surface coverage and therefore the competitive adsorption. This variable is expressed by partial pressure, which is calculated using $p_i = y_i P$ this formula, and since it is an ethylene rich feed, the partial pressure of ethylene will be much higher. This means that, especially when acetylene concentration gets low, when conversion approaches completion, ethylene can compete for the active sites of the catalyst which leads to the hydrogenation to ethane.[6,40]

However, not only is acetylene and ethylene adsorption important for this reaction, but hydrogen also needs to be adsorbed and activated before it can react with either of these molecules. If the partial pressure of hydrogen is too low, the formation of activated hydrogen gets too little, thus becoming the limiting rate for its reaction. But when the partial pressure is too high, the after hydrogenation, adsorbed ethylene still can receive another activated hydrogen molecule leading to the hydrogenation of ethylene to ethane.[4]

3.4. Kinetic modeling

Kinetic modeling is often used in research to get a better understanding of the reaction being researched, especially for our hydrogenation reaction because the reaction is determined not only by the conversion of acetylene, but also by its selectivity towards ethylene. So in our case, that is the hydrogenation of acetylene to ethane, at which we can look at in two steps. Step one is the desired hydrogenation of acetylene to ethylene (r_1), and the second step of the reaction is the undesired hydrogenation of ethylene to ethane (r_2). Through kinetic modeling we can get insight on the influence of operating conditions as temperature and pressure, but also on the influence hydrogen, acetylene and ethylene concentration have on the acetylene conversion and ethylene selectivity. Especially for tail-end conditions this is important, because in this part of the reactor acetylene concentrations becomes

quite low and therefore small changes in surface coverage can have a big influence on the balance between conversion and selectivity.[20,40]

3.4.1. Power law model

A kinetic model that is often used in the beginning is the relatively simple power law model. The power law model uses reaction orders to describe the change in hydrogenation rate when partial pressures are changed, in addition to the stoichiometry of the reaction these also take into account adsorption and surface coverage.[40] The general power law equations for this hydrogenation reaction are equations 3.2 and 3.3. Earlier research has shown that the reaction order of hydrogen has been reported to be positive, showing that hydrogen availability has an immediate influence on the hydrogenation rate.[40] For acetylene the reaction rates have been found to be near zero, or in some cases even negative, this shows the strong bond between the palladium and acetylene, but also shows that when the catalysts active sites are full the addition of extra acetylene does not influence the reaction.[40] The effect of ethylene is not really important for acetylene consumption, which explains the reaction order being zero, but ethylene does become more important for ethane formation so for the second reaction ethane usually has a reaction order between 0.3 and 1.[20] This shows that the concentration of ethylene does increase the rate of ethane formation, but not linearly.

$$r_1 = k_1(T)\, p_{\mathrm{C_2H_2}}^{a_1}\, p_{\mathrm{H_2}}^{b_1} \tag{3.2}$$

$$r_2 = k_2(T)\, p_{\mathrm{C_2H_4}}^{a_2}\, p_{\mathrm{H_2}}^{b_2} \tag{3.3}$$

3.4.2. Langmuir-Hinselwood-Hougen-Watson model

Compared with the power-law model, the LHHW model takes an adsorption term into account. These are the K-terms in the equations 3.4 and 3.5, and given that earlier studies have shown that the LHHW kinetic model is a representative way of looking at the dynamics, this model is most used in research on the kinetics of this reaction.[20] For the hydrogenation reaction, acetylene, ethylene, and hydrogen compete for the same active sites. This makes the adsorption term per molecule important for the kinetic model and its ability to predict the conversion and selectivity. In the denominator term, the competition between acetylene and ethylene can be seen by both terms being stated there. As described, the adsorption term of acetylene will be higher than that of ethylene, with both adsorption terms being much higher than that of hydrogen.[20] However, also in this kinetic model, it becomes clear that when the acetylene concentration drops, the reaction will shift towards r_2 and thus produce ethane as an end product.

$$r_1 = \frac{k_1 K_{\mathrm{C_2H_2}} p_{\mathrm{C_2H_2}} K_{\mathrm{H_2}} p_{\mathrm{H_2}}}{\left(1 + K_{\mathrm{C_2H_2}} p_{\mathrm{C_2H_2}} + K_{\mathrm{C_2H_4}} p_{\mathrm{C_2H_4}} + K_{\mathrm{H_2}} p_{\mathrm{H_2}}\right)^2} \tag{3.4}$$

$$r_2 = \frac{k_2 K_{\mathrm{C_2H_4}} p_{\mathrm{C_2H_4}} K_{\mathrm{H_2}} p_{\mathrm{H_2}}}{\left(1 + K_{\mathrm{C_2H_2}} p_{\mathrm{C_2H_2}} + K_{\mathrm{C_2H_4}} p_{\mathrm{C_2H_4}} + K_{\mathrm{H_2}} p_{\mathrm{H_2}}\right)^2} \tag{3.5}$$

3.4.3. Mass Transfer Limitations

However, there is still is a factor that both models do not take into account. Both models assume the measured reaction rates only reflect the intrinsic catalyst behavior. This assumption can only be made when mass transfer does not significantly limit the transport of reactants.[41] If external, or internal mass transfer becomes rate-limiting the observed reaction rate can, next to adsorption and surface reaction steps, also be determined by diffusion of reactants through the gas phase, the pores of the catalyst or inside the particle.[41] Mass transfer limitations can occur in different phases of the reaction, such as diffusion of the feed stream to the catalyst surface (external) and diffusion in the catalyst pores (internal).[41,42] Typical signs of mass transfer limitations include a significant dependence of the observed reaction rate on flow rate, and more active catalysts does not result in proportionally higher reaction rates.[42,43]

External mass transfer
External mass transfer limitations occurs when reactants are consumed at a rate at the catalyst surface faster than that supplied by the feed stream. In this case, the concentration of the feed stream is higher than at the catalyst surface.[43] Kinetic models such as the LHHW model may assume incorrect reactant concentrations at the active sites when limited external mass transfer occurs. The model can assume the concentration of acetylene at the catalyst surface is equal to the concentration in the bulk gas phase.[20] When internal mass-transfer limitations are present, this assumption is no longer valid and must be corrected.

Internal mass transfer
Internal mass transfer limitations occur when reactants cannot diffuse through the pores of catalyst particles fast enough to reach all internal active sites. As a result, the whole catalyst particle is not effectively used. The active sites of the catalyst near the surface are used mainly, while more internal active sides experience lower reactant concentrations.[44]

Correcting for mass transfer limitations and other physical effects
When mass transfer limitations are significant, the observed reaction rate no longer only reflects the intrinsic surface kinetics. In that case, a purely kinetic LHHW model may overpredict the reaction rate, especially under conditions where the intrinsic acetylene hydrogenation rate becomes very high. To account for this effect in a simplified way, an empirical rate cap can be introduced for the acetylene hydrogenation rate. This correction limits the rate predicted by the LHHW expression to a maximum observable rate, $(r_{1,\max})$, as shown in equation 3.6.[43] It should be noted that this rate cap should not be interpreted as representing only mass transfer limitations. Instead, it is an empirical correction that takes the maximal observed rate into account. Therefore is this cap a combination of several effects as reactor operation and catalyst bed behavior. But mass transfer limitations are expected to have a significant influence on this maximum observed rate, because of their possible restriction of reactants reaching the active sites. As a result, when the intrinsic reaction rate is low, $r_1 \approx r_{1,LHHW}$, the reaction is kinetically controlled. In contrast, when the intrinsic reaction rate becomes higher r_1 becomes closer to $r_{1,max}$, eventually reaching a maximum reaction rate controlled by other effects, like mass transfer limitations.

$$r_{1,obs} = \frac{r_{1,LHHW}}{1 + \frac{r_{1,LHHW}}{r_{1,max}}} \tag{3.6}$$

4

Materials and Methods

4.1. Catalyst preparation

All catalysts used in this work have all been prepared in the same way, with the incipient wetness co-impregnation method. Starting with 3 grams of SiO_2 as the support on which a predetermined amount of Pd and Ag is dropped. The amount of Pd and Ag loaded onto the SiO_2 support is based on the ratio we wanted to analyze. As a base however we ensured a Ag wt% of 5% and change the ratio by the amount of Pd getting loaded. We worked with ratios ranging from $PdAg_{50}$ to $PdAg_{200}$. For getting the right amount Pd loaded, $Pd(NO_3)_2 \cdot 2\,H_2O$ is used as the stock molecule and for Ag this is $AgNO_3$. After the weighting of both molecules has been done, 1 mL of Pd (stock solution) is added to the Ag solution. After this solution is shaken, 0.5 mL water is added. When this is shaken thoroughly again, half of this solution is added on the SiO_2 support dropwise. After mixing this for 5 minutes the other half of the solution is added in the same way and mixed for 10 minutes. Finally, the mixture is dried for 12 hours at $80\,°C$ so the mixture can calcinate. The prepared catalyst that have been tested can be found in Table 4.1.

Table 4.1: Prepared PdAg catalysts with each weight percentage and ratio

Catalyst	Pd wt%	Ag wt%	Ratio (Ag:Pd)
Pd_0Ag	0.025	5	200
Pd_2Ag	0.05	5	100
Pd_3Ag	0.1	5	50

4.2. Reactor Configuration and Catalytic Testing

To test the selective hydrogenation of acetylene, a certain setup was used, for start a fixed-bed tubular reactor was used. In this glass reactor quartz wool is put inside on both ends, while having the catalyst loaded in between. The glass reactor then gets put inside the experimental setup, where it is tested for leaks with a flow of He and Snoop. When there are no more bubbles forming, meaning there are no leakages, the pretreatment of the catalyst can start.

This means the reactor needs to be heated up to $250\,°C$ under a flow of He and H_2 at 10% and 40% respectively with a ramp rate of $10\,°C\,min^{-1}$. When the temperature has reached $250\,°C$, it stays here for 1 hour, after which the catalyst gets cleaned. This is done by turning the flow of He up to 80% and turning the flow of H_2 off and keeping these conditions for 10 minutes. After these 10 minutes, the temperature is put to $80\,°C$ and when the reactor is cooled down the experiment can start, when the experiment has started the reactor temperature is maintained at $80\,°C$.

For the running of the experiment, a recipe is used to automatically switch every half an hour between flows of H_2, He, C_2H_2 and C_2H_4. When all recipes are started correctly, wait 7 minutes for it to stabilize, after which the gas chromatograph (GC) is started. The concentration of each material is based on the area's of C_2H_2, C_2H_4 and C_2H_6. This is done by a formula obtained with the calibration of the mass flow controller (MFC). These formula's look as follows:

$$C_{C_2H_2} = 1.034518 \times 10^{-8} \cdot \text{GC area}_{C_2H_2} \tag{4.1}$$
$$C_{C_2H_4} = 1.5234839 \times 10^{-8} \cdot \text{GC area}_{C_2H_4} \tag{4.2}$$
$$C_{C_2H_6} = 1.52348391 \times 10^{-8} \cdot \text{GC area}_{C_2H_6} \tag{4.3}$$

After performing these concentration calculations for every area that results from GC, we can calculate the specific conversion of acetylene and selectivity towards ethylene using the formulas 4.4 and 4.5.

$$\text{conversion} = \frac{C_2H_{2_{\text{feed}}} - C_2H_2}{C_2H_{2_{\text{feed}}}} \times 100\% \tag{4.4}$$
$$\text{selectivity} = \left(1 - \frac{C_2H_6 + 2C_{4+}}{C_2H_{2_{\text{feed}}} - C_2H_2}\right) \times 100\% \tag{4.5}$$

4.3. Kinetic modeling

The first part to setup for the kinetic model is the reactor model. The packed bed reactor is the reactor used throughout the regular experiments, and therefore the best to use for the reactor model. However, most packed bed reactors describe flow over catalyst weight, but for our research we want to see how the composition of the flow changes over time, showing that the semi hydrogenation took place, or that over-hydrogenation happened. Before we can set up this model, it is important to know which reaction is which. Therefore we define the hydrogenation of acetylene as r_1 and the hydrogenation of ethylene as r_2, as seen below.

$$C_2H_2 + H_2 \xrightarrow{r_1} C_2H_4 \qquad \Delta H = -172\ \text{kJ mol}^{-1}$$
$$C_2H_4 + H_2 \xrightarrow{r_2} C_2H_6 \qquad \Delta H = -136\ \text{kJ mol}^{-1}$$

With this in mind, the differential molar balance equations can be determined. Given that the reaction is quite simple, the balance equations are also quite simple, as can be seen in the reaction formula's below:

$$\frac{dF_{C_2H_2}}{dz} = -r_1 \rho_{\text{support}} A \tag{4.6}$$
$$\frac{dF_{C_2H_4}}{dz} = (r_1 - r_2) \rho_{\text{support}} A \tag{4.7}$$
$$\frac{dF_{C_2H_6}}{dz} = r_2 \rho_{\text{support}} A \tag{4.8}$$
$$\frac{dF_{H_2}}{dz} = (-r_1 - r_2) \rho_{\text{support}} A \tag{4.9}$$

There are 2 common kinetic models in use, namely the power law and the Langmuir-Hinselwood-Hougen-Watson (LHHW) model. The power law is a simpler model that does not need as much computing effort, while the LHHW model is a more complex model one that takes into account competitive adsorption of ethylene and acetylene , while also considering dissociative H_2 adsorption. With these models the reaction rates for both reactions can be calculated, for the power law this is done through equations 4.10 and 4.11. In these formula's a_1, b_1, a_2, and b_2 describe the reaction order.

$$r_1 = k_1(T)\, p_{C_2H_2}^{a_1}\, p_{H_2}^{b_1} \tag{4.10}$$
$$r_2 = k_2(T)\, p_{C_2H_4}^{a_2}\, p_{H_2}^{b_2} \tag{4.11}$$

As stated before, the LHHW model is a more realistic kinetic model commonly used to describe the kinetics of the semi hydrogenation of acetylene. Therefore the reaction rates for r_1 and r_2 need to

be calculated in a different way, that can be found in equations 4.12 and 4.13. In these equations the denominator describes the active area of the catalyst, and shows that if acetylene is present in a large enough quantity, it vastly suppresses r_2 and thus prevents the over-hydrogenation to ethane.

$$r_1 = \frac{k_1 K_{C_2H_2} p_{C_2H_2} K_{H_2} p_{H_2}}{\left(1 + K_{C_2H_2} p_{C_2H_2} + K_{C_2H_4} p_{C_2H_4} + K_{H_2} p_{H_2}\right)^2} \tag{4.12}$$

$$r_2 = \frac{k_2 K_{C_2H_4} p_{C_2H_4} K_{H_2} p_{H_2}}{\left(1 + K_{C_2H_2} p_{C_2H_2} + K_{C_2H_4} p_{C_2H_4} + K_{H_2} p_{H_2}\right)^2} \tag{4.13}$$

Given the fact that the LHHW model gives a better grip on reality, this reactor model is the model used throughout the rest of the research. Important to note is the fact that the addition of another metal, like silver, does not change these models but only changes the kinetic parameters within these models.

When both these models have ben set up, the selectivity and conversion functions can be set up through the formula's 4.4 and 4.5. Based on the modeled conversion and selectivity the error between the simulated and experimental data can be calculated. The error is determined through the root mean square error (RMSE), 4.14. This RMSE is calculated per experiment after which, the overall RMSE is calculated by dividing it by the total number of experiments, as can be seen in 4.15.

$$\mathrm{RMSE} = \sqrt{\frac{\left(\mathrm{conv}_{\mathrm{sim}} - \mathrm{conv}_{\mathrm{exp}}\right)^2 + \left(\mathrm{selec}_{\mathrm{sim}} - \mathrm{selec}_{\mathrm{exp}}\right)^2}{2}} \tag{4.14}$$

$$RMSE_{overall} = \frac{1}{N} \sum_{i=1}^{N} RMSE_i \tag{4.15}$$

If the error calculation is set up, a way is needed to find the best kinetic parameters fit. This is done through a technique called differential evolution (DE), this optimization technique is commonly used in kinetic modeling. It is commonly used because it is a global optimizer, which is nice for kinetic modeling because these models can have a lot local minima. The advantage this technique has is that it does not need any start values, and only needs bounds to operate in. The target of the DE is to minimize the RMSE between the experimental and simulated conversion and selectivity. It does so by creating a population of possible parameter combinations which then are recombined to new candidate solutions. This ultimately gives us a minimum error, and therefore the best fit of parameters. However, algorithms like DE do not necessary guarantee the optimal solution to be found, but with the bounds chosen right, has a very high chance to do so.

4.3.1. Modifications of LHHW model

Simplification

The first modification done to the standard LHHW is to simplify the LHHW model. This is done to evaluate the hydrogen parameter inside the kinetic model. If removal of the hydrogen adsorption term results in a similar or better fit, this indicates that the fitted model is not strongly dependent on this term. Therefore, the hydrogen adsorption term may be left out to simplify the LHHW expression. However, this does not necessarily mean that hydrogen adsorption is physically absent, only that its contribution is not required to describe the experimental data within the tested model structure. This simplification results in the rate equations below.

$$r_1 = \frac{k_1 K_{C_2H_2} p_{C_2H_2}}{\left(1 + K_{C_2H_2} p_{C_2H_2} + K_{C_2H_4} p_{C_2H_4}\right)^2} \tag{4.16}$$

$$r_2 = \frac{k_2 K_{C_2H_4} p_{C_2H_4}}{\left(1 + K_{C_2H_2} p_{C_2H_2} + K_{C_2H_4} p_{C_2H_4}\right)^2} \tag{4.17}$$

Inhibition term

As stated before, acetylene forms a stronger bond with the catalyst active sites. Therefore, acetylene can inhibit the hydrogenation of ethylene when it remains on the catalyst surface. The addition of this term can mainly improve the model's selectivity because of its inhibitory effect on r_2. The rate equations the addition gives us are Equations 4.18 and 4.19 for the reaction rates.

$$r_1 = \frac{k_1 K_{C_2H_2} p_{C_2H_2} K_{H_2} p_{H_2}}{\left(1 + K_{C_2H_2} p_{C_2H_2} + K_{C_2H_4} p_{C_2H_4} + K_{H_2} p_{H_2}\right)^2} \tag{4.18}$$

$$r_2 = \frac{k_2 K_{C_2H_4} p_{C_2H_4} K_{H_2} p_{H_2}}{\left(1 + K_{C_2H_2} p_{C_2H_2} + K_{C_2H_4} p_{C_2H_4} + K_{H_2} p_{H_2}\right)^2 \left(1 + K_{inh} p_{C_2H_2}\right)} \tag{4.19}$$

Inhibition term squared

In addition to just the inhibition term as a term in the LHHW model, the model's fit with the inhibition term squared was also calculated. The reason to test the inhibition term squared is to know whether, for some catalysts, the regular inhibition term is too weak to describe the drop in ethylene hydrogenation. This could improve the model because it allows the model to describe a situation where acetylene inhibition is weak at low acetylene concentration, but becomes significantly stronger with increasing concentrations. The rate equations that follow from this modification can be found in Equation 4.20 and 4.21.

$$r_1 = \frac{k_1 K_{C_2H_2} p_{C_2H_2} K_{H_2} p_{H_2}}{\left(1 + K_{C_2H_2} p_{C_2H_2} + K_{C_2H_4} p_{C_2H_4} + K_{H_2} p_{H_2}\right)^2} \tag{4.20}$$

$$r_2 = \frac{k_2 K_{C_2H_4} p_{C_2H_4} K_{H_2} p_{H_2}}{\left(1 + K_{C_2H_2} p_{C_2H_2} + K_{C_2H_4} p_{C_2H_4} + K_{H_2} p_{H_2}\right)^2 \left(1 + K_{inh} p_{C_2H_2}\right)^2} \tag{4.21}$$

Final rate equations

As a final modification, a combined LHHW-based model was tested. In this model, the first hydrogenation step was limited using an $r_{1,\max}$ term, while the second hydrogenation step included a flexible acetylene inhibition term. The acetylene hydrogenation rate was first calculated as a regular LHHW-type rate, after which it was corrected by the maximum-rate expression. This correction prevents the model from predicting unrealistically high acetylene consumption rates when $r_{1,\mathrm{raw}}$ becomes large. The rate cap therefore improves the physical meaning of the conversion fit, but should not be interpreted as a description of a specific physical phenomena but more the collective of non-mentioned influential phenomena. By the addition of this $r_{1,max}$ an extra equation is created to the model. Besides this addition, another parameter was added, n_{inh}. This makes the model more flexible than the previous inhibition models. In those models, n_{inh} is fitted linear or straight, allowing the strength of acetylene inhibition to change depending on the catalyst. All the final rate equations can be found in Equations 4.22, 4.23 and 4.24.

$$r_{1,\mathrm{raw}} = \frac{k_1 K_{C_2H_2} p_{C_2H_2} K_{H_2} p_{H_2}}{\left(1 + K_{C_2H_2} p_{C_2H_2} + K_{C_2H_4} p_{C_2H_4} + K_{H_2} p_{H_2}\right)^2} \tag{4.22}$$

$$r_1 = \frac{r_{1,\mathrm{raw}}}{1 + \frac{r_{1,\mathrm{raw}}}{r_{1,\max}}} \tag{4.23}$$

$$r_2 = \frac{k_2 K_{C_2H_4} p_{C_2H_4} K_{H_2} p_{H_2}}{\left(1 + K_{C_2H_2} p_{C_2H_2} + K_{C_2H_4} p_{C_2H_4}\right)\left(1 + K_{\mathrm{inh}} p_{C_2H_2}\right)^{n_{\mathrm{inh}}}} \tag{4.24}$$

4.4. Characterization

4.4.1. Powder XRD

Powder X-Ray Diffraction (XRD) is used to identify the crystalline phases in the prepared catalysts. The measurements were obtained through a *Bruker D8 Advanced Diffractometer* equipped with a copper source. The measured samples were in powder form, meaning particle sizes below $100\,\mu\mathrm{m}$. Total measurement times per sample were 30 min, spanning a 2θ range between $20°$ and $90°$.

4.4.2. H_2-TPR

H_2-Temperature Programmed Reduction (TPR) experiments were carried out using a custom built setup equipped with a H_2-TCD. In each experiment, $150\,\mathrm{mg}$ of sample was physically mixed with $300\,\mathrm{mg}$ of SiC and loaded into a tubular quartz reactor. The packed bed was kept in place by a quartz wool plug, and at the top a thermocouple was placed. A ramp rate of $10\,°\mathrm{C\,min^{-1}}$ was used until a temperature of $600\,°\mathrm{C}$ was reached. No pretreatment of the samples was carried out, and the reduction took place under a flow of $30\,\mathrm{mL/min}$, with 10 vol.% H_2 concentration in Ar.

5

Characterization

5.1. XRD

The results of the performed XRD can be found in Figure 5.1. This figure shows that the overall diffraction patterns are very similar for all catalysts. The first broad bump for every catalyst at ~22-25° 2θ, which can be attributed to amorphous SiO_2.[8] One can expect to see palladium oxide reflections around ~30° and ~46°.[8] In contrast, at these angles no distinct palladium silver reflections can be resolved above the broad located at 33°. These reflections are only visible for the two catalysts with highest Pd amount. However, their band is broad, indicating very small particle sizes which origin at the low palladium loading (0.025 - 0.1 wt%).

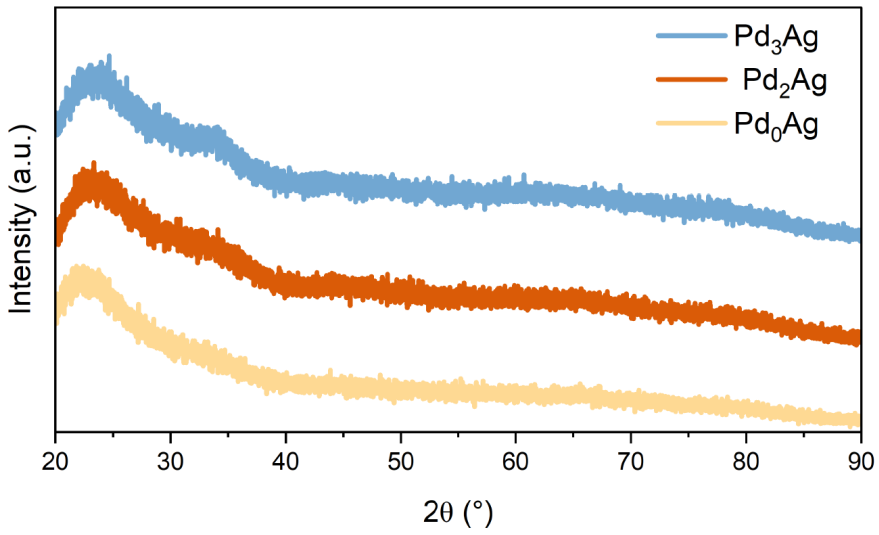


Figure 5.1: XRD of the different prepared catalysts

5.2. Hydrogen Temperature-Programmed-Redcution

H_2-TPR experiments were conducted to study the degree of reducibility of the metal species within the catalysts. As shown in Figure 5.2, the Pd and Ag sites reduce prior to 250°C, which is typical of Pd and Ag-based materials. Moreover, the intensity in H_2 consumption positively varies depending on loading, directly related to the amount of sites capable of consuming H_2 during reduction.

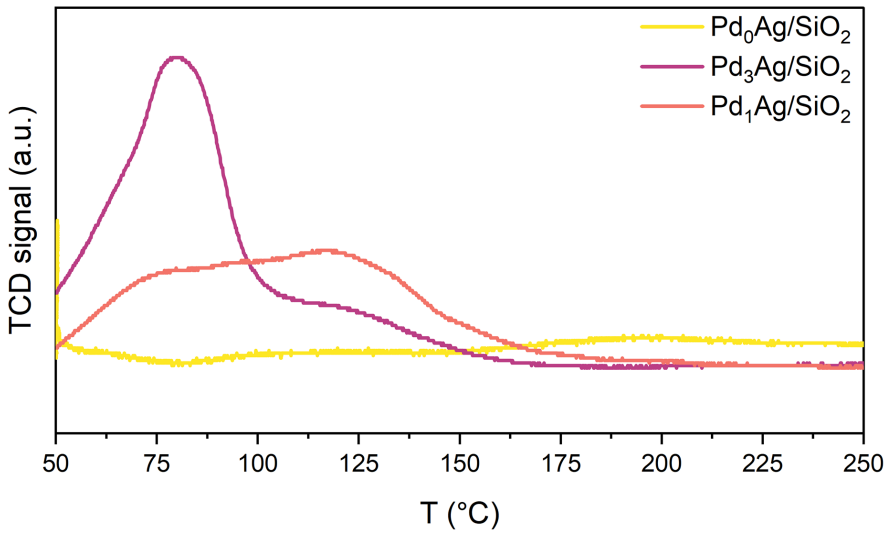


Figure 5.2: TPR of the different prepared catalysts

6

Results & Discussion

6.1. Experimental

To test the effect of each parameter on each catalysts conversion of acetylene and selectivity of ethylene, several experiments have been done on the catalysts. The catalytic performance of each catalyst was tested against different parameters, these parameters being the flow rate, hydrogen concentration, ethylene concentration, and acetylene concentration. All experiments carried out have been performed under the conditions of $100\,°\mathrm{C}$ and $1000\,\mathrm{mbar}$, as can be seen in Table 6.1.

Table 6.1: Catalyst conditions

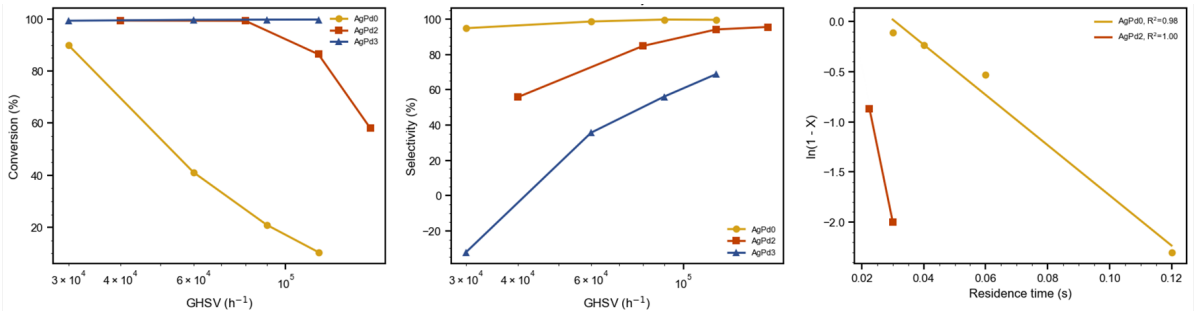
Catalyst	Catalyst mass (mg)	Operating temperature (°C)	Operating pressure (mbar)
Pd_0Ag	100	100	1000
Pd_2Ag	75	100	1000
Pd_3Ag	100	100	1000

6.1.1. Effect of flow rate

First, the effect of the flow rate on the performance of the catalysts was tested. This was done with a composition of 1 vol.% acetylene, 20 vol.% ethylene, 20 vol.% hydrogen, and balance helium. To be able to make a good comparison between the different catalysts and the possible difference in catalyst mass used, a mass normalized flow rate is used as a parameter (GHSV). GHSV is calculated by dividing the flow rate by the mass of the catalyst, which gives this quantity units of $cm^3h^{-1}g_{cat}^{-1}$. Each catalyst is tested at total flows of 50, 100, 150 and $200\,\mathrm{mL\,min^{-1}}$, the GHSV per catalyst can be found in Table 6.2. The effect of GHSV on conversion, selectivity and the first-order behavior is shown in Figure 6.1.

Table 6.2: Mass normalized space velocity calculated for each total flow rate and catalyst.

Flow (mL/min)	$\mathrm{GHSV}_{\mathrm{Pd_0Ag}}$ $(cm^3h^{-1}g_{cat}^{-1})$	$\mathrm{GHSV}_{\mathrm{Pd_2Ag}}$ $(cm^3h^{-1}g_{cat}^{-1})$	$\mathrm{GHSV}_{\mathrm{Pd_3Ag}}$ $(cm^3h^{-1}g_{cat}^{-1})$
50	30000	40000	30000
100	60000	80000	60000
150	90000	120000	90000
200	120000	160000	120000



(a) (b) (c)

Figure 6.1: Influence of flow per catalyst, shown with: (a) conversion plotted against GHSV, (b) selectivity plotted against GHSV, (c) first order check

Conversion

In general, increasing the mass-normalized space velocity leads to a decrease in acetylene conversion, as can be seen in the graph in Figure 6.1(a). This can be explained by the shorter residence time of the gas over the catalyst surface. At higher flow rates, acetylene molecules have less time to adsorb on the catalyst surface and react with hydrogen before leaving the reactor. As a result, more unconverted acetylene remains in the product stream.

Both Pd_0Ag and Pd_2Ag follow this trend, while the conversion of Pd_3Ag remains close to 100% with higher flow rates. This is consistent with the higher palladium loading of Pd_3Ag, which likely provides more available active sites for acetylene hydrogenation. As a result, this catalyst can compensate better for the shorter residence time. For Pd_0Ag and Pd_2Ag, the lower palladium loading likely results in fewer active palladium sites. Therefore, these catalysts cannot compensate for the shorter residence time, leading to a decrease in conversion as the flow rate increases.

The first order check in Figure 6.1(c) was used to evaluate whether the decrease in conversion follows first order behavior. For Pd_0Ag, the data shows a clear linear trend, indicating that the conversion decrease is consistent with first order acetylene consumption. Unfortunately there are only 2 data points for Pd_2Ag, therefore not much can be said about the reaction order. However, first-order behavior too can be assumed for this catalyst.

Selectivity

In contrast to conversion, ethylene selectivity generally increases with increasing flow rates, as shown in Figure 6.1(b). The results show that the selectivity for ethylene increases with increasing GHSV. This trend can be explained by the same residence-time principle used to describe the conversion results. Newly formed ethylene has less time to readsorb on the catalyst surface and undergo further hydrogenation to ethane. Therefore, a larger fraction of the converted acetylene leaves the reactor as ethylene.

For Pd_0Ag, the ethylene selectivity is already high over the full GHSV range and only slightly increases with increasing flow rate. This indicates that further hydrogenation of ethylene is already largely suppressed for this catalyst, even at lower flow rates. Therefore, increasing the flow rate has only a limited additional effect on its selectivity.

For Pd_2Ag a much stronger increase in selectivity is observed, from ~55 to ~99%. This suggests that for this catalyst, at lower flow rates ethylene hydrogenation is more allowed. But with an increase in flow rate this problem can be strongly reduced, resulting in a selectivity of almost 100%.

In contrast, Pd_3Ag shows negative ethylene selectivity at low flow rates. This indicates net ethylene consumption, meaning that the ethylene from the feed and/or newly formed ethylene is further hydrogenated to ethane. This behavior is consistent with the higher palladium loading, which makes it more active for hydrogenation. Even at the highest flow rate, the selectivity remains relatively low compared with other catalysts, showing that this catalyst is too active for selective ethylene formation under these conditions.

Conclusion on GHSV effect

Overall, the flow rate experiments show a clear trade-off between acetylene conversion and ethylene selectivity. Increasing the flow rate decreases the residence time, which can lower acetylene conversion for certain catalysts. At the same time, increased flow rate prevents ethylene hydrogenation, thereby improving selectivity. This trade-off has been found to strongly depend on palladium loading. Pd_0Ag shows the highest selectivity over the full GHSV range, but its lower loading limits its ability to maintain high conversion at higher flow rates. In contrast, Pd_3Ag maintains a high conversion even as the flow rate increases, but shows significantly lower ethylene selectivity than the other catalysts due to stronger overhydrogenation. Pd_2Ag shows the clearest improvement in selectivity with increasing flow rate, suggesting that its performance is especially sensitive to the residence time.

6.1.2. Effect of acetylene

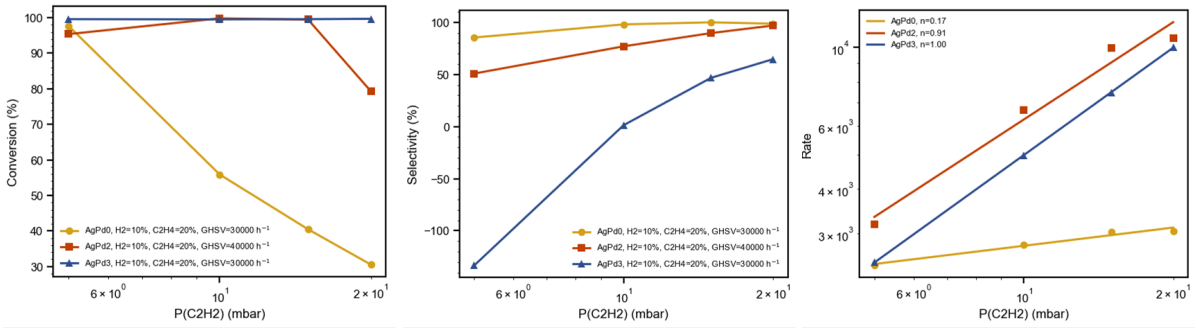
Next, the effect of the acetylene concentration on catalyst performance was investigated. Five sets of experiments were carried out. For each set of experiments, the acetylene concentration was increased from 0.5 to 2 vol.% with steps of 0.5 vol.%.

Five sets of experiments were carried out under different flow rates, hydrogen concentration and ethylene concentration. Sets 1 and 2 were performed at a total flow rate of $100\,\mathrm{mL\,min^{-1}}$, while sets 3 to 5 were performed at $50\,\mathrm{mL\,min^{-1}}$. The corresponding space velocities and gas composition are shown in Table 6.3.

Table 6.3: Experimental conditions used to investigate the influence of acetylene concentration on catalyst performance.

Set	Catalyst	Flow rate (mL/min)	GHSV $(cm^3h^{-1}g_{cat}^{-1})$	vol.% C_2H_4	vol.% H_2	Set	Catalyst	Flow rate (mL/min)	GHSV $(cm^3h^{-1}g_{cat}^{-1})$	vol.% C_2H_4	vol.% H_2
1	Pd_0Ag	100	60000	10	20	3	Pd_0Ag	50	30000	30	20
	Pd_2Ag	100	80000	10	20		Pd_2Ag	50	40000	30	20
	Pd_3Ag	100	60000	10	20		Pd_3Ag	50	30000	30	20
2	Pd_0Ag	100	60000	20	20	4	Pd_0Ag	50	30000	20	10
	Pd_2Ag	100	80000	20	20		Pd_2Ag	50	40000	20	10
	Pd_3Ag	100	60000	20	20		Pd_3Ag	50	30000	20	10
						5	Pd_0Ag	50	30000	20	5
							Pd_2Ag	50	40000	20	5
							Pd_3Ag	50	30000	20	5

The results of set 4 are selected to be discussed in detail because this set shows a clear and generally consistent trend over the investigated acetylene range. In addition, this set was carried out at intermediate ethylene concentration and relatively low hydrogen concentration, making it useful for evaluating the influence of acetylene on both conversion and selectivity. The influence of acetylene partial pressure on conversion, ethylene selectivity, and the acetylene reaction order for set 4 is shown in Figure 6.2. And the results of the other sets of experiments are included in Appendix **??**.



(a) (b) (c)

Figure 6.2: Influence of acetylene concentration on catalyst performance for set 4: (a) acetylene conversion as a function of acetylene partial pressure, (b) ethylene selectivity as a function of acetylene partial pressure, and (c) acetylene reaction order fit.

Conversion

The acetylene conversion as a function of acetylene concentration is shown in Figure 6.2(a). In general, increasing acetylene concentration has the strongest effect on catalysts with lower palladium loading. The conversion of Pd_0Ag decreases strongly with increasing acetylene concentration, while Pd_2Ag only

shows a clear decrease at the highest acetylene concentration. In contrast, Pd_3Ag maintains almost complete conversion over the full investigated range.

For Pd_3Ag, the acetylene remains above 99% over the whole range of acetylene concentration. This indicates that the catalyst has sufficient hydrogenation capacity under these conditions to convert nearly all acetylene, even when more acetylene is in the feed stream.

Pd_2Ag also maintains a good conversion for the lower and medium concentrations, but decreases at higher acetylene concentrations. This suggests that the catalyst can accommodate the increase in acetylene concentration up to a certain point, after which the hydrogenation capacity becomes limiting.

The decrease is much more pronounced for Pd_0Ag. At the highest acetylene concentration, the conversion decreases to ~30%. This shows that this catalyst is less able to maintain high acetylene conversion when acetylene concentration is increased in the feed.

The decrease in conversion at higher acetylene concentrations can be explained by the lower hydrogen to acetylene ratio. Because hydrogen concentration is kept constant while acetylene concentration is increased, more acetylene has to be converted with the same amount of hydrogen. In addition, more acetylene molecules compete for the available active sites. If the catalyst has a limited amount of active sites, the fraction of acetylene that is converted decreases. The acetylene reaction order in Figure 6.2(c) supports this observation. Pd_3Ag shows an acetylene reaction order close to 1, indicating that the reaction rate increases strongly with acetylene partial pressure. This is consistent with the maintained high conversion observed for this catalyst. For Pd_2Ag, the reaction order is also high, but slightly lower than for Pd_3Ag, which agrees with the observed decrease in conversion at the maximum acetylene concentration. In contrast, Pd_0Ag shows a much lower reaction order, indicating that the reaction rate changes only weakly with increasing acetylene concentration. This suggests that the catalyst becomes limited under these conditions, most likely because the available number of active sites is too low to convert the increased acetylene feed.

Selectivity

The effect of acetylene concentration on ethylene selectivity is shown in Figure 6.2(b). For all catalysts, the ethylene selectivity increases with increasing acetylene concentration. This increase is minimal for Pd_0Ag, because the selectivity is already at ~85% at the lowest acetylene concentration. For Pd_2Ag and Pd_3Ag, the increase is much more pronounced.

The minimal increase for Pd_0Ag indicates that overhydrogenation is already strongly limited for this catalyst.

For Pd_2Ag, a stronger increase in selectivity is observed, showing that higher acetylene concentrations strongly suppress the further hydrogenation of ethylene.

The largest increase is for Pd_3Ag, this is mainly because the catalyst shows negative selectivity at low acetylene concentration. This means that mainly ethylene is consumed for the hydrogenation towards ethane. At higher acetylene concentrations, this overhydrogenation is reduced, resulting in a strong increase in selectivity.

Increase in ethylene selectivity can be explained by competitive adsorption between acetylene and ethylene. Acetylene adsorbs more strongly on palladium than ethylene. Therefore, at higher acetylene concentrations, more active sites are occupied by acetylene, making it more difficult for ethylene to readsorb and further hydrogenate to ethane. In addition, the decreasing hydrogen-acetylene ratio with increasing acetylene concentrations can reduce overhydrogenation because relatively less hydrogen is available per hydrogenation reaction.

Conclusion on acetylene effect

Generally, increasing the acetylene concentration has the strongest influence on conversion of the less active catalysts. This effect is especially clear for Pd_0Ag, where the conversion decreases strongly at higher acetylene concentrations. In contrast, Pd_3Ag maintains almost complete acetylene conversion over the full investigated range. This behavior is consistent with its higher palladium loading, which likely provides a larger number of available active sites.

In contrast to conversion, ethylene selectivity increases with increasing acetylene concentration for all catalysts. This can be explained by stronger competitive adsorption of acetylene compared with

ethylene. At higher acetylene concentrations, acetylene occupies more active sites, which suppresses ethylene readsorption and further hydrogenation to ethane. Therefore, increasing the acetylene concentration improves ethylene selectivity, but can lower acetylene conversion if the catalyst does not have enough active sites to hydrogenate the higher concentration of acetylene.

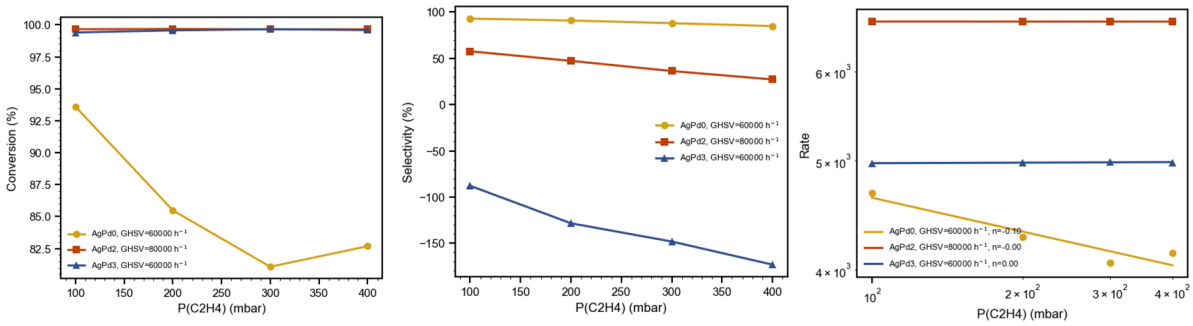
6.1.3. Effect of ethylene

To investigate the effect that the change of ethylene has on the catalyst performance, experiments were carried out at two different flow rates. Set 1 was performed at $50\,\mathrm{mL/min}$, while set 2 was performed at $100\,\mathrm{mL/min}$. Because different catalyst masses were used, the corresponding GHSV values differ between the catalysts, as shown in Table 6.4. During these experiments, the acetylene and hydrogen concentrations were kept constant, while the concentration of ethylene increased from 10 to 40 vol.%, with steps of 10 vol.%.

Table 6.4: Experimental conditions used to investigate the influence of ethylene concentration on catalyst performance.

Catalyst	Set	Flow rate (mL/min)	GHSV $(cm^3h^{-1}g_{cat}^{-1})$	vol.% C_2H_2	vol.% H_2
Pd_0Ag	1	50	30000	1	20
	2	100	60000	0.5	20
Pd_2Ag	1	50	40000	1	20
	2	100	80000	0.5	20
Pd_3Ag	1	50	30000	1	20
	2	100	60000	0.5	20

Set 2 was selected for the main discussion because the trends in conversion and selectivity are more pronounced at the higher flow rates. This makes it easier to compare the influence of ethylene concentration between the different catalysts. The influence of ethylene partial pressure on acetylene conversion, ethylene selectivity, and the ethylene reaction order fit for set 2 is shown in Figure 6.3, while the results of set 1 can be found in the appendix **??**.



(a) (b) (c)

Figure 6.3: Influence of ethylene concentration on catalyst performance for set 2: (a) acetylene conversion as a function of ethylene partial pressure, (b) ethylene selectivity as a function of ethylene partial pressure, and (c) ethylene reaction-order fit.

Conversion

The acetylene conversion as a function of ethylene partial pressure is shown in Figure 6.3(a). The increasing of ethylene concentration has only a limited effect on the conversion of Pd_2Ag and Pd_3Ag. Both these catalysts maintain a high conversion over the whole ethylene concentration range. In contrast, Pd_0Ag shows a more notable decrease in conversion with increasing ethylene concentration.

This trend can also be seen in the ethylene reaction order fit shown in Figure 6.3(c). Both Pd_2Ag and Pd_3Ag show an ethylene reaction order near zero. This indicates that under these conditions, the acetylene hydrogenation rate is nearly independent of ethylene partial pressure. For Pd_0Ag, a light negative reaction order is found. This is consistent with the decrease in acetylene conversion at higher ethylene concentrations.

The decrease in conversion for Pd_0Ag can be explained by the competitive adsorption between ethylene and acetylene for the catalysts active sites. Although acetylene adsorbs stronger than ethylene,

the much higher ethylene partial pressure can compensate for this, increasing the surface coverage of ethylene. As a result, fewer active sites are available for acetylene to adsorb on, causing a weak inhibiting effect. For Pd_0Ag this effect is most visible, because a lower palladium loading likely results in fewer available palladium active sites. Blocking even a small fraction of these sites can result in a larger relative effect on acetylene conversion.

Selectivity

The influence of ethylene concentration on ethylene selectivity is shown in Figure 6.3(b). Every catalyst follows the same trend, increasing ethylene concentration leads to a decrease in selectivity. This indicates that a higher ethylene partial pressure promotes further hydrogenation of ethylene to ethane.

The decrease in selectivity is strongest for Pd_3Ag. At low ehtylene concentration this catalyst already shows a relatively low selectivity, and the selectivity decreases further when the ethylene concentration is increased. This indicates that this catalyst has a strong tendency to hydrogenate ethylene, especially at higher ethylene partial pressures.

Pd_2Ag shows a more moderate decrease. This indicates that the catalyst has a lower tendency for over-hydrogenation to ethane than Pd_3Ag, while still being sensitive to changes in ethylene partial pressure.

For Pd_0Ag, the decrease in selectivity is the smallest. The selectivity remains relatively high over the investigated ethylene concentration range. This shows that further hydrogenation is suppressed the strongest by this catalyst.

The decrease in ethylene selectivity with increasing ethylene concentration can be explained by the higher probability of ethylene readsorption by the catalyst. This effect is especially strong for catalysts with higher palladium loading, like Pd_3Ag, because more active sites are available for the ethylene to be hydrogenated. In contrast, for catalysts with lower palladium loading, such as Pd_0Ag, the loading likely limits the number of active sites available for hydrogenation, which helps maintain a higher selectivity.

Conclusion on ethylene effect

In general, increasing the ethylene concentration has only a limited effect on the conversion for Pd_2Ag and Pd_3Ag. For these catalysts, the reaction order with respect to ethylene is close to zero, indicating the acetylene hydrogenation rate is virtually independent of ethylene partial pressure under these conditions. The effect on selectivity is much stronger for all catalysts. Increasing the ethylene concentration decreases ethylene selectivity, as can be seen for every catalyst. This shows that higher ethylene partial pressure promotes further hydrogenation to ethane. This effect has been found to be the strongest for Pd_3Ag, which remains highly active, and the weakest for Pd_0Ag, where the lower palladium loading likely limits the further hydrogenation.

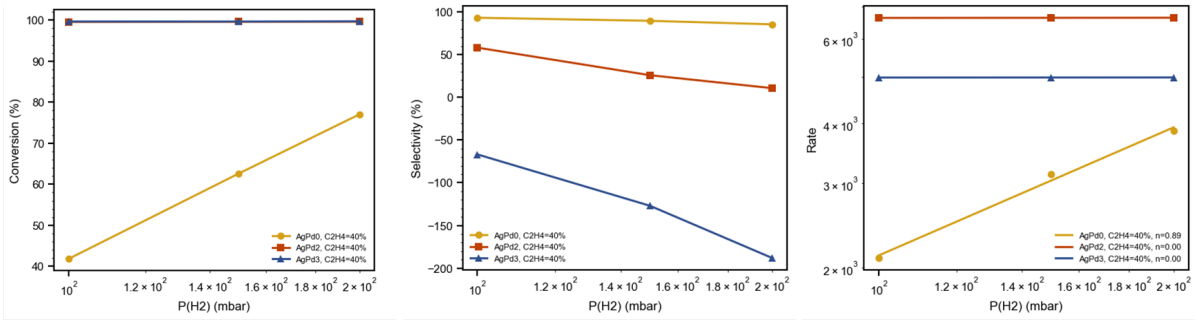
6.1.4. Effect of hydrogen

The effect of hydrogen concentration on the catalyst conversion and selectivity was also investigated. For each catalyst this was done at a total flow rate of $100\,\mathrm{mL/min}$. In set 1, the ethylene concentration was kept at 20 vol.%, while for set 2 the ethylene concentration was kept at 40 vol.%. The remaining reaction conditions were kept constant and are shown in Table 6.5. For set 1 the experiments were carried out with a hydrogen concentration range of 5 to 25 vol.%, while for set 2 it was varied from 5 to 20 vol.%.

Table 6.5: Experimental conditions used to investigate the influence of hydrogen concentration on catalyst performance.

Catalyst	Set	Flow rate (mL/min)	GHSV ($cm^3h^{-1}g_{cat}^{-1}$)	vol.% C_2H_2	vol.% C_2H_4
Pd_0Ag	1	100	60000	0.5	20
	2	100	60000	0.5	40
Pd_2Ag	1	100	80000	0.5	20
	2	100	80000	0.5	40
Pd_3Ag	1	100	60000	0.5	20
	2	100	60000	0.5	40

The results from set 2 were selected for the main discussion because the higher ethylene concentration makes the influence of hydrogen on conversion, selectivity and overhydrogenation more clearly visible. The influence of hydrogen partial pressure on conversion, selectivity, and the hydrogen reaction order fit for set 2 is shown in Figure 6.4, and the results of set 1 are included in Appendix **??**.



(a) (b) (c)

Figure 6.4: Influence of H_2 per catalyst, set 2, shown with: (a) conversion plotted against P_{H_2}, (b) selectivity plotted against P_{H_2}, (c) rate order fit

Conversion

The influence of the hydrogen concentration on acetylene conversion is shown in Figure 6.4(a). For all catalysts, increasing the hydrogen concentration has a positive effect on acetylene conversion. This effect is most clearly visible for Pd_0Ag, where the conversion increases strongly with increasing hydrogen concentration. In contrast, the conversion for Pd_2Ag and Pd_3Ag remains close to 100% over the investigated hydrogen concentration range.

For Pd_0Ag, the conversion increases significantly, which shows that hydrogen concentration has a strong influence on the activity of this catalyst. At lower hydrogen concentrations the catalyst is not able to convert all acetylene, while at higher concentrations the catalyst increasingly is able to.

For both Pd_2Ag and Pd_3Ag, the conversion remains near 100% conversion. This indicates that these catalysts are active enough on their own to convert virtually all acetylene. Therefore, increasing the hydrogen concentration barely influences the conversion for these catalysts.

The hydrogen reaction order fit in Figure 6.4(c) supports these observations. For Pd_0Ag, the reaction order is ~0.89, which is close to first order. This indicates that the acetylene hydrogenation rate strongly depends on the partial pressure of hydrogen. This is consistent with the strong increase in conversion for this catalyst.

In contrast, the reaction order for Pd_2Ag and Pd_3Ag is close to zero. This indicates that, under these condition, the acetylene hydrogenation rate is nearly independent of hydrogen partial pressure, agreeing with the constant conversion of both catalysts.

The positive effect of hydrogen concentration on conversion can be explained by the increased availability of hydrogen on the catalyst surface. At higher concentrations, hydrogen partial pressure increases, meaning more hydrogen can adsorb on palladium sites, making more activated hydrogen available for acetylene hydrogenation.

For Pd_0Ag, this effect is particularly important because of its lower palladium loading. This likely limits the number of sites available to activate hydrogen on at lower hydrogen concentrations. However, when hydrogen partial pressure does increase, more hydrogen can adsorb on the active sites making more hydrogen available for the hydrogenation reaction. This limitation is less visible for Pd_2Ag and Pd_3Ag, these catalyst already achieve nearly complete conversion at low hydrogen concentration suggesting both catalysts have enough active sites to activate enough hydrogen for the hydrogenation of acetylene.

Selectivity

The influence of hydrogen on the selectivity of the catalysts can be seen in Figure 6.4(b). In this case, every catalyst follows the same general trend, showing a decrease in selectivity with increasing hydro-

gen concentration. This shows that a higher hydrogen partial pressure also promotes hydrogenation of ethylene.

The decrease is smallest for Pd_0Ag. Even at higher hydrogen concentrations, the selectivity remains relatively high. This suggests that hydrogenation of ethylene is largely suppressed for this catalyst.

For Pd_2Ag, the decrease in selectivity is more pronounced. At higher hydrogen concentrations the selectivity approaches slightly negative values. This shows that ethylene hydrogenation is becoming increasingly important.

The most significant decrease can be found for Pd_3Ag. This catalyst shows strong negative selectivity at higher hydrogen concentrations, indicating strong over-hydrogenation towards ethane. From this can be found that the catalyst is too active for selective hydrogenation in hydrogen rich conditions.

the decrease in ethylene selectivity with increasing hydrogen concentration can be explained by the increase of activated hydrogen available on the catalyst surface. A higher hydrogen partial pressure increases the probability that when ethylene is adsorbed it undergoes hydrogenation to ethane. This effect is strongest for catalysts with higher palladium loading, such as Pd_3Ag, because more active sites are available for hydrogen activation and ethylene hydrogenation. In contrast, for lower palladium loaded catalysts, like Pd_0Ag, the number of active sites is less making it harder for the overhydrogenation. As a result, the ethylene selectivity remains relatively high.

Conclusion for hydrogen

In general, increasing the hydrogen concentration has a positive effect on the conversion of Pd_0Ag, while the conversion of Pd_2Ag and Pd_3Ag remains almost unchanged. This is because these catalysts already reach nearly complete conversion at the lowest hydrogen concentration. For Pd_0Ag, the positive reaction order with respect to hydrogen shows that the reaction rate is strongly influenced by hydrogen concentration.

However, the effect on selectivity is negative for every catalyst. Increasing the hydrogen concentration promotes further hydrogenation of ethylene towards ethane, causing ethylene selectivity to decrease. Pd_0Ag remains the most selective catalyst, while Pd_3Ag shows the strongest decrease in selectivity due to severe overhydrogenation. This shows that a higher hydrogen concentration can be beneficial for conversion for catalysts with low palladium loadings, but reduces selectivity, especially for catalysts with a higher palladium loading.

6.2. Kinetic modeling

6.2.1. Standard LHHW model

The LHHW model is often used to describe hydrogenation kinetics, but the standard form of the model does not describe all catalysts with the same accuracy, as can be seen in the results for each catalyst in Figure 6.5.

For Pd_0Ag, the model gives a reasonable description of the experimental data, as shown in Figure 6.5(a), with an overall RMSE of 17.54 percentage points. Especially the selectivity is fitted well with 100% of the points in the $\pm$10% band. In contrast, the model does not give a good fit for conversion. All data points are widely scattered, with a small number of points falling within the $\pm$10% band. This also explains the high RMSE of 30.36 percentage points. This suggests that for Pd_0Ag the standard LHHW model gives a good description of selectivity, but its conversion fit is relatively weak over the whole experimental range.

As can be seen in Figure 6.5(b) the standard LHHW model seems to make a better fit for Pd_2Ag, with an overall RMSE value of 11.98 percentage points. Especially the acetylene conversion seems better fitted for this catalyst, with 83.3% within the $\pm$10% band. However, the model predicts $\sim$100% conversion for every condition, which suggests the data is better fitted than it actually is since a lot of experimental data is also near 100%. However, this should be interpreted carefully because every point is predicted near 100%, making it more likely that the fitted parameters are physically meaningless. In contrast, the modeled selectivity for this catalyst is poorly described with only 28.6% inside the $\pm$10% band. At higher experimental selectivity the model appears to underestimate the selectivity, while at the lower experimental selectivity the simulated selectivity is overestimated. This indicates that

the standard LHHW model does not fully capture the competition between r_1 and r_2 for Pd_2Ag.

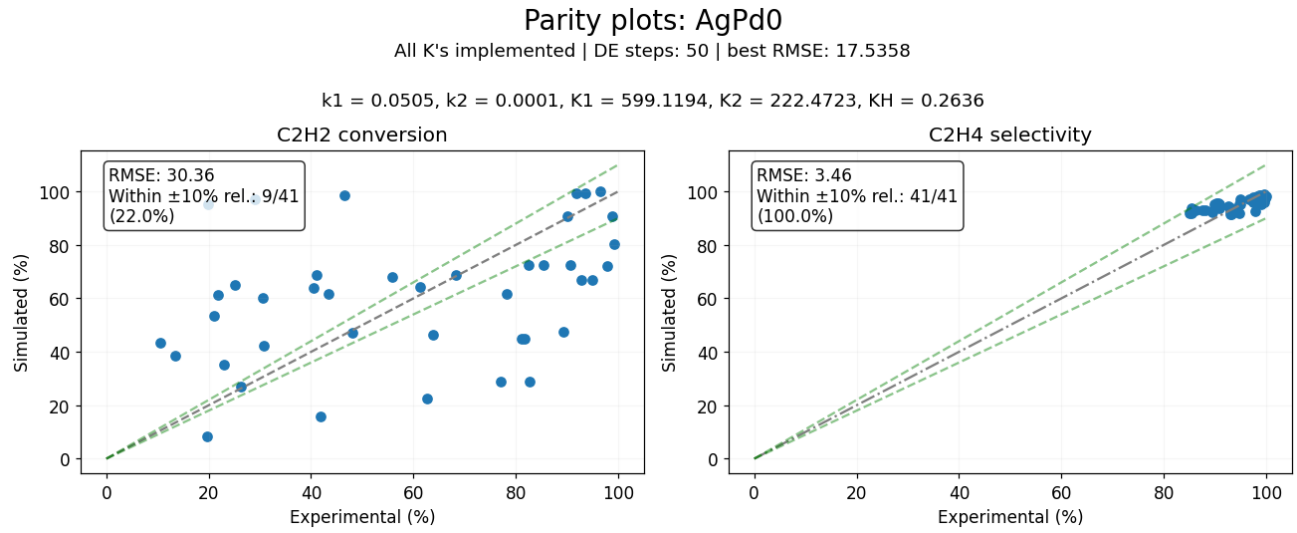
Finally, the fit for Pd_3Ag can be found in Figure 6.5(c), for this catalyst the overall RMSE is 43.08 percentage points, significantly worse than the other catalysts. Acetylene conversion seems to have been described reasonably well with 81.8% of the points within the $\pm$10% band. However, for this catalyst too all simulated conversion points are at $\sim$100% even though not every experimental value is $\sim$100%. Once again making physically meaningless parameters very likely. For ethylene selectivity it becomes clear this model performs poorly, confirmed with a selectivity RMSE of 81.82 percentage points. This shows that for Pd_3Ag the LHHW model also cannot accurately describe the competition between r_1 and r_2.

Shortcomings standard LHHW model

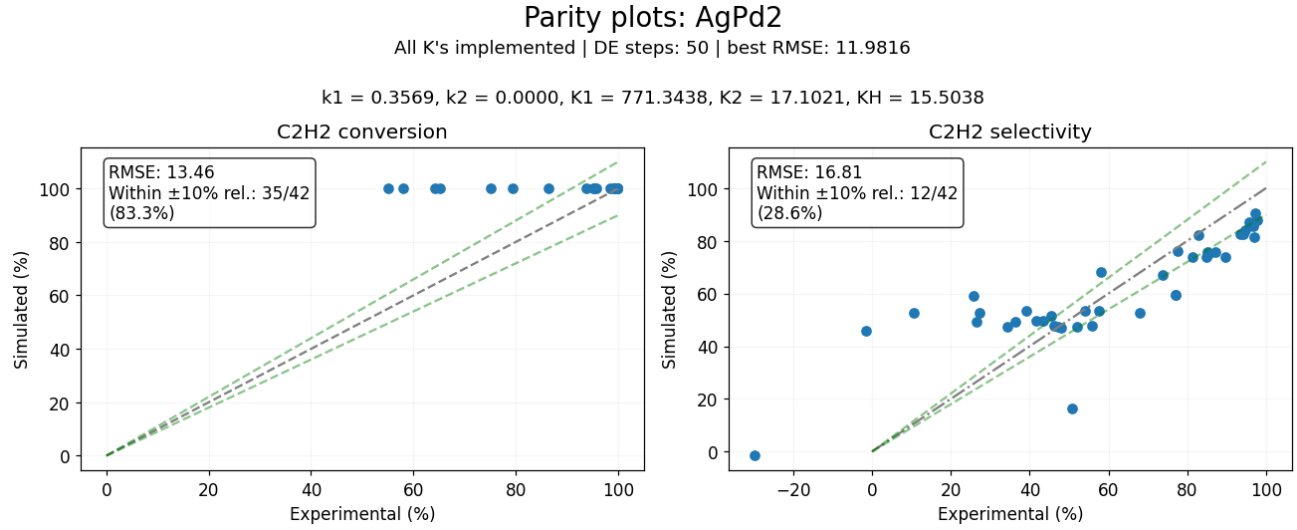
The standard LHHW model has shown that for the Pd_0Ag catalyst it gives a mixed description of the experimental data. The quality differs strongly between conversion and selectivity. The selectivity is described well, while the conversion fit is less accurate. At lower and intermediate conversion, the model over- and under predicts data points indicating that it does not capture the correct dependence of conversion. At higher experimental conversion, several points are simulated too low, meaning that the model also struggles to reproduce conditions where the catalyst experimentally reaches high conversion. Therefore, the standard LHHW model captures the conversion trend only weakly, but it is not flexible enough to accurately simulate the full conversion behavior. From Figure 6.1(a) the conversion does decrease with increasing flow rate. This suggests that mass transfer limitations are not limiting for this catalyst, since mass transfer limited reactions would show little sensitivity to flow rate. Nevertheless, this bad fit suggests that the fitted parameters are not really physically meaningful.

In contrast, for Pd_2Ag it seems the model describes the catalyst relatively well. Especially acetylene conversion is suggested to be simulated quite well, but every simulated conversion is $\sim$100%. In contrast, the selectivity fit does not seem so good. At lower and intermediate experimental values the model tends to overestimate it. This indicates that the standard LHHW model does not accurately describe the balance between r_1 and r_2. This balance can come from the same mistake that ensures every simulated conversion is $\sim$100%. A possible explanation could be parameter compensation of k_1 and K_1. It is plausible that these parameters are artificially high to fit most of the experimental $\sim$100% results rather than correctly describing the kinetics of the catalyst. This parameter compensation can also be the reason the ethylene selectivity fit is off. An artificially high k_1 might force the model toward high acetylene conversion, which can indirectly affect the predicted balance between ethylene formation and ethane formation. Compensation of K_1 increases surface coverage of acetylene, this can suppress r_2 and increase simulated selectivity. This can explain the overestimation of selectivity at low experimental selectivity and underestimation at higher experimental selectivity. In Figure 6.1(a) in the beginning the conversion is not influenced by increasing GHSV, consistent with possible external mass transfer limitations. However, after a certain point the conversion does drop, which is expected for a catalyst without physical limitations. Therefore, even though the overall RMSE suggests a relatively well fit, it is very likely that the parameters are physically meaningless and the good fit is artificially created.

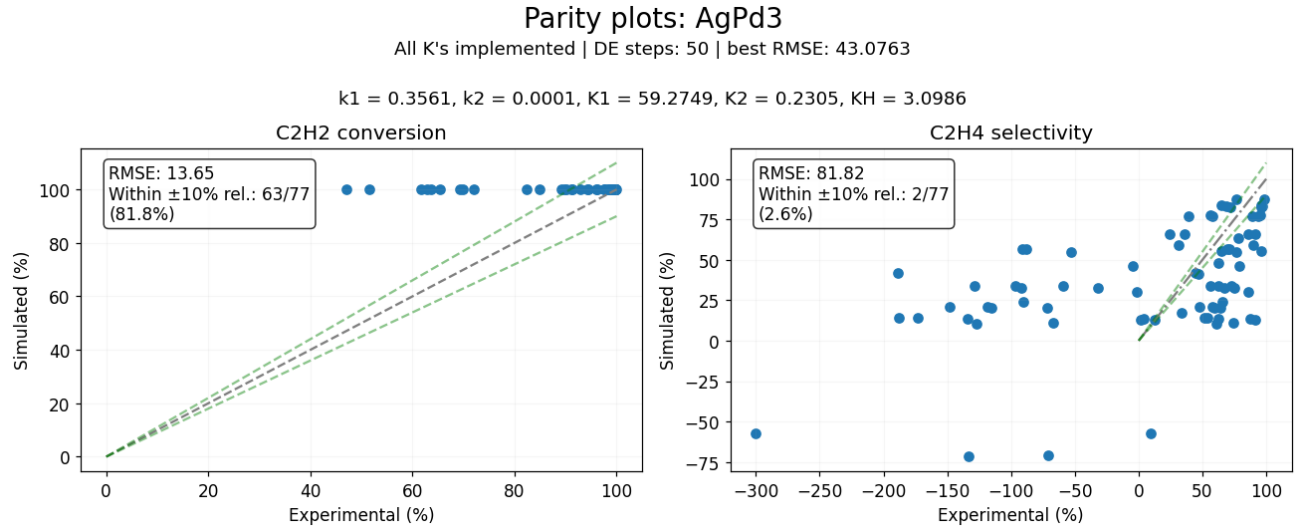
The LHHW model does significantly show its limitations for the Pd_3Ag catalyst. Although the simulated conversion seems relatively well fit, this is again because the model predicts $\sim$100% conversion for every condition. Once again, this too suggests the idea of parameter compensation. The modeled selectivity values similarly deviate substantially from experiment, showing that the LHHW model requires modification to accurately describe the competition between r_1 and r_2. For this catalyst Figure 6.1(a) does suggest mass transfer limitations need to be taken into account. In this figure the conversion of Pd_3Ag does not decrease with increasing GHSV. This behavior is consistent with possible external mass transfer limitations. However, because the conversion remains close to 100%, this trend alone is not sufficient to prove mass transfer limitations.



(a)



(b)



(c)

Figure 6.5: Fit of the standard LHHW model, with fitted k_1, k_2, K_1, K_2, K_H for: (a) Pd_0Ag,(b) Pd_2Ag, (c) Pd_3Ag

6.2.2. LHHW model modification

As mentioned the standard LHHW model was used as a starting point for the kinetic modeling (Equation 6.1 and 6.2). Especially for Pd_2Ag and Pd_3Ag the model has shown to not describe either experimental conversion or selectivity. Therefore several modifications were made to the LHHW model, resulting in different variants of the LHHW model. For every catalyst the same kinetic model changes have been made. The different variants of the LHHW model were modeled stepwise. First the standard LHHW model was stripped down, in which specific adsorption terms, or partial pressures were removed. After this, different models were tested with a r_2 inhibition constant, linear and squared.

The parameters used in each variant can be found in Appendix tables E.1 to E.3, together with every calculated RMSE, $\text{RMSE}_{conversion}$ and $\text{RMSE}_{selectivity}$ values. The target of these different variations was not only to reduce the RMSE, but also to evaluate whether the fitted parameters are physically meaningful.

$$r_1 = \frac{k_1 K_{C_2H_2} p_{C_2H_2} K_{H_2} p_{H_2}}{\left(1 + K_{C_2H_2} p_{C_2H_2} + K_{C_2H_4} p_{C_2H_4} + K_{H_2} p_{H_2}\right)^2} \tag{6.1}$$

$$r_2 = \frac{k_2 K_{C_2H_4} p_{C_2H_4} K_{H_2} p_{H_2}}{\left(1 + K_{C_2H_2} p_{C_2H_2} + K_{C_2H_4} p_{C_2H_4} + K_{H_2} p_{H_2}\right)^2} \tag{6.2}$$

Simplified LHHW model (v4a)

The simplification by removal of the hydrogen parameters resulted in v4a. The rate equations for this variant are rate equations 6.3 and 6.4. The fit this resulted in for each catalyst can be found in Appendix G.1.

$$r_1 = \frac{k_1 K_{C_2H_2} p_{C_2H_2}}{\left(1 + K_{C_2H_2} p_{C_2H_2} + K_{C_2H_4} p_{C_2H_4}\right)^2} \tag{6.3}$$

$$r_2 = \frac{k_2 K_{C_2H_4} p_{C_2H_4}}{\left(1 + K_{C_2H_2} p_{C_2H_2} + K_{C_2H_4} p_{C_2H_4}\right)^2} \tag{6.4}$$

For Pd_0Ag, the overall RMSE increases with $\sim$4.2 percentage points. Although the selectivity is described very well with an RMSE of only 3.45 percentage points and 100% of all points plotted within the $\pm$10% band, the conversion fit is worse. By removing it the RMSE increased to 35.62 percentage points. This is also supported by the hydrogen order having a value between 0.84 and 0.89 in Figure 6.4(c), suggesting hydrogen does influence the acetylene hydrogenation rate. From this we can conclude that the hydrogen parameter in the LHHW model is needed for this catalyst.

In addition, the overall RMSE for Pd_2Ag increases with $\sim$2.6 percentage points with the removal of hydrogen parameters. This increase can also be seen for the selectivity RMSE. This increases with $\sim$3.66 percentage points. For the simulated conversion fitted points, the model does suggest that the newly fitted parameters are more physically meaningful because not every simulated conversion is $\sim$100% anymore.

The hydrogen order analysis supports this observation. The apparent hydrogen orders of $\sim$0 in Figure 6.4(c) show that the acetylene conversion rate is almost independent of hydrogen and partial pressure under the investigated conditions. This suggests that hydrogen does not strongly control the acetylene consumption rate for Pd_2Ag. Generally speaking, the removal of the hydrogen parameters barely influences the overall RMSE, suggesting that the hydrogen parameters are not necessarily needed in the adsorption term. At the same time, the simulated conversion data does become physically more meaningful, so a better conclusion from these plots would be that the model fit does not strongly require explicit hydrogen dependence under these conditions.

The removal of hydrogen parameters for Pd_3Ag shows a slight improvement in overall RMSE of $\sim$0.3 percentage points. However, for this catalyst, this improvement also seems negligible when looking at the graph in Figure G.1(c). The conversion seems to not change at all, maintaining the same RMSE. This is supported by the hydrogen order for Pd_3Ag being $\sim$0 in Figure 6.4(c). Indicating that

the acetylene consumption rate is independent of hydrogen. Nevertheless, removing the hydrogen parameters has not made the simulated data for conversion physically meaningful, every point still reaches ~100% conversion suggesting parameter compensation of k_1 and K_1. Even though the selectivity RMSE decreases, the RMSE value of 76.29 percentage points is still too high to be meaningful. Therefore, the removal of the hydrogen adsorption term gives a numerical improvement in the overall RMSE, but does not result in a physically reliable model for Pd_3Ag.

The simplification of the standard LHHW model by the removal of hydrogen parameters shows us that for Pd_0Ag it is important to keep the parameters within the model. This is confirmed by a clear worsening of the conversion fit, while also confirmed by the hydrogen reaction order. The results for Pd_2Ag suggest that the hydrogen parameters can probably be left out of the denominator. The general fit barely changes, and the hydrogen reaction order is ~0. This does not mean that hydrogen is unimportant for the reaction but that hydrogen parameters are not needed in the denominator to describe the reaction. In contrast, for Pd_3Ag the hydrogen parameters are needed to give a meaningful result. The removal did not result in meaningfull simulated conversion across the whole region. In short, for Pd_0Ag and Pd_3Ag the hydrogen parameters are necessary, while for Pd_2Ag it is not in the denominator but does give a slightly worse fit when included.

LHHW model with acetylene inhibition (v5)

The next modification made to the LHHW model is the addition of an acetylene inhibition term in the denominator of the rate equation for the hydrogenation of ethylene to ethane, resulting in v5. The rate equations this addition gives us can be found in Equation 6.5 and 6.6. The resulting fit for every catalyst is found in Appendix fig. H.1.

$$r_1 = \frac{k_1 K_{C_2H_2} p_{C_2H_2} K_{H_2} p_{H_2}}{\left(1 + K_{C_2H_2} p_{C_2H_2} + K_{C_2H_4} p_{C_2H_4} + K_{H_2} p_{H_2}\right)^2} \tag{6.5}$$

$$r_2 = \frac{k_2 K_{C_2H_4} p_{C_2H_4} K_{H_2} p_{H_2}}{\left(1 + K_{C_2H_2} p_{C_2H_2} + K_{C_2H_4} p_{C_2H_4} + K_{H_2} p_{H_2}\right)^2 \left(1 + K_{inh} p_{C_2H_2}\right)} \tag{6.6}$$

Figure fig. H.1(a) shows that for Pd_0Ag the addition of this inhibition term quite significantly improves the overall fit compared to the regular LHHW model. The overall RMSE decreases by ~11.10 percentage points. The strongest improvement can be found for conversion. The RMSE for conversion decreases to 9.73 percentage points. This shows that the regular LHHW model is not able to describe the acetylene conversion well, while the inhibition term gives the model more flexibility to follow the conversion trend. The selectivity was already described pretty well by the regular LHHW model. The RMSE increases by 1.02 percentage points, showing a worse fit with the addition of the inhibition term. However, this decrease is small compared to the strong improvement in the conversion fit. This suggest that acetylene inhibition plays an important role for Pd_0Ag. Chemically, this sounds plausible because acetylene forms a stronger bond with palladium, inhibiting the hydrogenation of ethylene. As a result the addition of the inhibition term helps improve the conversion fit because the model gains more flexibility to describe both hydrogenation steps separate. The increase for selectivity is not really meaningful, because the fit is still good. Therefore, the conclusion can be made that the addition of the inhibition term overall improves the models fit.

In contrast, the addition of the inhibition term barely influences the fit for Pd_2Ag. The overall RMSE decreases from 12.75 to 12.72 percentage points. For the conversion, RMSE does not change between the models, and the selectivity RMSE of the model slightly decreases with 0.33 percentage points. Which shows that for this catalyst the inhibition term likely is not strong enough. This suggests that the addition of a linear inhibition term does not significantly influence the models fit of the catalyst.

For Pd_3Ag the addition of the inhibition term slightly improves the fit of the model. The overall, conversion and selectivity RMSE values decrease, with changes of ~1.07, ~0.89 and ~9.15 percentage points, respectively. This suggests that the addition of an inhibition term helps the model predict the catalyst behavior, but the overall fit is still far off. Therefore, inhibition of acetylene in the second

hydrogenation step is an improvement, but not significant enough to meaningfully describe Pd_3Ag.

The influence of the addition of the acetylene inhibition term has been shown to be catalyst dependent. For Pd_0Ag the inhibition term improves the conversion fit, decreasing the overall RMSE. This suggests that acetylene inhibition can help describe the catalysts activity. However, for Pd_2Ag, the addition of the term only slightly improves the models fit. This indicates that acetylene inhibition can help describe the catalysts behavior, but does not significantly improve it. In addition, for Pd_3Ag the addition barely shows any change to the fit, only slightly improving the fit. This means that for this catalyst too the addition of the inhibition term can help describe the catalyst, but not significantly enough to have a meaningful fit. In short, the addition of the inhibition term significantly improves the Pd_0Ag fit, while only very slightly improving the fit for Pd_2Ag and Pd_3Ag.

LHHW model with acetylene inhibition squared (v5d)
In addition to just the inhibition term as a term in the LHHW model, the models fit with the inhibition term squared was also calculated, resulting in v5d and reaction rate equations 6.7, 6.8. The results for this updated model can be found in Appendix I.1.

$$r_1 = \frac{k_1 K_{C_2H_2} p_{C_2H_2} K_{H_2} p_{H_2}}{\left(1 + K_{C_2H_2} p_{C_2H_2} + K_{C_2H_4} p_{C_2H_4} + K_{H_2} p_{H_2}\right)^2} \tag{6.7}$$

$$r_2 = \frac{k_2 K_{C_2H_4} p_{C_2H_4} K_{H_2} p_{H_2}}{\left(1 + K_{C_2H_2} p_{C_2H_2} + K_{C_2H_4} p_{C_2H_4} + K_{H_2} p_{H_2}\right)^2 \left(1 + K_{inh} p_{C_2H_2}\right)^2} \tag{6.8}$$

For Pd_0Ag, the addition of the squared inhibition term has increased the overall RMSE compared to the regular LHHW model and linear inhibition term. This model shows an overall RMSE of 18.07 percentage points, this increase overall RMSE can mainly be attributed to the worse conversion fit. The conversion RMSE increases with ∼21.02 percentage points compared to the linear inhibition term. The selectivity fit too becomes slightly worse but is minimal compared to the decrease for conversion. Chemically this suggests that the influence of acetylene inhibition is described strong enough by linear inhibition. By squaring the term, the inhibition term has become too strong resulting in a worse fit for Pd_0Ag.

The squaring of the inhibition term for Pd_2Ag barely influences the models fit. The overall RMSE is slightly better than that of the linear inhibition term. The conversion RMSE remains the same, and the selectivity RMSE decreases slightly. Chemically this means that for this catalyst the linear inhibition term is sufficient to describe the acetylene inhibition. This means that squaring the inhibition term for this catalyst does not give a meaningful benefit, suggesting that a strong inhibition effect is not necessarily needed to model for Pd_2Ag.

The graph in Figure 6.6(c) shows no significant difference compared to the linear inhibition term for Pd_3Ag. The overall RMSE slightly increases, while both the conversion and selectivity RMSE show a negligible increase. This means that for this catalyst increasing the influence of the inhibition effect does not improve its fitting. This confirms that very likely for Pd_3Ag the inhibition term is not the effect that explains the relatively poor model fit.

Overall, the effect of squaring the inhibition term has again shown to be catalyst dependent. For Pd_0Ag, squaring the term has shown a worse fit to the model. This suggests that for this catalyst acetylene inhibition may still be important, but a linear inhibition term appears to describe it better than a squared term. The linear inhibition term has been proven to be enough for Pd_2Ag. The model barely changes by squaring the term, suggesting that the reaction is not very sensitive to whether the inhibition term is linear or squared. Pd_3Ag has shown no improvement by adding the squared inhibition term, suggesting that this catalyst is not strongly governed by nonlinear acetylene inhibition.

Overview of LHHW model modifications
In total, three modifications to the standard LHHW model were evaluated to determine which physical effects are needed to best describe acetylene hydrogenation over the different PdAg catalysts. The

standard LHHW model includes adsorption terms for acetylene, ethylene and hydrogen in the denominator to describe competitive adsorption on the catalyst active sites. This means that the model does not only describe the reaction rate through the rate constants, but also through the surface coverage of the different adsorbing species.

The first modifications were made to see whether the LHHW model could be simplified. This was done by removing the hydrogen terms in the model, evaluating the hydrogen importance in the model. If the simplified model gave a similar or better fit, this would suggest that the hydrogen terms are not necessary to describe the data. However, this does not mean hydrogen has no influence on the reaction.

Secondly, an inhibition term was added to the rate equation for ethylene hydrogenation. This was added because acetylene can inhibit the second hydrogenation step by blocking active catalyst sites through its stronger adsorption on palladium active sites. The addition of this term therefore makes it possible to describe a situation where ethylene hydrogenation is suppressed by the presence of acetylene. This can improve the selectivity fit because it suppresses the modeled formation of ethane.

Finally, the inhibition term for ethylene hydrogenation was squared. This modification increases the strength of the inhibition effect at higher acetylene concentrations. Chemically, this can represent stronger site blocking or a more pronounced suppression of ethylene hydrogenation when the surface is highly covered by acetylene. This modification can therefore be found useful when the linear inhibition term is too weak to describe the observed decrease in ethylene hydrogenation.

The Pd_0Ag catalyst was already described quite well by the original LHHW model. In particular the selectivity is fitted well, while the conversion is less accurate, especially conversion at lower rates is not simulated accurately. Removal of the hydrogen term worsens the fit significantly by increasing the overall RMSE, even though the selectivity remains described well. This suggests that for this catalyst the hydrogen term should stay in the model. The addition of the inhibition term shows a decrease in overall RMSE, especially when the inhibition term is squared. This suggests that acetylene inhibition is important for this catalyst and that this inhibition becomes stronger at higher acetylene coverage. Therefore, for Pd_0Ag, hydrogen should be retained in the model and among the tested variants, and the LHHW model with the squared acetylene inhibition term gives the most suitable description of the data.

For Pd_2Ag, the regular LHHW model seems to describe the reaction reasonably well. However, all the simulated conversion points are $\sim$100% meaning that the values are not fully physically meaningful. In addition, the selectivity is described less well, showing a relatively large spread throughout the selectivity range. This combination suggests that the regular LHHW model can describe the overall trends to some extent, but that the conversion fit should be interpreted carefully because the model tends to over predict conversion. The test in which the hydrogen terms were omitted shows a slightly worse fit. This suggests that the hydrogen terms are not dominant for Pd_2Ag, but because the fit becomes slightly worse, it is still preferable to continue with the regular LHHW model. Adding the inhibition term improves the overall model fit, mainly by providing a better selectivity fit. The squared inhibition term gives a similar fit as the linear inhibition term. This is consistent with the chemical explanation that acetylene inhibits the further hydrogenation of ethylene by blocking active palladium sites. The effect of the inhibition term appears less pronounced than for Pd_0Ag, which may be related to the higher palladium content and the resulting difference in available active sites. Therefore, for Pd_2Ag, the LHHW model with the linear acetylene inhibition term gives the most suitable description of the data, but the conversion fit should still be interpreted carefully because the model tends to over predict conversion.

The regular LHHW model describes Pd_3Ag reasonably well based on the overall RMSE. However, the simulated conversion values are often close to complete conversion, suggesting that these predictions are not fully physically meaningful. The main limitation of the model is found in the selectivity description, where the graphs show a very large spread. The hydrogen-removal test shows that simplifying the model by omitting the hydrogen terms does not improve the fit. The addition of either the linear or squared inhibition term also does not improve the model fit. This could mean that the reaction for this catalyst is not mainly controlled by acetylene inhibition of ethylene hydrogenation. Therefore, for Pd_3Ag, the regular LHHW model remains limited, and neither the linear nor the squared inhibition

term improves the description sufficiently. This suggests that the current LHHW modifications are not sufficient for Pd_3Ag, and a different model structure or additional physical effects may be needed to describe this catalyst properly.

6.2.3. Final extended LHHW model

From all the information these LHHW model modifications gave on improving the models fit to describe conversion and selectivity a new, final, kinetic base model was created for every catalyst. This model takes the hydrogen terms and inhibition term into account. This addition was done to compensate for the often simulated unrealistic ~100% conversion. Besides both these terms the model also limits the reaction rate through a $r_{1,cap}$, while the second hydrogenation step included a flexible inhibition term. The addition of these terms give the final reaction rate equations in Equations 6.9, 6.10 and 6.11. This model was fitted for every catalyst having Figure 6.6 as result.

$$r_{1,\mathrm{raw}} = \frac{k_1 K_{C_2H_2} p_{C_2H_2} K_{H_2} p_{H_2}}{\left(1 + K_{C_2H_2} p_{C_2H_2} + K_{C_2H_4} p_{C_2H_4} + K_{H_2} p_{H_2}\right)^2} \tag{6.9}$$

$$r_1 = \frac{r_{1,\mathrm{raw}}}{1 + \frac{r_{1,\mathrm{raw}}}{r_{1,\max}}} \tag{6.10}$$

$$r_2 = \frac{k_2 K_{C_2H_4} p_{C_2H_4} K_{H_2} p_{H_2}}{\left(1 + K_{C_2H_2} p_{C_2H_2} + K_{C_2H_4} p_{C_2H_4}\right)\left(1 + K_{\mathrm{inh}} p_{C_2H_2}\right)^{n_{\mathrm{inh}}}} \tag{6.11}$$

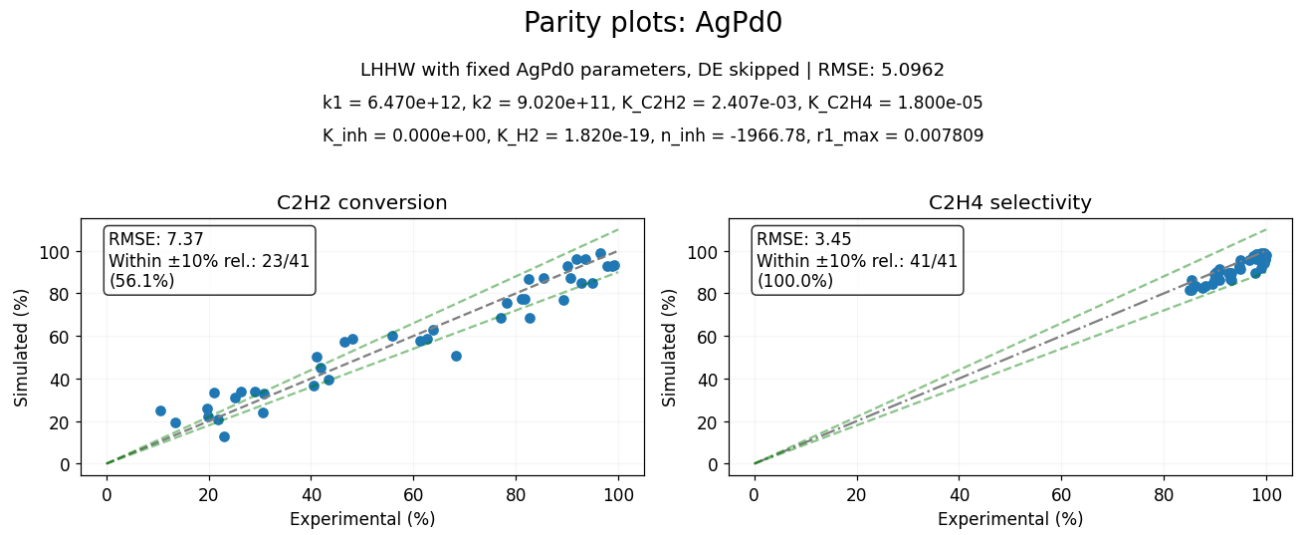
This final model shows a good fit for all catalysts. With an overall RMSE of 5.10, 5.43, and 7.14 for Pd_0Ag, Pd_2Ag, and Pd_3Ag, respectively.

For Pd_0Ag, the model gives a good description of the catalysts selectivity. The selectivity RMSE remains low, showcasing that the model can describe the selectivity well, even in the high selectivity region. The conversion is also described reasonably, although some deviations remain significant, especially at lower conversion.

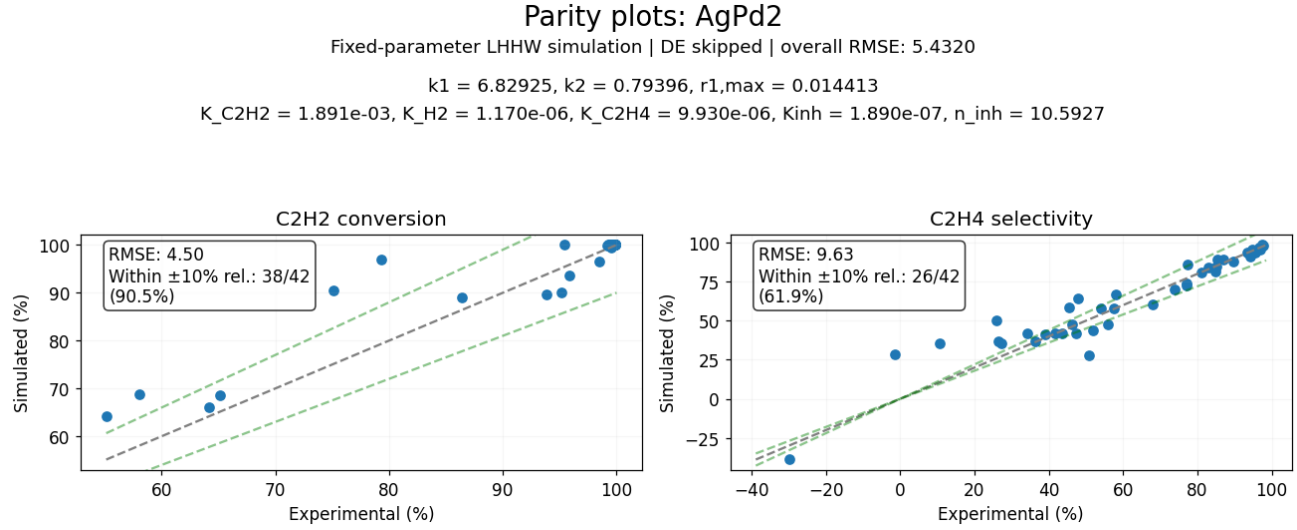
Compared to previous modifications the fit improvements are limited. This is to be expected because the selectivity of Pd_0Ag was already described well by previous models. However, the addition of a $r_{1,cap}$ and flexible inhibition do not strongly change the interpretation of the catalyst, but does make the model more flexible. This allows the taking the conversion limitations into account without losing the good selectivity fit.

The final kinetic model does show significant improvement for Pd_2Ag. In particular does the fit for conversion improve for this catalyst, shown by the significantly lower conversion RMSE. This indicates that the introduction of the maximal reaction rate for acetylene hydrogenation helps to correct the over prediction previous models have shown. The selectivity fit of the model also increases significantly compared to the previous models, although some scatter can still be found in the plot.

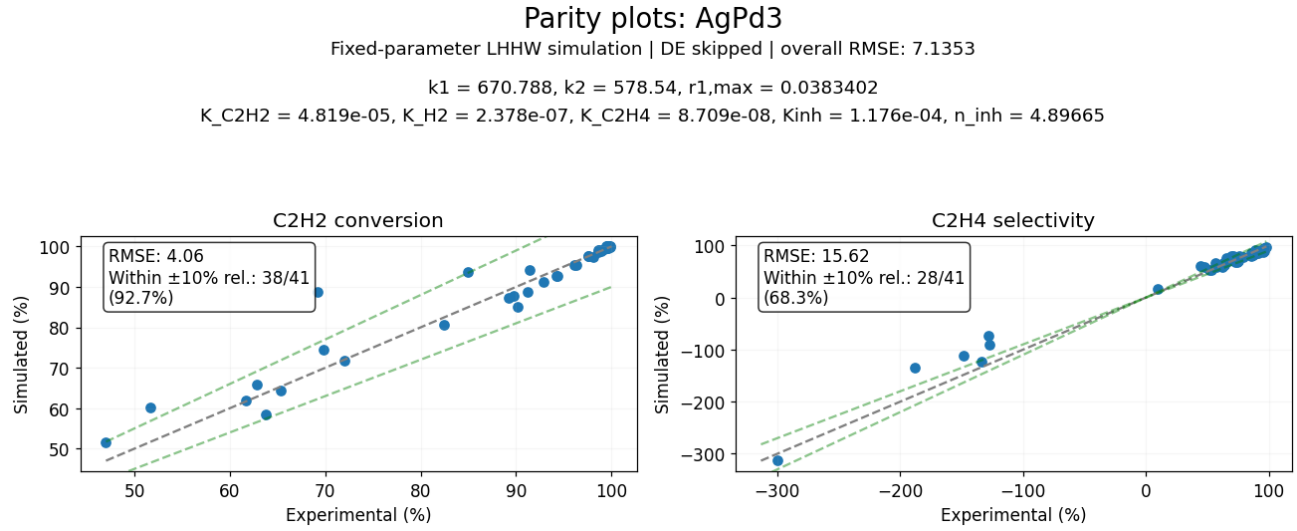
Compared to the regular and hydrogen-less LHHW models, this final model gives a more complete description because it keeps the dependence on adsorption and hydrogen availability, while also preventing unrealistically high acetylene consumption rates. Compared to the LHHW models with the linear and squared inhibition terms, fitted n_{inh} component provides more freedom for the model to find the best fit by determining the right strength for this term. This suggests that acetylene hydrogenation and ethylene hydrogenation are affected by different limitations: the first step is mainly limited by the maximum attainable acetylene consumption rate, while the second step is strongly influenced by acetylene inhibition of ethylene hydrogenation.



(a)



(b)



(c)

Figure 6.6: Addition of $r_{1,cap}$ and fitted n_{inh} inhibition term in LHHW model: (a) Pd_0Ag, (b) Pd_2Ag, (c) Pd_3Ag

6.2.4. Cross-catalyst comparison

For every catalyst, the final extended LHHW kinetic model gave the best fit. However, from this one cannot conclude that every catalyst has the same kinetics because different parameters were fitted for every catalyst. Using the same final kinetic model allows the fitted parameter sets to be compared, which gives insight into how the catalysts differ kinetically.

In this model the rate constants k_1 and k_2 describe their contribution to their representative hydrogenation reaction. The rate constant k_1 relates to the hydrogenation of acetylene to ethylene, while k_2 relates to the over-hydrogenation of ethylene to ethane. For selectivity, the relative balance between k_1 and k_2 is especially important, while the absolute values also influence the predicted conversion and overall reaction rate. A relatively large k_1 compared to k_2 indicates that the semi hydrogenation is favored, resulting in higher ethylene selectivity. In contrast, a stronger contribution of k_2 suggests a higher tendency for overhydrogenation. This trend can be connected to the ensemble effect. Larger palladium loading usually leads to larger palladium ensembles which are more active in hydrogenation because they provide neighboring palladium sites for adsorption and hydrogen activation. However, these larger ensembles also increase the probability that ethylene stays adsorbed and hydrogenates to ethane. Therefore, an increase in palladium loading usually coincides with a lower ratio k_1: k_2, showing the increased chance of over-hydrogenation. This can also clearly be seen in the ratio for Pd_0Ag and Pd_3Ag, explaining the tendency of Pd_3Ag to over hydrogenate and therefore its lower selectivity.

The fitted adsorption parameters $K_{C_2H_2}$, $K_{C_2H_4}$ and K_{H_2} describe how these reactants compete for the catalyst surface. The strongest interaction is between acetylene and the catalyst surface. This can be explained by its ability of forming such a strong bond with palladium. This results in the adsorption constant of acetylene being the largest of the three, and therefore more strongly dominates the catalyst surface. This increases the selectivity because this prevents active sites adsorbing ethylene and over hydrogenating. The ethylene adsorption constant has the opposite effect. If this parameter increases, the probability of ethylene readsorption increases, lowering the catalysts selectivity. Therefore, the ratio between $K_{C_2H_2}$ and $K_{C_2H_4}$ contributes to the predicted selectivity, because it describes the extent to which acetylene can compete with ethylene for surface sites. However, this ratio should be interpreted together with k_2, K_{H_2}, and the inhibition term to be able to describe the difference between the catalysts quality.

The hydrogen adsorption constant represents the role of hydrogen availability in the rate expressions. Because hydrogen is needed for both hydrogenation reactions, a higher hydrogen contribution can increase conversion and may also enhance the second hydrogenation reaction. Therefore, this constant can promote the conversion of acetylene, but can also decrease selectivity.

The inhibition parameters K_{inh} and n_{inh} describe the suppression of ethylene hydrogenation by acetylene. Both these terms are only included in r_2 to emphasize the competitive adsorption between acetylene and the other reactants. These terms represents how, when acetylene is adsorbed on the catalyst surface it strongly inhibits the second hydrogenation reaction.

For Pd_0Ag, K_{inh} is essentially zero; this means that for this final model inhibition has almost no effect on the catalyst. This suggests that the high selectivity of this catalyst is mainly described by the low effective contribution of ethylene and the limited hydrogenation capacity of the catalyst. For Pd_2Ag, this term has a small, but significant value. The value of K_{inh} is very small, but through the n_{inh} it still has a significant influence on the model. Through these fitted parameters the balance between acetylene and ethylene hydrogenation is better described in the model. The inhibition term has the biggest influence on Pd_3Ag. This catalyst on its own has a strong tendency to over hydrogenate. The fitted inhibition contribution helps reduce this over-hydrogenation when acetylene is present. By adding this term, the model has significantly improved the models selectivity fit, but it still shows that this catalyst has a high tendency to over hydrogenate.

The parameter $r_{1,max}$ caps the maximum acetylene hydrogenation rate. The addition of this term prevents the model over predicting acetylene conversion to unrealistic values. Without this term, meaningless values would be fitted. This capacity can be influenced by the number of active sites, palladium loading, but also by the physical limitations when the reaction becomes fast. Therefore, this cap is es-

pecially important for highly active catalysts, the observed conversion may be affected by mass transfer limitations affecting how efficiently reactants reach the active sites. That is why the fitted rate should be interpreted as the maximum observable acetylene hydrogenation capacity under the tested conditions, and not as a directly chemically meaningful parameter.

Overall, the fitted parameters show how the same final extended LHHW model is able to describe the kinetic behavior of each catalyst well. For Pd_0Ag, the fitted model mainly reflects a limited contribution of the second hydrogenation step, agreeing with the high selectivity of the catalyst. The parameters of Pd_2Ag show a more balanced system. Acetylene conversion stays high, while ethylene hydrogenation is still partly suppressed. Pd_3Ag shows that the fitted model needs a higher acetylene hydrogenation capacity. Additionally, it also needs a stronger contribution of ethylene hydrogenation. Therefore, the parameter comparison shows how the balance between the first and second hydrogenation step changes with palladium loading. However, it is important to keep in mind that all parameters are fitted simultaneously, and therefore the individual values are not really meaningful. It is mainly the combination that can be used to compare the catalysts kinetics.

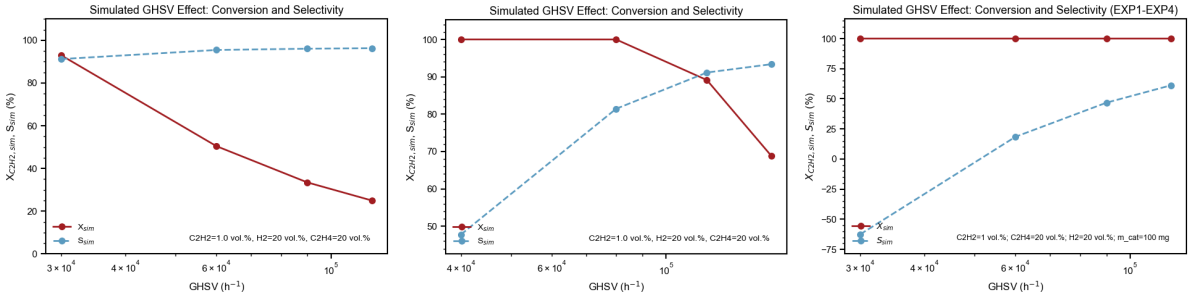
6.2.5. Comparison with experimental data

The same conversion and selectivity analysis as in Chapter 6.1 has been done on the simulated conversion and selectivity. By reviewing the experimental and simulated conversion and selectivity trends, the extent to which the kinetic model captures the observed catalytic behavior can be evaluated. In this section only the influence of GHSV will be discussed and can be found in Figure 6.7. The remaining plots can be found in Appendix chapters B to D.

For Pd_0Ag the simulated conversion and selectivity trend follows the experimental values quite well. The decrease in conversion with increasing GHSV is slightly less pronounced in the simulation, but the overall trend is reproduced well. The same can be said about the simulated selectivity trend. The simulation shows the small increase selectivity with increasing GHSV, and in addition starts at a similar selectivity. Chemically, this suggests that for this catalyst the fitted parameters describe the dominant kinetic trends reasonably well.

When comparing the simulated and experimental values of Pd_2Ag it also appears to match well. The decrease in conversion begins at approximately the same GHSV in both the experimental and simulated data, indicating that the model captures the general effect of residence time on acetylene conversion. The selectivity of the simulated data appears to be $\sim$10% lower than that is calculated based on the experimental data. This suggests that chemically the model slightly overestimates the rate of the secondary hydrogenation reaction, or that it underestimates the inhibiting effect of acetylene on ethylene on active sites. Therefore, although the conversion behavior of Pd_2Ag is captured well, the difference for selectivity suggests the model does not fully describe the competitive adsorption on the catalyst surface.

The results for Pd_3Ag require a careful interpretation. The conversion is almost indistinguishable between experimental and simulated data. However, as stated before the conversion is $\sim$100% over the whole flow rate range. Because the conversion remains close to complete conversion, it provides limited sensitivity for evaluating the fitted kinetic parameters. In contrast, the selectivity seems to not be reproduced well. The simulated selectivity for this catalyst starts $\sim$40% below the experimental value at that flow rate. This suggests that the model still has difficulty describing the balance between r_1 and r_2. Therefore, for Pd_3Ag, the final extended LHHW model does not provide a satisfactory description of the experimental selectivity trend.



(a) **(b)** **(c)**

Figure 6.7: Conversion and selectivity plotted against GHSV. (a) - Pd_0Ag, (b) - Pd_2Ag, (c) - Pd_3Ag

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Conclusion

This research investigated the selective hydrogenation of acetylene in an ethylene rich feed stream over $PdAg/SiO_2$ catalysts with different palladium loadings (Pd:Ag loadings of 1:200 Pd_0Ag, 1:100 Pd_2Ag, and 1:50 Pd_3Ag). The main aim was to investigate how different compositions of palladium silver catalysts influence the conversion and selectivity, combining experimental testing with kinetic modeling to describe the experimentally observed trends.

The experimental results clearly demonstrate the effect of palladium loading on each catalysts conversion and selectivity. Higher palladium loading generally shows near complete conversion, which can be explained by the larger number of active palladium sites, but this is often accompanied with low selectivity. In contrast, catalysts with lower palladium maintain a higher selectivity, while its loading limited its hydrogenation capacity. The mediate loading of Pd_2Ag emerged as promising catalyst composition. The catalyst maintained high conversion across most tested conditions, while achieving significantly better selectivity than catalysts with higher palladium loading.

The role of silver, as diluting metal, was confirmed by the experimental trends on the catalyst behavior. Palladium is necessary for hydrogenation, but large palladium ensembles promote over hydrogenation too. Through the addition of silver, palladium is diluted and reduces the number of active palladium sites. As a result, semi hydrogenation can still take place, while the undesired hydrogenation is suppressed. The observed differences between the catalysts therefore suggest that the interaction between the palladium loading and silver dilution significantly determines the final catalytic performance.

The standard LHHW model was not able to correctly describe the experiment data. The model consistently predicted near complete acetylene conversion regardless of conditions, suggesting that the fitted parameters were physically meaningless. Furthermore, the standard model too failed to describe the competition between acetylene and ethylene hydrogenation, leading to a poor selectivity fit.

The important improvements made to the standard model are the introduction of a maximum conversion rate term for acetylene hydrogenation, combined with a flexible inhibition term. The introduction of a maximum rate prevents the model from over predicting conversion. Instead of parameters compensating for this, the maximum rate term provides a hard limit to the conversion of acetylene. While the flexible inhibition term also improved the description of the catalysts selectivity. This term gives the model the freedom to fit the right strength of the inhibition term. This improves the fit because acetylene acts as not only a reactant but also as an inhibitor of over hydrogenation.

The final extended LHHW model incorporating both modifications achieved overall RMSE values of 5.10, 5.43 and 7.14 for Pd_0Ag, Pd_2Ag and Pd_3Ag respectively, representing significant improvement. Especially for Pd_2Ag and Pd_3Ag the additions improved the overall model's fit.

The effect this final, improved model has been proven to be catalyst depended. For Pd_0Ag, the improvement was relatively limited because the selectivity was already well described by previous models. In this case, the added flexibility mainly allowed the model to account for conversion limitations without

losing the good selectivity fit.

Overall this work shows that the performance of $PdAg/SiO_2$ catalysts for acetylene hydrogenation is strongly affected by palladium loading and silver dilution. Higher palladium loading increases acetylene conversion, while silver maintains ethylene selectivity by suppressing over hydrogenation. The final kinetic model, significantly improved the models fit through the addition of an artificial conversion rate cap and a flexible inhibition term. This suggests that this model was able to capture the most significant experimental trends relatively accurate for most catalysts. This suggests that acetylene and ethylene hydrogenation are affected by different physical limitations; acetylene hydrogenation is mainly limited by the maximum attainable consumption rate, while ethylene hydrogenation is strongly influenced by acetylene inhibition. Therefore, the final model provides a useful tool for comparing the catalytic behavior and understanding the balance between activity and selectivity in acetylene hydrogenation for Pd_0Ag and Pd_2Ag, while for Pd_3Ag the model needs to be improvised further before it becomes useful.

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A

Results non mentioned sets

A.1. Influence ethylene

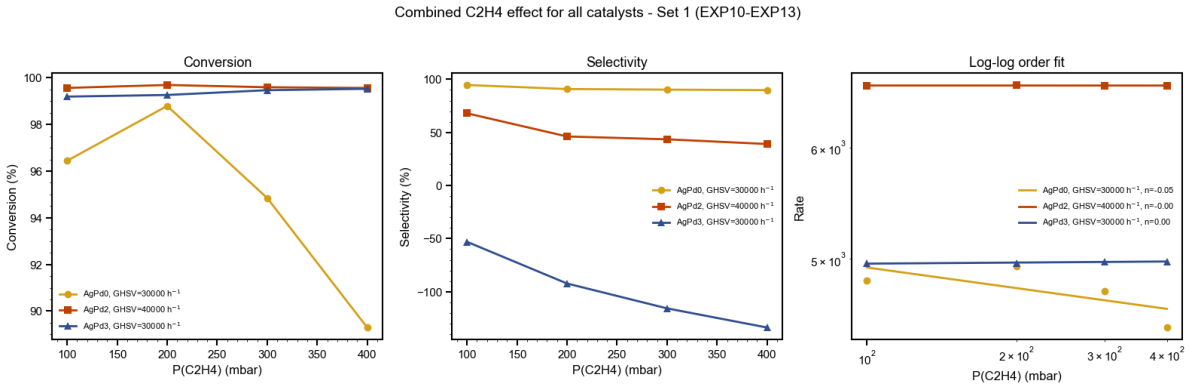


Figure A.1: Combined influence ethylene, set 1

A.2. Influence hydrogen

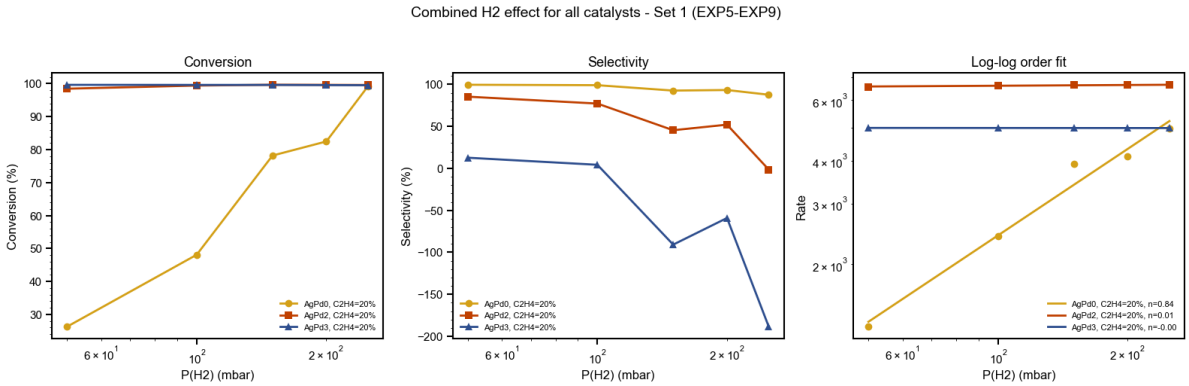
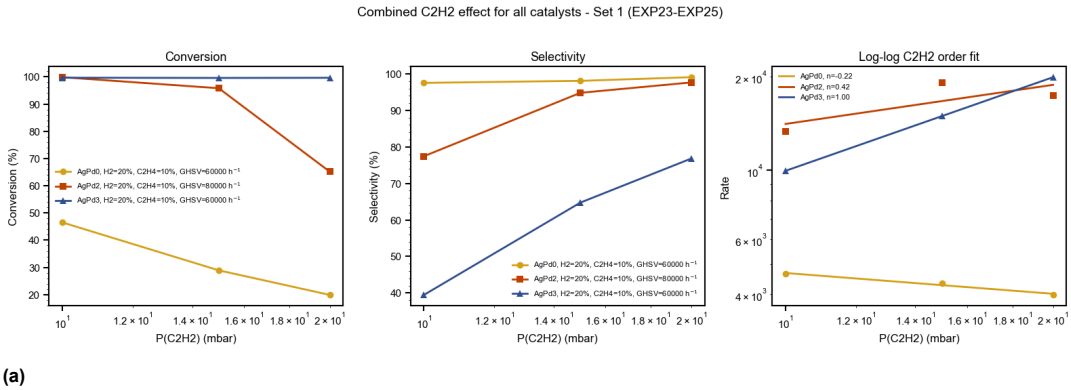
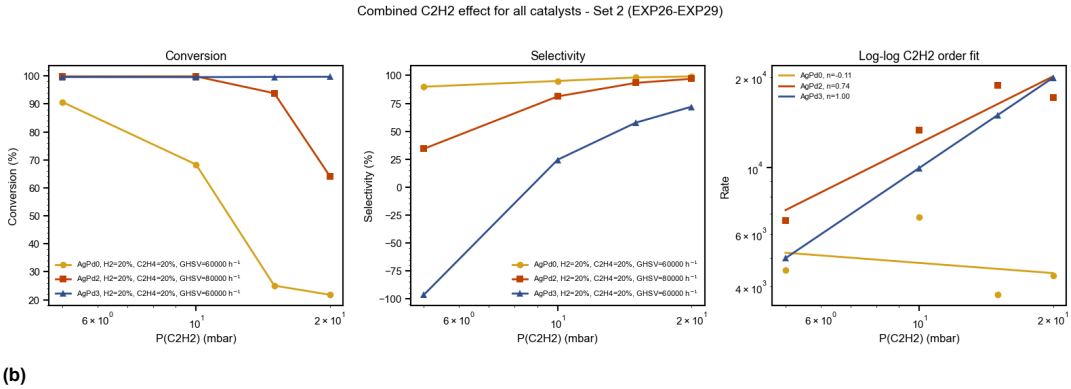


Figure A.2: Combined influence hydrogen, set 1

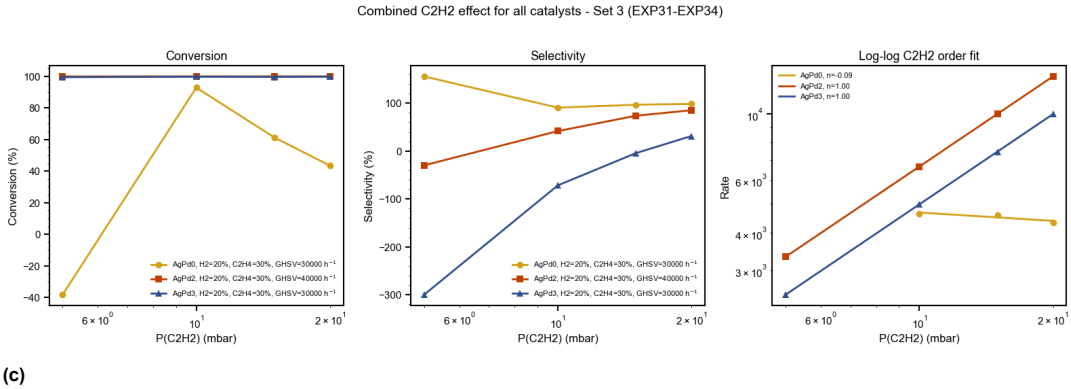
A.3. Influence acetylene



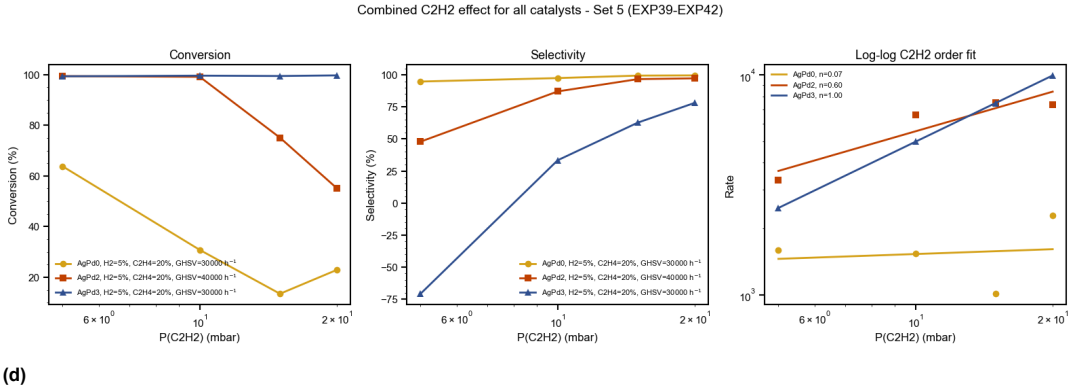
(a)



(b)



(c)



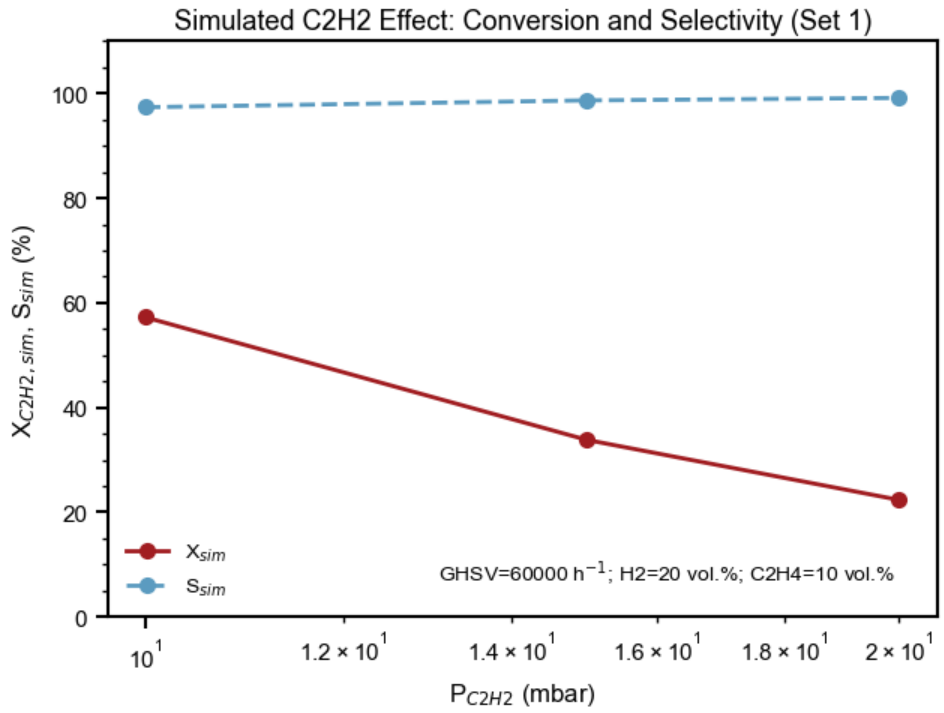
(d)

Figure A.3: Combined influence acetylene, set 1-3 and 5

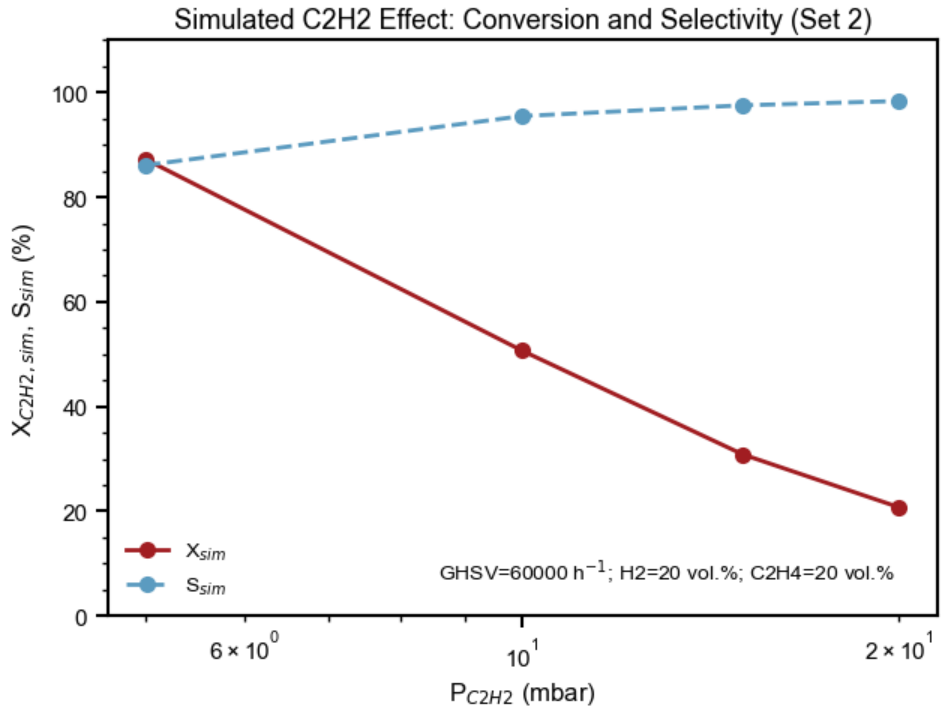
B

Conversion and Selectivity plotted against changing parameter Pd_0Ag

B.1. Conversion + Selectivity against $p_{C_2H_2}$

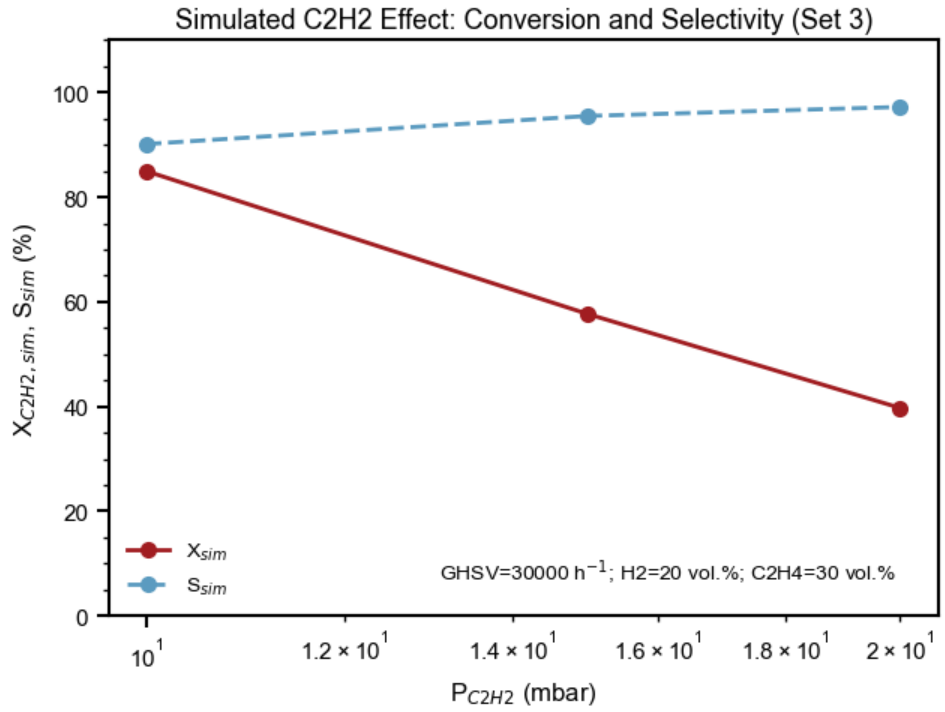


(a)

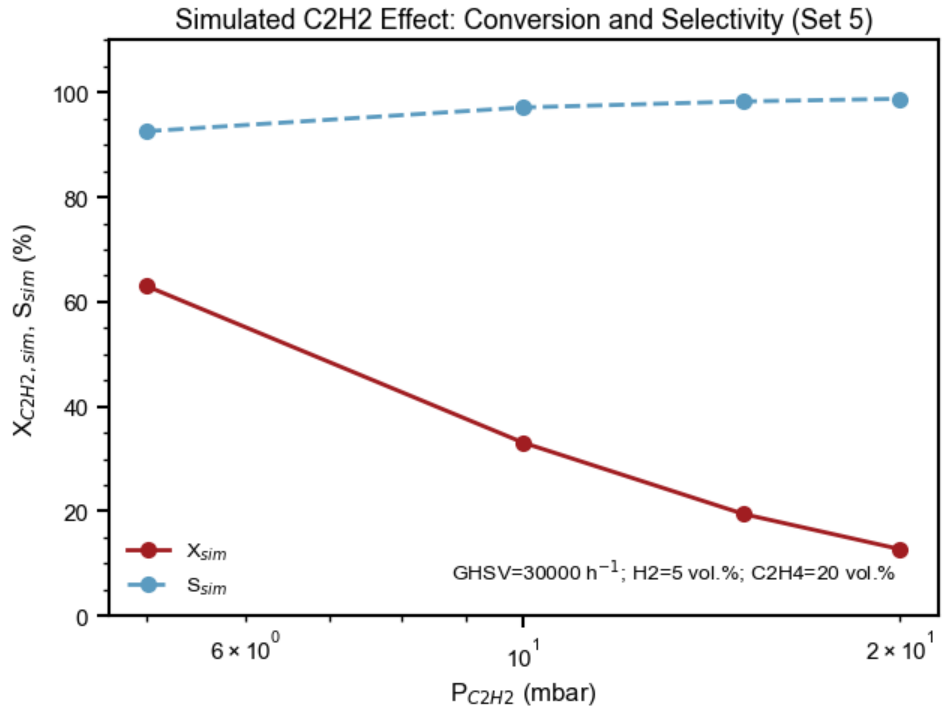


(b)

Figure B.1: Conversion and selectivity, plotted against influence acetylene, set 1 and 2



(a)



(b)

Figure B.2: Conversion and selectivity, plotted against influence acetylene, set 3 and 5

B.2. Conversion + Selectivity against $p_{C_2H_4}$

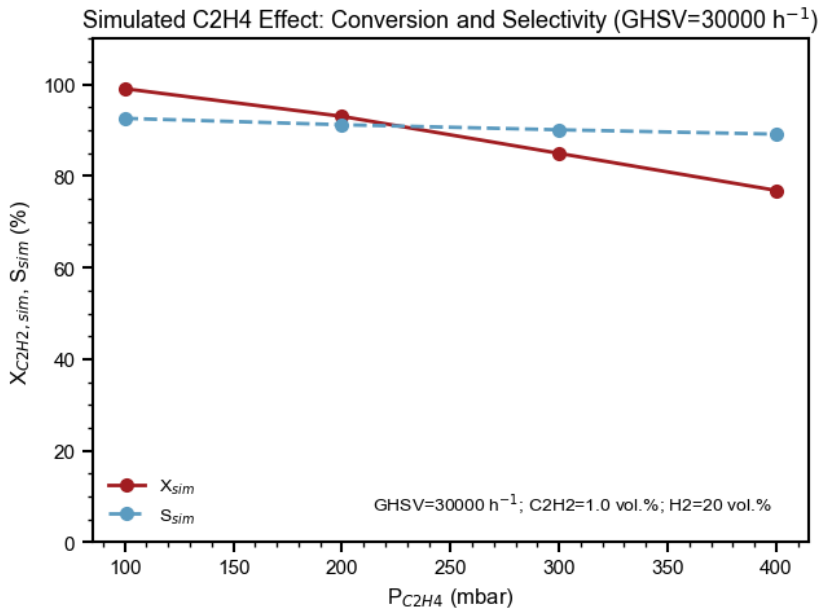


Figure B.3: Conversion and selectivity plotted against the ethylene partial pressure, set 1

B.3. Conversion + Selectivity against p_{H_2}

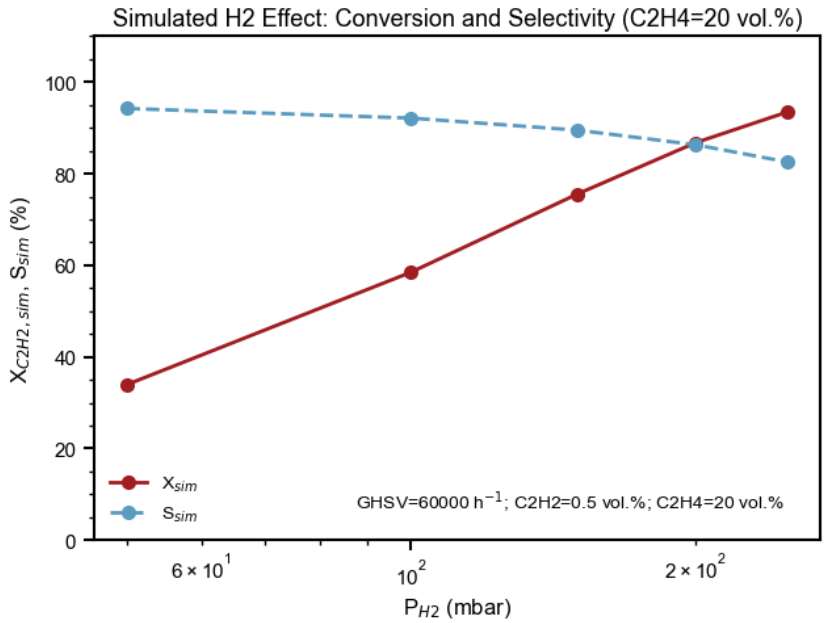
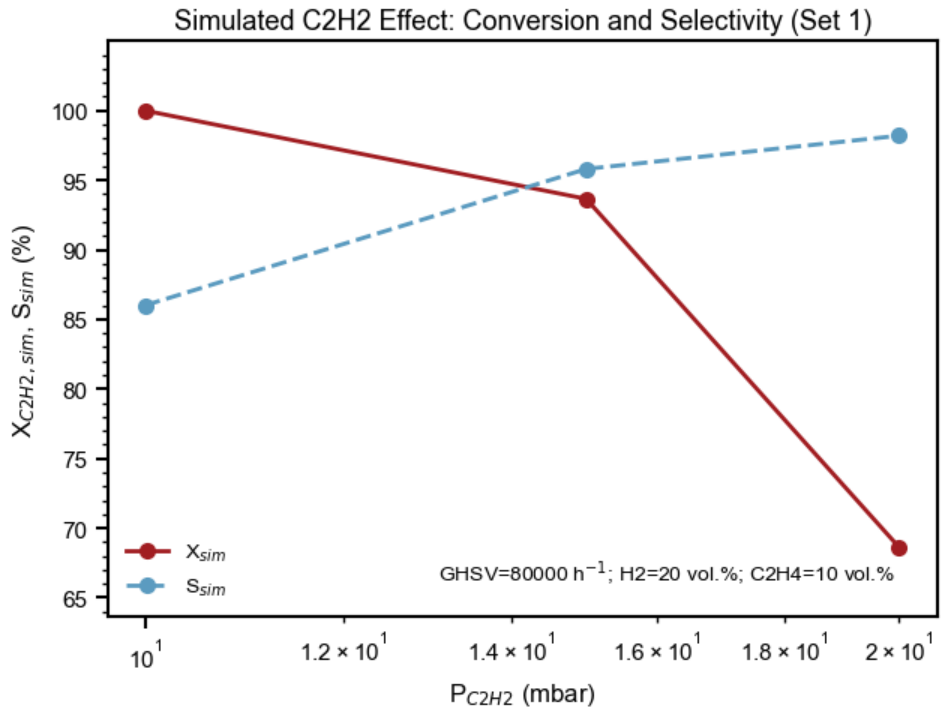


Figure B.4: Conversion and selectivity plotted against the hydrogen partial pressure, set 1

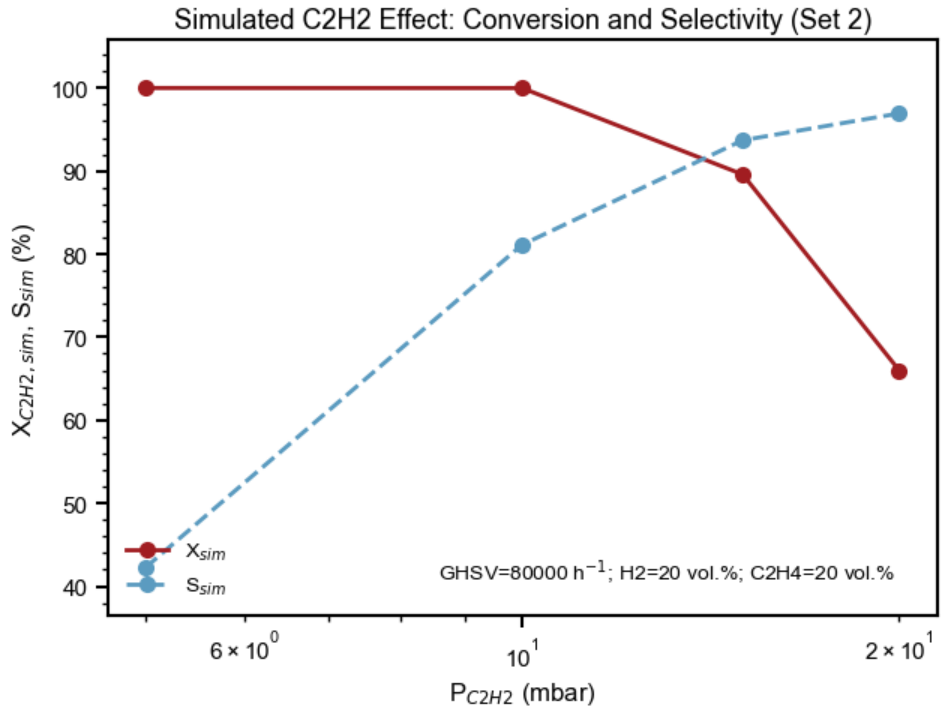
C

Conversion and Selectivity plotted against changing parameter Pd_2Ag

C.1. Conversion + Selectivity against $p_{C_2H_2}$

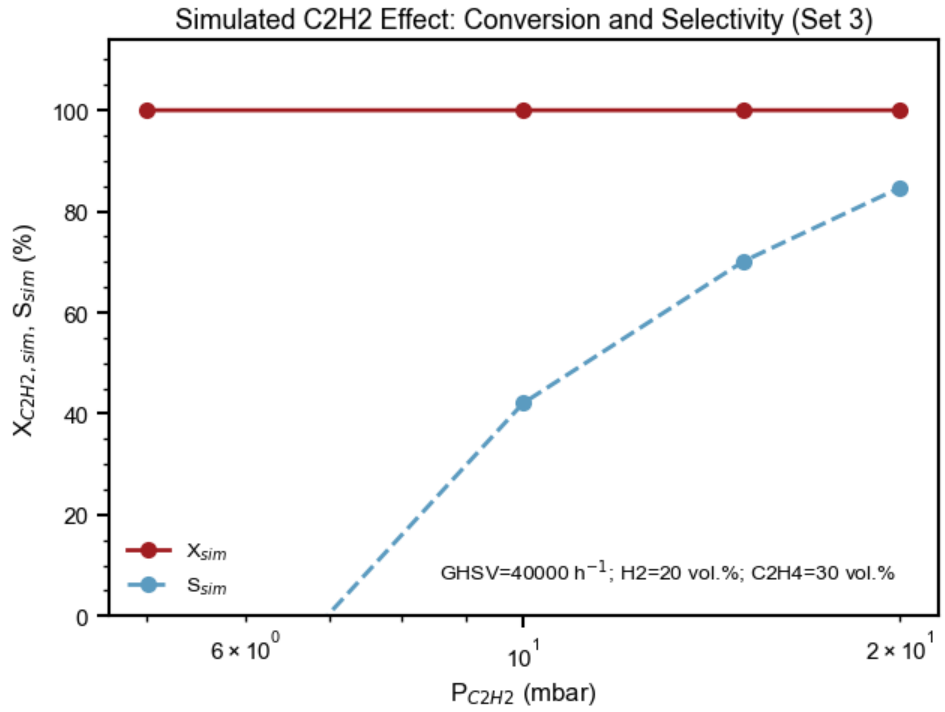


(a)

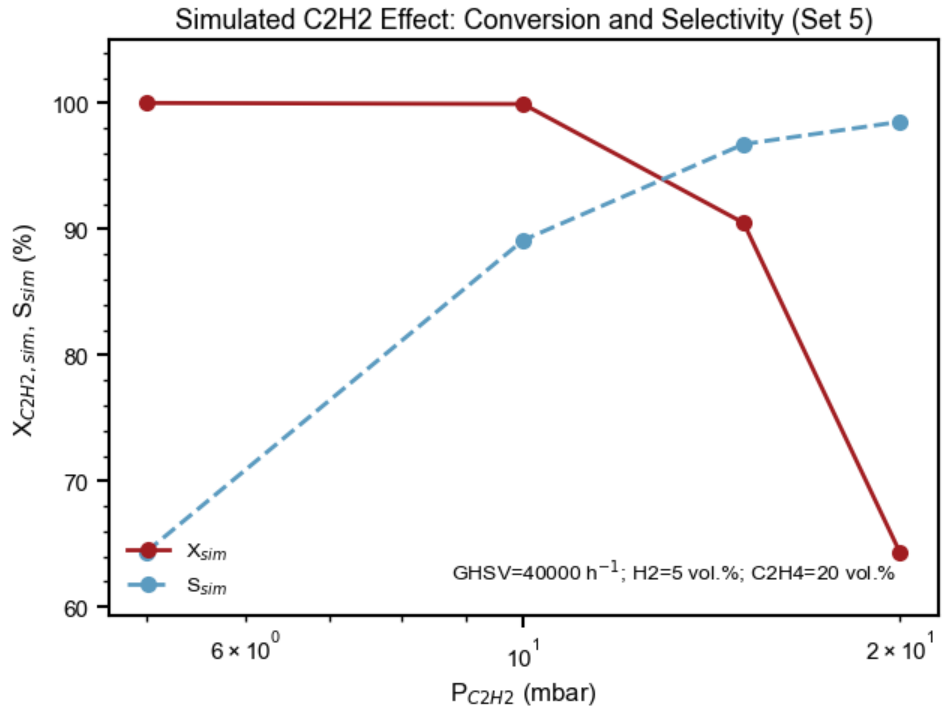


(b)

Figure C.1: Conversion and selectivity, plotted against influence acetylene, set 1 and 2



(a)



(b)

Figure C.2: Conversion and selectivity, plotted against influence acetylene, set 3 and 5

C.2. Conversion + Selectivity against $p_{C_2H_4}$

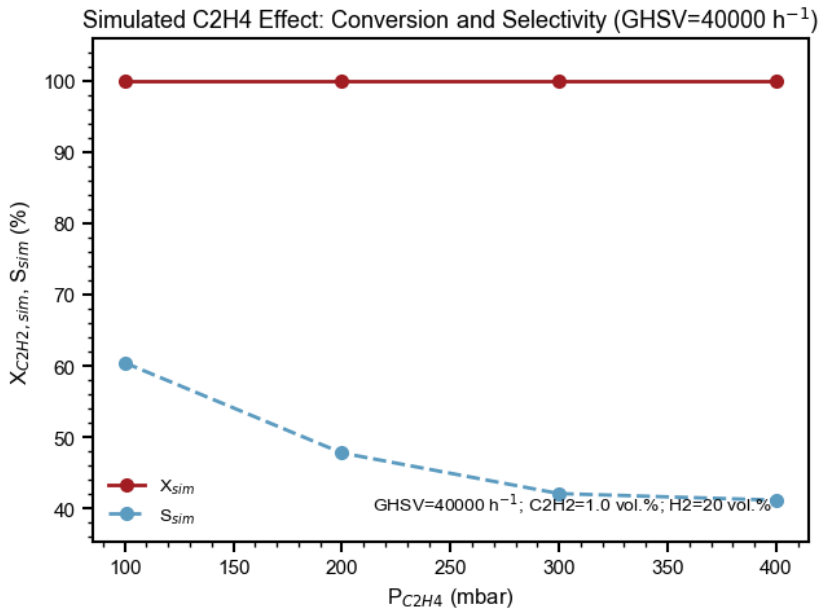


Figure C.3: Conversion and selectivity plotted against the ethylene partial pressure, set 1

C.3. Conversion + Selectivity against p_{H_2}

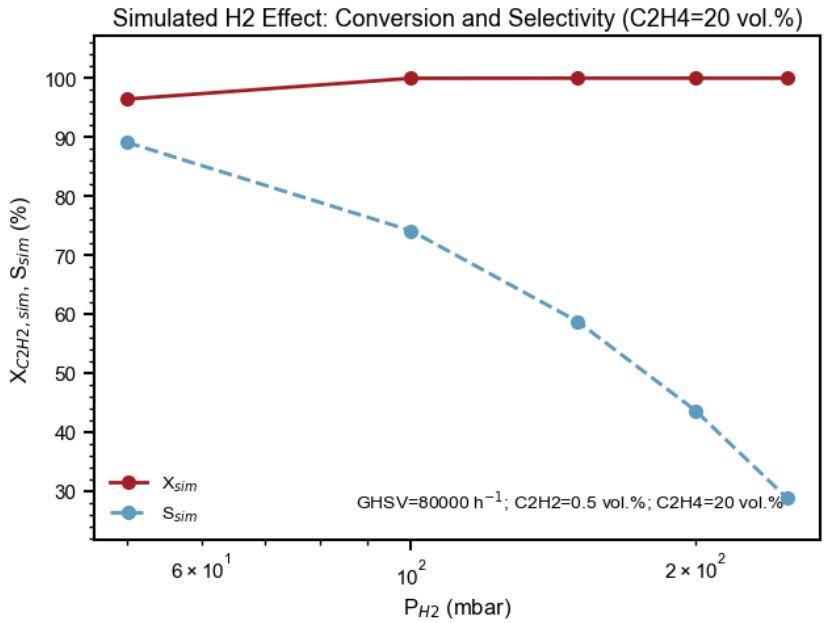
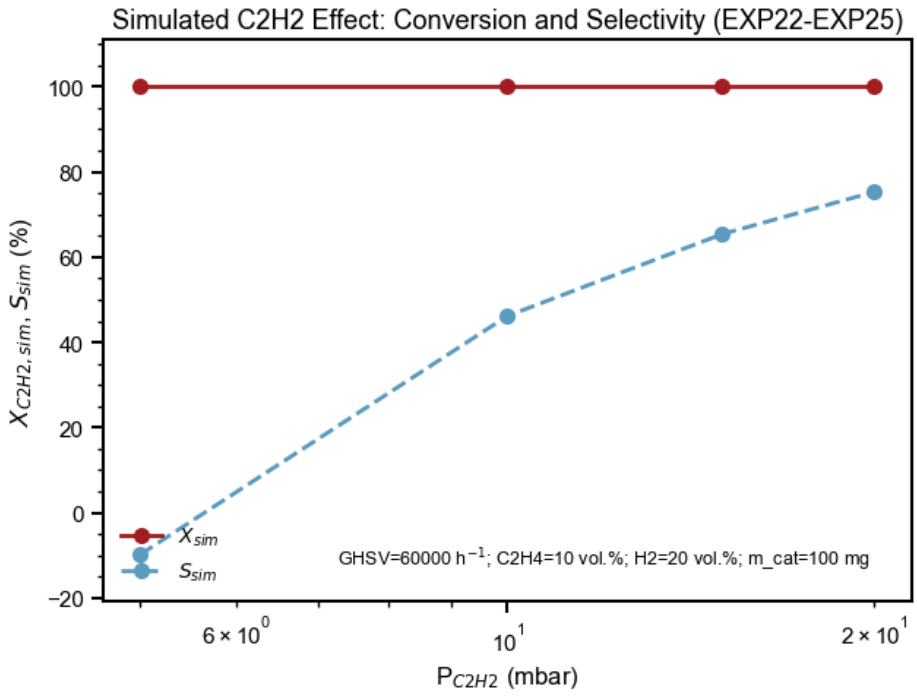


Figure C.4: Conversion and selectivity plotted against the hydrogen partial pressure, set 1

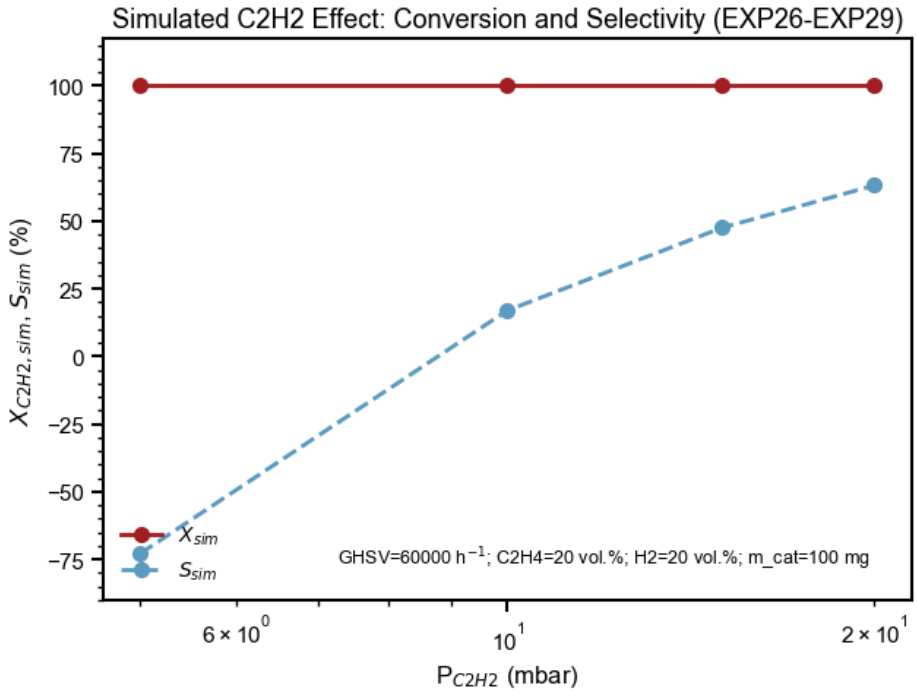
D

Conversion and selectivity plotted against changing parameter Pd_3Ag

D.1. Conversion + selectivity against $p_{C_2H_2}$

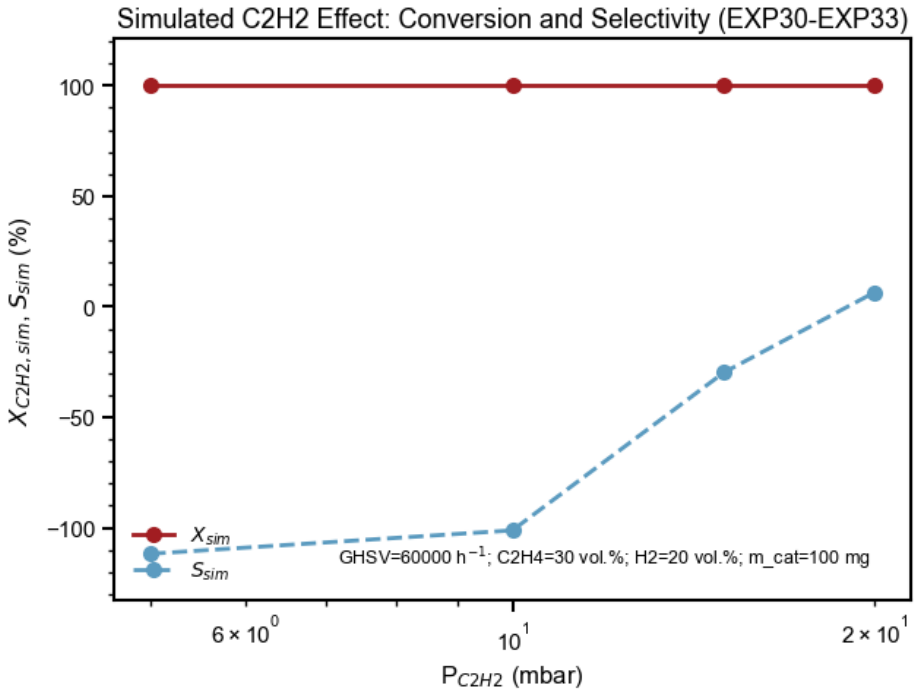


(a) Set 1



(b) Set 2

Figure D.1: Conversion and selectivity plotted against the acetylene partial pressure for Pd_3Ag, sets 1 and 2.



(a) Set 3

Figure D.2: Conversion and selectivity plotted against the acetylene partial pressure for Pd_3Ag, set 3.

D.2. Conversion + Selectivity against $p_{C_2H_4}$

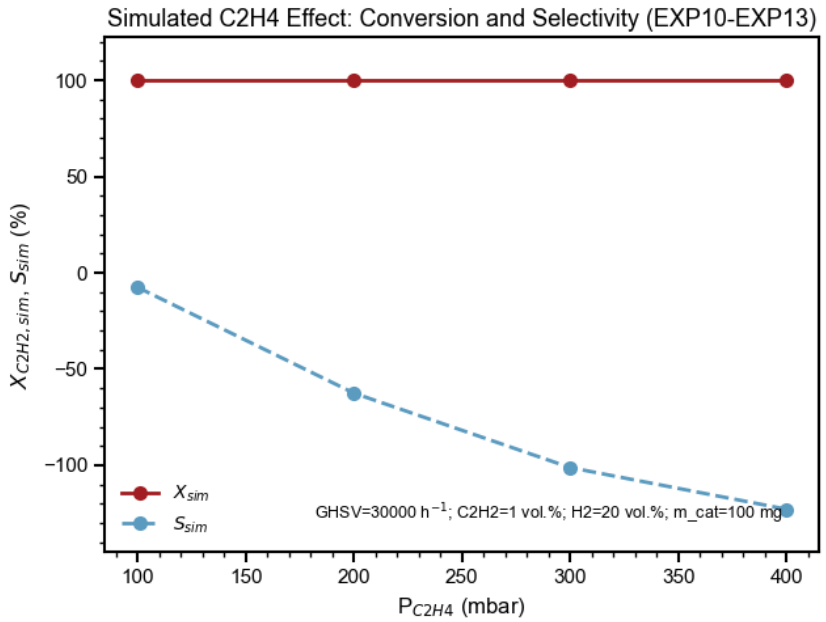


Figure D.3: Conversion and selectivity plotted against the ethylene partial pressure, set 1

D.3. Conversion + Selectivity against p_{H_2}

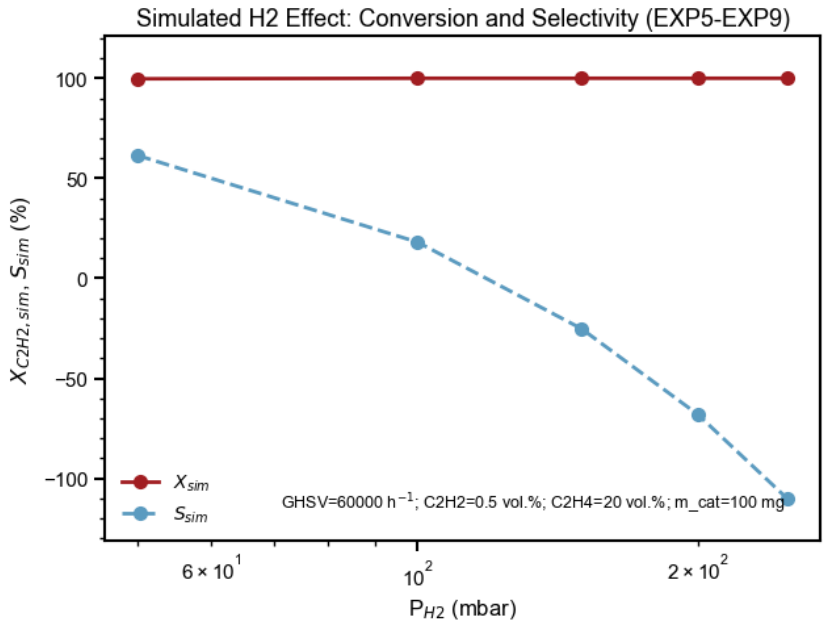


Figure D.4: Conversion and selectivity plotted against the hydrogen partial pressure, set 1

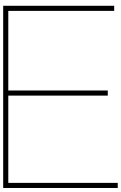


Table tests

E.1. Table per cat. with every LHHW model variance

E.1.1. Pd_0Ag

Table E.1: Overview of the tested kinetic model variants for Pd_0Ag, if ✓it is a variable in the LHHW model and if ✗the term is let out of the LHHW model

Variant	H_2-term	C_2H_4-term	Inhibition term	K_{H_2}	Power Law	Overall RMSE	$RMSE_{conversion}$	$RMSE_{selectivity}$
1	✓	✓	✓	✓	✗	42.3166	66.51	4.31
2	✓	✓	✗	✗	✗	15.2241	27.12	3.52
2a	✗	✓	✗	✗	✗	17.2401	28.02	3.21
2b	✓	✗	✗	✗	✗	28.6856	49.81	3.21
2c	✗	✗	✗	✗	✗	29.5322	49.81	5.78
3	✓	✓	✓	✗	✗	13.8472	24.88	3.71
3a	✗	✓	✓	✗	✗	17.4908	28.53	4.10
3b	✓	✗	✓	✗	✗	10.7848	18.38	3.67
3c	✗	✗	✓	✗	✗	2338.0896	49.81	3987.61
4	✓	✓	✗	✓	✗	17.5358	30.36	3.46
$4_{\text{smaller bounds}}$	✓	✓	✗	✓	✗	11.4083	18.45	6.56
4a	✗	✓	✗	✗	✗	32.7246	35.62	3.45
4b	✓	✗	✗	✓	✗	–	–	–
4c	✗	✗	✗	✗	✗	29.5322	49.81	5.78
5	✓	✓	✓	✓	✗	6.4395	9.73	4.55
$5_{\text{smaller bounds}}$	✓	✓	✓	✓	✗	8.7065	13.89	5.81
5a	✗	✓	✓	✗	✗	17.4908	28.53	4.10
5b	✓	✗	✓	✓	✗	28.6857	49.81	3.21
5c	✗	✗	✓	✗	✗	2338.0896	49.81	3987.61
5d	✓	✓	✓	✓	✗	18.0673	30.75	5.51
6	✓	✓	✗	✗	✓	8.4021	13.95	3.53
$6_{\text{smaller bounds}}$	✓	✓	✗	✗	✓	7.0357	12.64	3.34
7	✓	✓	✓	✗	✓	7.6719	12.63	5.59
8	✓	✓	✗	✓	✓	15.1870	24.02	3.54
9	✓	✓	✓	✓	✓	14.5374	23.22	2.73
$9_{\text{smaller bounds}}$	✓	✓	✓	✓	✓	7.0172	12.64	3.16
10	✓	✓	✓	✓	✓	8.2459	11.39	6.63

E.1.2. Pd_2Ag

Table E.2: Overview of the tested kinetic model variants for Pd_2Ag, if ✓it is a variable in the LHHW model and if ✗the term is let out of the LHHW model

Variant	H_2-term	C_2H_4-term	Inhibition term	K_{H_2}	Power Law	Overall RMSE	$RMSE_{conversion}$	$RMSE_{selectivity}$
1	✓	✓	✓	✓	✗	73.2983	94.85	46.38
2	✓	✓	✗	✗	✗	12.8855	13.46	18.36
2a	✗	✓	✗	✗	✗	14.5874	13.97	20.47
2b	✓	✗	✗	✗	✗	9.4210	9.36	15.66
2c	✗	✗	✗	✗	✗	2361.6036	13.46	4014.28
3	✓	✓	✓	✗	✗	12.7469	13.46	18.26
3a	✗	✓	✓	✗	✗	14.3602	13.46	20.97
3b	✓	✗	✓	✗	✗	10.3796	11.39	15.88
3c	✗	✗	✓	✗	✗	54.1860	75.43	22.03
4	✓	✓	✗	✓	✗	11.9816	13.46	16.81
$4_{\text{smaller bounds}}$	✓	✓	✗	✓	✗	12.7543	13.44	18.30
4a	✗	✓	✗	✗	✗	14.5874	13.97	20.47
4b	✓	✗	✗	✓	✗	9.4182	9.38	15.65
4c	✗	✗	✗	✗	✗	2361.6036	13.46	4014.28
5	✓	✓	✓	✓	✗	12.0874	13.46	16.97
$5_{\text{smaller bounds}}$	✓	✓	✓	✓	✗	14.1928	13.46	20.77
5a	✗	✓	✓	✗	✗	14.3602	13.46	20.97
5b	✓	✗	✓	✓	✗	12.7298	13.46	18.23
5c	✗	✗	✓	✗	✗	54.1860	75.43	22.03
5d	✓	✓	✓	✓	✗	12.1767	13.46	17.15
6	✓	✓	✗	✗	✓	9.8769	4.38	19.15
$6_{\text{smaller bounds}}$	✓	✓	✗	✗	✓	9.4047	3.97	18.98
7	✓	✓	✓	✗	✓	10.4806	5.62	19.62
8	✓	✓	✗	✓	✓	12.7709	10.71	19.37
9	✓	✓	✓	✓	✓	13.1385	11.55	21.30
$9_{\text{smaller bounds}}$	✓	✓	✓	✓	✓	10.9095	7.61	18.91
10	✓	✓	✓	✓	✓	25.2860	13.46	41.70

E.1.3. Pd_3Ag

Table E.3: Overview of the tested kinetic model variants for Pd_3Ag, if ✓it is a variable in the LHHW model and if ✗the term is let out of the LHHW model

Variant	H_2-term	C_2H_4-term	Inhibition term	K_{H_2}	Power Law	Overall RMSE	$RMSE_{conversion}$	$RMSE_{selectivity}$
2	✓	✓	✗	✗	✗	42.6540	13.65	73.31
2a	✗	✓	✗	✗	✗	42.7758	13.65	76.29
2b	✓	✗	✗	✗	✗	42.4521	13.65	72.90
2c	✗	✗	✗	✗	✗	2392.2863	13.65	4070.10
3	✓	✓	✓	✗	✗	42.6557	13.65	73.31
3a	✗	✓	✓	✗	✗	42.7382	13.65	74.90
3b	✓	✗	✓	✗	✗	42.6281	13.65	73.27
3c	✗	✗	✓	✗	✗	2392.2830	13.65	4070.09
4	✓	✓	✗	✓	✗	43.0763	13.65	81.82
$4_{\text{smaller bounds}}$	✓	✓	✗	✓	✗	46.9826	13.96	92.28
4a	✗	✓	✗	✗	✗	42.7758	13.65	76.29
4b	✓	✗	✗	✓	✗	808.4597	13.65	1276.75
4c	✗	✗	✗	✗	✗	2392.2870	13.65	4070.10
5	✓	✓	✓	✓	✗	42.1434	12.76	72.67
$5_{\text{regular LHHW model}}$	✓	✓	✓	✓	✗	46.9933	13.93	92.32
5a	✗	✓	✓	✗	✗	42.7382	13.65	74.90
5b	✓	✗	✓	✓	✗	42.6548	13.65	73.31
5c	✗	✗	✓	✗	✗	2392.2830	13.65	4070.09
5d	✓	✓	✓	✓	✗	42.2333	12.95	73.09
6	✓	✓	✗	✗	✓	41.8279	11.96	73.49
$6_{\text{smaller bounds}}$	✓	✓	✗	✗	✓	41.8141	11.70	73.48
7	✓	✓	✓	✗	✓	41.7086	11.81	72.75
8	✓	✓	✗	✓	✓	41.9421	11.70	73.57
9	✓	✓	✓	✓	✓	42.1767	13.46	74.86
$9_{\text{smaller bounds}}$	✓	✓	✓	✓	✓	41.7004	11.84	72.80
10	✓	✓	✓	✓	✓	55.2482	13.65	111.18

F

Standard LHHW model code

This is the base code used for every created kinetic model. The only difference between the models is the formula for each (variant) of the LHHW model.

Kinetic modeling Acetylene semihydrogenation

Import of all needed extensions, and plot standard view

In [1]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from scipy.integrate import solve_ivp
from scipy.optimize import differential_evolution, minimize
from scipy.stats import t as student_t
import warnings
warnings.filterwarnings("ignore", category=RuntimeWarning)

plt.rcParams.update({
    "figure.dpi": 110,
    "font.size": 11,
    "axes.grid": True,
    "grid.alpha": 0.3,
})
```

Import data from excel.

Through pandas. Each different exp. having unique sheet, and name.

In [2]:
```
data_AgPd0 = pd.read_excel("/Users/Victor/Desktop/Uni/MST 4e jaar/BEP/Acetylene_Data.xlsx", sheet_name="AgPd0_SiO2_Copy", usecols="A, B, C, F, G, H, I, J, L, AH, AI, AJ")
data_AgPd2 = pd.read_excel("/Users/Victor/Desktop/Uni/MST 4e jaar/BEP/Acetylene_Data.xlsx", sheet_name="AgPd2_SiO2", usecols="A, B, C, F, G, H, I, J, L, AH, AI, AJ")
data_AgPd3 = pd.read_excel("/Users/Victor/Desktop/Uni/MST 4e jaar/BEP/Acetylene_Data.xlsx", sheet_name="AgPd3_SiO2", usecols="A, B, C, F, G, H, I, J, L, AH, AI, AJ")

# Exclude experiment 34 (Excel row 36) from the AgPd0 dataset.
# The experiment labels are text values such as "EXP34", so first extract the number.
exp_number_AgPd0 = pd.to_numeric(
    data_AgPd0.iloc[:, 0].astype(str).str.extract(r"(\d+)", expand=False),
    errors="coerce"
)

data_AgPd0 = data_AgPd0[
    exp_number_AgPd0.ne(34) | exp_number_AgPd0.isna()
].reset_index(drop=True)
```

Save needed columns and its data.

1. AgPd2

In [3]:
```
exp_numb_AgPd2 = data_AgPd2.iloc[:,0]          #exp. number AgPd2
flow_AgPd2 = data_AgPd2.iloc[:,1]              #flow per exp. number AgPd2
WHSV_AgPd2 = data_AgPd2.iloc[:,2]              #Contact time per exp. number AgPd2
vol_percent_C2H2_AgPd2 = data_AgPd2.iloc[:,3] #vol.% C2H2 AgPd2
vol_percent_C2H4_AgPd2 = data_AgPd2.iloc[:,4] #vol.% C2H4 AgPd2
vol_percent_H2_AgPd2 = data_AgPd2.iloc[:,5]   #vol.% H2 AgPd2
vol_percent_He_AgPd2 = data_AgPd2.iloc[:,6]   #vol.% He AgPd2
vol_percent_CO_AgPd2 = data_AgPd2.iloc[:,7]   #vol.% CO AgPd2
m_cat_AgPd2 = data_AgPd2.iloc[:,8]            #mass cat. AgPd2
conv_AgPd2 = data_AgPd2.iloc[:,9]             #conversion of AgPd2 through experiment
selec_AgPd2 = data_AgPd2.iloc[:,10]           #selectivity of AgPd2 through experiment
C_balans_AgPd2 = data_AgPd2.iloc[:,11]        #selectivity of AgPd2 through experiment
```

Check of tabel data

In [4]:
```
tabel_AgPd2 = data_AgPd2.iloc[:, [0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11]].copy()

tabel_AgPd2.columns = [
    "exp_numb",
    "flow (mL/min)",
    "WHSV (cm^3/ h/ g cat.)",
    "vol. % C2H2",
    "vol. % C2H4",
    "vol. % H2",
    "vol. % He",
    "vol. % CO",
    "mass cat. (mg)",
    "conv",
    "selec",
    "C-balans"
]

display(tabel_AgPd2)
```

	exp_numb	flow (mL/min)	WHSV (cm^3/ h/ g cat.)	vol. % C2H2	vol. % C2H4	vol. % H2	vol. % He	vol. % CO	mass cat. (mg)	conv	selec	C-balans
0	NaN	flow [mL/min]	WHSV [cm3/h/g_cat]	C2H2	C2H4	H2	He	CO	m_cat [mg]	X_C2H2 (%)	S (%)	C (%)
1	EXP1	200	160000	1	20	20	59	0	75	58.018657	95.669934	86.861662
2	EXP2	150	120000	1	20	20	59	0	75	86.417753	94.243281	87.317188
3	EXP3	100	80000	1	20	20	59	0	75	99.275357	84.886849	91.513288
4	EXP4	50	40000	1	20	20	59	0	75	99.349524	56.84652	90.843877
5	EXP5	100	80000	0.5	20	20	59.5	0	75	99.643373	52.018995	91.652849
6	EXP6	100	80000	0.5	20	10	69.5	0	75	99.428641	77.145141	90.40664
7	EXP7	100	80000	0.5	20	5	74.5	0	75	98.473103	85.357098	90.581642
8	EXP8	100	80000	0.5	20	25	54.5	0	75	99.603149	-1.455721	89.842034
9	EXP9	100	80000	0.5	20	15	64.5	0	75	99.672505	45.392802	93.824675
10	EXP10	50	40000	1	20	20	59	0	75	99.690002	46.175088	90.789918
11	EXP11	50	40000	1	30	20	58	0	75	99.592906	43.501811	95.123134
12	EXP12	50	40000	1	40	20	58	0	75	99.562281	39.081743	99.562015
13	EXP13	50	40000	1	10	20	58	0	75	99.5641	67.99235	84.862566
14	EXP14	100	80000	0.5	40	20	58	0	75	99.651391	27.155005	95.066297
15	EXP15	100	80000	0.5	10	20	58	0	75	99.659019	57.629123	83.283371
16	EXP16	100	80000	0.5	30	20	58	0	75	99.653853	36.294416	95.328362
17	EXP17	100	80000	0.5	20	20	58	0	75	99.681546	47.284657	96.457802
18	EXP18	100	80000	0.5	40	5	58	0	75	95.134499	82.876603	97.337821
19	EXP19	100	80000	0.5	40	15	58	0	75	99.602241	25.834125	98.12192
20	EXP20	100	80000	0.5	40	20	58	0	75	99.624922	10.648119	97.36907
21	EXP21	100	80000	0.5	40	10	58	0	75	99.548308	58.096788	96.180869
22	EXP22	100	80000	0.5	10	20	58	0	75	99.758811	54.078598	92.873962
23	EXP23	100	80000	2	10	20	58	0	75	65.146477	97.655965	86.238949

24	EXP24	100	80000	1.5	10	20	58	0	75	95.805658	94.809037	86.722084
25	EXP25	100	80000	1	10	20	58	0	75	99.866736	77.442457	70.077186
26	EXP26	100	80000	0.5	20	20	58	0	75	99.836984	34.318529	91.951498
27	EXP27	100	80000	2	20	20	58	0	75	64.205218	96.951742	91.431014
28	EXP28	100	80000	1.5	20	20	58	0	75	93.833853	93.315702	90.137674
29	EXP29	100	80000	1	20	20	58	0	75	99.844802	81.193792	101.40934
30	EXP30	100	80000	0.5	30	20	58	0	75	99.830618	26.425184	97.024715
31	EXP31	50	40000	2	30	20	58	0	75	99.869971	85.224109	97.391424
32	EXP32	50	40000	1.5	30	20	58	0	75	99.865303	73.736824	96.486999
33	EXP33	50	40000	1	30	20	58	0	75	99.869483	41.673898	95.86405
34	EXP34	50	40000	0.5	30	20	58	0	75	99.813089	-29.870214	95.999882
35	EXP35	50	40000	2	20	10	58	0	75	79.316932	96.905178	99.406619
36	EXP36	50	40000	1.5	20	10	58	0	75	99.487792	89.667645	92.990812
37	EXP37	50	40000	1	20	10	58	0	75	99.832456	76.951782	104.254605
38	EXP38	50	40000	0.5	20	10	58	0	75	95.392888	50.856992	90.573448
39	EXP39	50	40000	2	20	5	58	0	75	55.12919	97.319244	94.109775
40	EXP40	50	40000	1.5	20	5	58	0	75	75.098964	96.70324	99.565028
41	EXP41	50	40000	1	20	5	58	0	75	99.14635	87.0853	102.54149
42	EXP42	50	40000	0.5	20	5	58	0	75	99.431031	47.892041	91.4697

2. AgPd3

In [5]:
```
exp_numb_AgPd3 = data_AgPd3.iloc[:,0]          #exp. number AgPd3
flow_AgPd3 = data_AgPd3.iloc[:,1]              #flow per exp. number AgPd3
WHSV_AgPd3 = data_AgPd3.iloc[:,2]              #Contact time per exp. number AgPd3
vol_percent_C2H2_AgPd3 = data_AgPd3.iloc[:,3] #vol.% C2H2 AgPd3
vol_percent_C2H4_AgPd3 = data_AgPd3.iloc[:,4] #vol.% C2H4 AgPd3
vol_percent_H2_AgPd3 = data_AgPd3.iloc[:,5]   #vol.% H2 AgPd3
vol_percent_He_AgPd3 = data_AgPd3.iloc[:,6]   #vol.% He AgPd3
vol_percent_CO_AgPd3 = data_AgPd3.iloc[:,7]   #vol.% CO AgPd3
m_cat_AgPd3 = data_AgPd3.iloc[:,8]            #mass cat. AgPd3
conv_AgPd3 = data_AgPd3.iloc[:,9]   #conversion of AgPd3 through experiment
selec_AgPd3 = data_AgPd3.iloc[:,10]  #selectivity of AgPd3 through experiment
C_balans_AgPd3 = data_AgPd3.iloc[:,11]  #selectivity of AgPd3 through experiment
```

Check of tabel data

In [6]:
```
tabel_AgPd3 = data_AgPd3.iloc[:, [0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11]].copy()

tabel_AgPd3.columns = [
    "exp_numb",
    "flow (mL/min)",
    "WHSV (cm^3/ h/ g cat.)",
    "vol. % C2H2",
    "vol. % C2H4",
    "vol. % H2",
    "vol. % He",
    "vol. % CO",
    "mass cat. (mg)",
    "conv",
    "selec",
    "C-balans"
]

display(tabel_AgPd3)
```

	exp_numb	flow (mL/min)	WHSV (cm^3/ h/ g cat.)	vol. % C2H2	vol. % C2H4	vol. % H2	vol. % He	vol. % CO	mass cat. (mg)	conv	selec	C-balans
0	NaN	flow [mL/min]	WHSV [cm3/h/g_cat]	C2H2	C2H4	H2	He	CO	m_cat [mg]	X_C2H2 (%)	S (%)	C (%)
1	EXP1	200	120000	1	20	20	59	0	100	99.74857	68.89139	99.206749
2	EXP2	150	90000	1	20	20	59	0	100	99.714813	56.155749	102.694068
3	EXP3	100	60000	1	20	20	59	0	100	99.63909	35.640171	100.446485
4	EXP4	50	30000	1	20	20	59	0	100	99.306389	-32.203658	96.834521
...	...	...	...	...	...	...	...	...	...	...	...	...
73	EXP73	50	150000	2	30	20	58	0	20	91.372041	89.795865	102.690968
74	EXP74	50	150000	1.5	30	20	58	0	20	98.626359	79.081587	98.387508
75	EXP75	50	150000	1	30	20	58	0	20	99.55274	59.65292	97.59163
76	EXP76	50	150000	0.5	30	20	58	0	20	99.817177	9.62407	97.166481
77	EXP77	50	150000	2	20	10	58	0	20	69.231734	96.13066	99.103822

78 rows × 12 columns

Table data with all catalysts

0. AgPd0

In [7]:
```
exp_numb_AgPd0 = data_AgPd0.iloc[:,0]          #exp. number AgPd0
flow_AgPd0 = data_AgPd0.iloc[:,1]              #flow per exp. number AgPd0
WHSV_AgPd0 = data_AgPd0.iloc[:,2]              #Contact time per exp. number AgPd0
vol_percent_C2H2_AgPd0 = data_AgPd0.iloc[:,3] #vol.% C2H2 AgPd0
vol_percent_C2H4_AgPd0 = data_AgPd0.iloc[:,4] #vol.% C2H4 AgPd0
vol_percent_H2_AgPd0 = data_AgPd0.iloc[:,5]   #vol.% H2 AgPd0
vol_percent_He_AgPd0 = data_AgPd0.iloc[:,6]   #vol.% He AgPd0
vol_percent_CO_AgPd0 = data_AgPd0.iloc[:,7]   #vol.% CO AgPd0
m_cat_AgPd0 = data_AgPd0.iloc[:,8]            #mass cat. AgPd0
conv_AgPd0 = data_AgPd0.iloc[:,9]             #conversion of AgPd0 through experiment
selec_AgPd0 = data_AgPd0.iloc[:,10]           #selectivity of AgPd0 through experiment
C_balans_AgPd0 = data_AgPd0.iloc[:,11]        #carbon balance of AgPd0 through experiment
```

Check of tabel data

In [8]:
```
tabel_AgPd0 = data_AgPd0.iloc[:, [0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11]].copy()

tabel_AgPd0.columns = [
    "exp_numb",
    "flow (mL/min)",
    "WHSV (cm^3/ h/ g cat.)",
    "vol. % C2H2",
    "vol. % C2H4",
    "vol. % H2",
    "vol. % He",
    "vol. % CO",
    "mass cat. (mg)",
```

```
    "mass cat. (mg)",
    "conv",
    "selec",
    "C-balans"
]

display(tabel_AgPd3)
```

	exp_numb	flow (mL/min)	WHSV (cm^3/ h/ g cat.)	vol. % C2H2	vol. % C2H4	vol. % H2	vol. % He	vol. % CO	mass cat. (mg)	conv	selec	C-balans
0	NaN	flow [mL/min]	WHSV [cm3/h/g_cat]	C2H2	C2H4	H2	He	CO	m_cat [mg]	X_C2H2 (%)	S (%)	C (%)
1	EXP1	200	120000	1	20	20	59	0	100	10.462575	99.716055	103.118627
2	EXP2	150	90000	1	20	20	59	0	100	20.991291	99.858475	102.922557
3	EXP3	100	60000	1	20	20	59	0	100	41.14778	98.759845	98.042775
4	EXP4	50	30000	1	20	20	59	0	100	89.976409	94.960579	98.865154
5	EXP5	100	60000	0.5	20	20	59.5	0	100	82.46717	93.278888	95.638212
6	EXP6	100	60000	0.5	20	10	69.5	0	100	48.097891	99.018459	95.758961
7	EXP7	100	60000	0.5	20	5	74.5	0	100	26.268002	99.536018	99.202554
8	EXP8	100	60000	0.5	20	25	54.5	0	100	99.251859	87.648772	93.699723
9	EXP9	100	60000	0.5	20	15	64.5	0	100	78.19268	92.597644	95.425023
10	EXP10	50	30000	1	20	20	59	0	100	98.791597	90.843585	102.84236
11	EXP11	50	30000	1	30	20	58	0	100	94.846143	90.274211	97.025987
12	EXP12	50	30000	1	40	20	58	0	100	89.296816	89.835532	102.311236
13	EXP13	50	30000	1	10	20	58	0	100	96.442351	94.666778	102.029133
14	EXP14	100	60000	0.5	40	20	58	0	100	82.684527	84.948803	96.757535
15	EXP15	100	60000	0.5	10	20	58	0	100	93.611079	93.032813	100.078357
16	EXP16	100	60000	0.5	30	20	58	0	100	81.075988	88.054639	98.952162
17	EXP17	100	60000	0.5	20	20	58	0	100	85.488353	90.955986	95.055496
18	EXP18	100	60000	0.5	40	5	58	0	100	19.658322	97.8718	99.129158
19	EXP19	100	60000	0.5	40	15	58	0	100	62.632562	89.463707	97.943281
20	EXP20	100	60000	0.5	40	20	58	0	100	77.053764	85.348316	98.834894
21	EXP21	100	60000	0.5	40	10	58	0	100	41.838696	93.056003	98.297807
22	EXP22	100	60000	0.5	10	20	58	0	100	91.86307	93.217509	100.230661
23	EXP23	100	60000	2	10	20	58	0	100	19.894769	99.093028	101.813501
24	EXP24	100	60000	1.5	10	20	58	0	100	28.921287	98.090568	101.922444
25	EXP25	100	60000	1	10	20	58	0	100	46.5194	97.538916	98.645137
26	EXP26	100	60000	0.5	20	20	58	0	100	90.692707	89.875899	98.367993
27	EXP27	100	60000	2	20	20	58	0	100	21.782473	99.112238	100.032724
28	EXP28	100	60000	1.5	20	20	58	0	100	25.05544	98.164887	95.557869
29	EXP29	100	60000	1	20	20	58	0	100	68.35193	95.040856	96.784842
30	EXP30	100	60000	0.5	30	20	58	0	100	81.797062	86.166601	99.393256
31	EXP31	50	30000	2	30	20	58	0	100	43.369806	98.404244	98.713697
32	EXP32	50	30000	1.5	30	20	58	0	100	61.208434	96.725177	98.236953
33	EXP33	50	30000	1	30	20	58	0	100	92.784525	90.647609	97.973471
34	EXP35	50	30000	2	20	10	68	0	100	30.538149	98.660829	96.702082
35	EXP36	50	30000	1.5	20	10	68.5	0	100	40.469403	99.984229	97.831929
36	EXP37	50	30000	1	20	10	69	0	100	55.909774	98.006326	100.252049
37	EXP38	50	30000	0.5	20	10	69.5	0	100	97.752323	85.407464	99.608689
38	EXP39	50	30000	2	20	5	73	0	100	22.908086	99.599562	104.62659
39	EXP40	50	30000	1.5	20	5	73.5	0	100	13.471929	99.434254	101.782663
40	EXP41	50	30000	1	20	5	74	0	100	30.786808	97.504127	96.00494
41	EXP42	50	30000	0.5	20	5	74.5	0	100	63.78491	94.820135	98.3229

In [9]:
```
def prepare_table(tabel, catalyst_name):
    df_temp = tabel.copy()
    df_temp["catalyst"] = catalyst_name
    return df_temp

df_all = pd.concat([
    prepare_table(tabel_AgPd0, "AgPd0"),
    prepare_table(tabel_AgPd2, "AgPd2"),
    prepare_table(tabel_AgPd3, "AgPd3"),
], ignore_index=True)

display(df_all)
```

	exp_numb	flow (mL/min)	WHSV (cm^3/ h/ g cat.)	vol. % C2H2	vol. % C2H4	vol. % H2	vol. % He	vol. % CO	mass cat. (mg)	conv	selec	C-balans	catalyst
0	NaN	flow [mL/min]	WHSV [cm3/h/g_cat]	C2H2	C2H4	H2	He	CO	m_cat [mg]	X_C2H2 (%)	S (%)	C (%)	AgPd0
1	EXP1	200	120000	1	20	20	59	0	100	10.462575	99.716055	103.118627	AgPd0
2	EXP2	150	90000	1	20	20	59	0	100	20.991291	99.858475	102.922557	AgPd0
3	EXP3	100	60000	1	20	20	59	0	100	41.14778	98.759845	98.042775	AgPd0
4	EXP4	50	30000	1	20	20	59	0	100	89.976409	94.960579	98.865154	AgPd0
...	...	...	...	...	...	...	...	...	...	...	...	...	...
158	EXP73	50	150000	2	30	20	58	0	20	91.372041	89.795865	102.690968	AgPd3
159	EXP74	50	150000	1.5	30	20	58	0	20	98.626359	79.081587	98.387508	AgPd3
160	EXP75	50	150000	1	30	20	58	0	20	99.55274	59.65292	97.59163	AgPd3
161	EXP76	50	150000	0.5	30	20	58	0	20	99.817177	9.62407	97.166481	AgPd3
162	EXP77	50	150000	2	20	10	58	0	20	69.231734	96.13066	99.103822	AgPd3

163 rows × 13 columns

Constants

All the constants that can/ need be filled out beforehand, and turning DW [g] into Dz [m]. Through density of Al2O3 = 3,987 g/cm^3

In [10]:
```
# Constants
```

```
R = 8.314                                   #J /mol /K
D = 0.5                                     #diameter; cm
rho = 2.65                                  #Density SiO2; g /cm^3
m_cat_AgPd0 = data_AgPd0.iloc[2,8]          #mass of AgPd0 catalyst
m_cat_AgPd2 = data_AgPd2.iloc[2,8]          #mass of AgPd2 catalyst
m_cat_AgPd3 = data_AgPd3.iloc[2,8]          #mass of AgPd3 catalyst
T = 373                                     #K, 100 celcius
P_total = 100000                            #Pa

A = (np.pi *D**2)/4                         #Area reactor; cm^2

dW_dz = rho * A * 100                       #g/m

m_cat_AgPd0 = pd.to_numeric(
    data_AgPd0.iloc[:, 8].astype(str).str.replace(",", ".", regex=False),
    errors="coerce"
).dropna().iloc[0]

m_cat_AgPd2 = pd.to_numeric(
    data_AgPd2.iloc[:, 8].astype(str).str.replace(",", ".", regex=False),
    errors="coerce"
).dropna().iloc[0]

m_cat_AgPd3 = pd.to_numeric(
    data_AgPd3.iloc[:, 8].astype(str).str.replace(",", ".", regex=False),
    errors="coerce"
).dropna().iloc[0]

W_total_AgPd0 = m_cat_AgPd0 / 1000          #mg to g
W_total_AgPd2 = m_cat_AgPd2 / 1000          #mg to g
W_total_AgPd3 = m_cat_AgPd3 / 1000          #mg to g

L_total_AgPd0 = W_total_AgPd0/dW_dz         #calculation of reactor length AgPd0; m
L_total_AgPd2 = W_total_AgPd2/dW_dz         #calculation of reactor length AgPd2; m
L_total_AgPd3 = W_total_AgPd3/dW_dz         #calculation of reactor length AgPd3; m

# Reproducibility
RNG = np.random.default_rng(seed=42)

print("The totale square of the reactor is:",A,"cm2")
print("The total length of the catalyst AgPd0 is:",L_total_AgPd0, "m")
print("The total length of the catalyst AgPd2 is:",L_total_AgPd2, "m")
print("The total length of the catalyst AgPd3 is:",L_total_AgPd3, "m")
```

```
The totale square of the reactor is: 0.19634954084936207 cm2
The total length of the catalyst AgPd0 is: 0.0019218710109210008 m
The total length of the catalyst AgPd2 is: 0.0014414032581987504 m
The total length of the catalyst AgPd3 is: 0.0019218710109210008 m
```

Reactor model

Then the function of the packed-bed reactor, with dW changed to dz. Do this with dW = density (rho) area dz. Giving dF/dz = dF/dW *rho* A.

In [11]:

```
def pfr_rhs(z, F, LHHW, T, P_total, dW_dz, k1, k2, K1, K2, Kinh, KH):
    F = np.maximum(F, 0.0)                          #numerical safety; negative flows are unphysical
    F_tot = F.sum()
    if F_tot <= 0:
        return np.zeros_like(F)

    y = F / F_tot                                   #mol fraction
    p_C2H2, p_C2H4, p_C2H6, p_H2, p_He, p_CO = y * P_total          #Partial pressures (Pa)

    r1, r2 = LHHW(p_C2H2, p_C2H4, p_H2, T, k1, k2, K1, K2, Kinh, KH)

    dFdz = dW_dz * np.array([
        -r1,            #C2H2
        +r1 - r2,       #C2H4
        +r2,            #C2H6
        -r1 - r2,       #H2
        0.0,            #He
        0.0             #CO
    ])
    return dFdz
```

Solve PFR formula, change W_total etc. to L_total so it integrates over L = 0 to L = L_total instead of W = 0 to W = W_total.

In [12]:

```
def solve_pfr(F0, L_total, LHHW, T, P_total, dW_dz, k1, k2, K1, K2, Kinh, KH, n_points=50):
    z_eval = np.linspace(0.0, L_total, n_points)
    sol = solve_ivp(
        fun=lambda z, F: pfr_rhs(z, F, LHHW, T, P_total, dW_dz, k1, k2, K1, K2, Kinh, KH),
        t_span=(0.0, L_total),
        y0=F0,
        method="LSODA",                                 #stiff-aware; safer for kinetics
        t_eval=z_eval,
        rtol=1e-9, atol=1e-14,
    )
    if not sol.success:
        raise RuntimeError(f"PFR integration failed: {sol.message}")
    return sol.t, sol.y                                 #(n_points,), (4, n_points)
```

Create function of Power Law and LHHW model

For both these models, r1 is the wanted reaction and r2 is the unwanted reaction (over-hydrogenation)

The LHHW model describes the competitive adsorption on the catalyst between C2H2 and C2H4.

$$r_1 = \frac{k_1 K_{C_2H_2} p_{C_2H_2} K_{H_2} p_{H_2}}{(1 + K_{C_2H_2} p_{C_2H_2} + K_{C_2H_4} p_{C_2H_4})^2}$$

$$r_2 = \frac{k_2 K_{C_2H_4} p_{C_2H_4} K_{H_2} p_{H_2}}{(1 + K_{C_2H_2} p_{C_2H_2} + K_{C_2H_4} p_{C_2H_4})^2 (1 + K_{inh} p_{C_2H_2})}$$

In [13]:

```
# ----- LHHW model -----
def LHHW(p_C2H2, p_C2H4, p_H2, T, k1, k2, K1, K2, Kinh, KH):
        pA = max(p_C2H2, 0.0) / 1e5                 #pressure from Pa to bar
        pE = max(p_C2H4, 0.0) / 1e5                 #pressure from Pa to bar
        pH = max(p_H2,   0.0) / 1e5                 #pressure from Pa to bar

        denom = (1.0 + K1 * pA + K2 * pE + KH * pH)**2

        r1 = k1 * K1 * pA * KH * pH / denom
        r2 = k2 * K2 * pE * KH * pH / (denom * (1+Kinh*pA))
        return r1, r2
```

Simulation (check)

In [14]:

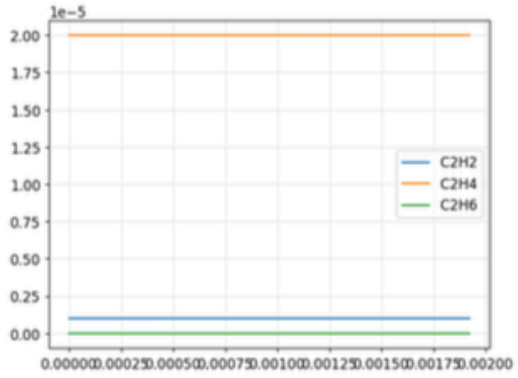
```
k1_test = 1.5e-6          #bar
```

```
k2_test = 1e-6              #bar
K1_test = 100               #bar
K2_test = 0.1               #bar
Kinh_test = 3               #bar
K6_test = 1                 #bar

z, f = solve_pfr([1e-6, 20e-6, 0.0, 10e-6, 69e-6, 0.0], L_total=L_total_AgPd0, LHHW=LHHW, T=T, P_total=P_total, dW_dz=dW_dz, k1= k1_test, k2=k2_test, K1=K1_test, K2= K2

ax = plt.gca()
ax.plot(z,f[0],label='C2H2')
ax.plot(z,f[1],label='C2H4')
ax.plot(z,f[2],label='C2H6')
plt.legend()
```

Out[14]: <matplotlib.legend.Legend at 0x1772f9c90>



Converting to mol/s

Converting inlet volume fraction and flow rate to mol/ s. Can be done through the ideal gas law

In [15]:
```
def V_frac_flow(df, R, T, P, flow_col):
    df = df.copy()

    # remove repeated text rows
    df = df[df[flow_col] != "flow (mL/min)"].reset_index(drop=True)

    # flow from mL/min to mol/s
    df["V_m3_s"] = df[flow_col] * 1e-6 / 60
    df["F_tot_mol_s"] = P * df["V_m3_s"] / (R * T)

    # vol.% to fraction
    df["y_C2H2"] = df["vol. % C2H2"] / 100
    df["y_C2H4"] = df["vol. % C2H4"] / 100
    df["y_H2"]   = df["vol. % H2"] / 100
    df["y_He"]   = df["vol. % He"] / 100
    df["y_CO"]   = df["vol. % CO"] / 100

    # components in mol/s
    df["F_C2H2_in"] = df["y_C2H2"] * df["F_tot_mol_s"]
    df["F_C2H4_in"] = df["y_C2H4"] * df["F_tot_mol_s"]
    df["F_H2_in"]   = df["y_H2"]   * df["F_tot_mol_s"]
    df["F_He_in"]   = df["y_He"]   * df["F_tot_mol_s"]
    df["F_CO_in"]   = df["y_CO"]   * df["F_tot_mol_s"]
    df["F_C2H6_in"] = 0.0

    return df
```

Add to tabel with all exp. data

In [16]:
```
df_all = V_frac_flow(
    df_all,
    R,
    T,
    P_total,
    flow_col="flow (mL/min)"
)

display(df_all)
```

	exp_numb	flow (mL/min)	WHSV (cm^3/ h/ g cat.)	vol. % C2H2	vol. % C2H4	vol. % H2	vol. % He	vol. % CO	mass cat. (mg)	conv	...	y_C2H4	y_H2	y_He	y_CO	F_C2H2_in	F_C2H4_in	F_H2_in	F_He_in	F_CO_in	F_C2H6_in
0	EXP1	200	120000	1	20	20	59	0	100	10.462575	...	0.2	0.2	0.59	0.0	0.000001	0.000021	0.000021	0.000063	0.0	0.0
1	EXP2	150	90000	1	20	20	59	0	100	20.991291	...	0.2	0.2	0.59	0.0	0.000001	0.000016	0.000016	0.000048	0.0	0.0
2	EXP3	100	60000	1	20	20	59	0	100	41.14778	...	0.2	0.2	0.59	0.0	0.000001	0.000011	0.000011	0.000032	0.0	0.0
3	EXP4	50	30000	1	20	20	59	0	100	89.976409	...	0.2	0.2	0.59	0.0	0.0	0.000005	0.000005	0.000016	0.0	0.0
4	EXP5	100	60000	0.5	20	20	59.5	0	100	82.46717	...	0.2	0.2	0.595	0.0	0.0	0.000011	0.000011	0.000032	0.0	0.0
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
155	EXP73	50	150000	2	30	20	58	0	20	91.372041	...	0.3	0.2	0.58	0.0	0.000001	0.000008	0.000005	0.000016	0.0	0.0
156	EXP74	50	150000	1.5	30	20	58	0	20	98.626359	...	0.3	0.2	0.58	0.0	0.0	0.000008	0.000005	0.000016	0.0	0.0
157	EXP75	50	150000	1	30	20	58	0	20	99.55274	...	0.3	0.2	0.58	0.0	0.0	0.000008	0.000005	0.000016	0.0	0.0
158	EXP76	50	150000	0.5	30	20	58	0	20	99.817177	...	0.3	0.2	0.58	0.0	0.0	0.000008	0.000005	0.000016	0.0	0.0
159	EXP77	50	150000	2	20	10	58	0	20	69.231734	...	0.2	0.1	0.58	0.0	0.000001	0.000005	0.000003	0.000016	0.0	0.0

160 rows × 26 columns

Convert fraction and flow for each tabel

In [17]:
```
data_AgPd0_mol_s = V_frac_flow(tabel_AgPd0, R, T, P_total, flow_col="flow (mL/min)")
data_AgPd2_mol_s = V_frac_flow(tabel_AgPd2, R, T, P_total, flow_col="flow (mL/min)")
data_AgPd3_mol_s = V_frac_flow(tabel_AgPd3, R, T, P_total, flow_col="flow (mL/min)")
```

Conversion and selectivity calculation

Conversion and selectivity calculation

Calculate conversion and selectivity at each of our experimental conditions. Store array X_sim and S_sim and compare with experimental. With for loop for i <= exp.num. Store in array and calculate error towards experimental.

Conversion and selectivity function

In [18]:
```
def conv_selec_sim (F0, F_out, F_C4 = 0):
    F_C2H2_feed = F0[0]          #in
    F_C2H4_feed = F0[1]          #in
    F_C2H6_feed = F0[2]          #in

    F_C2H2_out = F_out[0]        #out
    F_C2H4_out = F_out[1]        #out
    F_C2H6_out = F_out[2]        #out
    F_H2_out = F_out[3]          #out


    C2H2_consumed = F_C2H2_feed - F_C2H2_out
    C2H6_formed = F_C2H6_out - F_C2H6_feed

    conversion_sim = C2H2_consumed/F_C2H2_feed * 100
    selectivty_sim = (1-(C2H6_formed+ 2*F_C4)/C2H2_consumed) *100
    return conversion_sim, selectivty_sim, C2H2_consumed, C2H6_formed
```

In [19]:
```
sim_results =[]

for i in range(0,len(df_all)):
    row = df_all.iloc[i]

    F0 = [row["F_C2H2_in"],row["F_C2H4_in"],row["F_C2H6_in"],row["F_H2_in"],row["F_He_in"],row["F_CO_in"]]                    #Inlet fractions, mol /s

    z, F = solve_pfr(F0=F0, L_total=L_total_AgPd0, LHHW=LHHW, T=T, P_total=P_total, dW_dz=dW_dz, k1=k1_test,k2=k2_test,K1=K1_test,K2=K2_test, Kinh=Kinh_test, KH=KH_test

    F_out = F[:,-1]                          #get outletstreams from sim

    conversion_sim, selectivity_sim, C2H2_consumed, C2H6_formed = conv_selec_sim(F0=F0,F_out=F_out,F_C4=0)  #calculation of X, S, etc.

    sim_results.append({
        "exp_numb": row["exp_numb"],
        "catalyst": row["catalyst"],
        "conv_sim": conversion_sim,
        "selec_sim": selectivity_sim
    })

sim_results_df = pd.DataFrame(sim_results)

display(sim_results_df)
```

	exp_numb	catalyst	conv_sim	selec_sim
0	EXP1	AgPd0	0.566098	98.701536
1	EXP2	AgPd0	0.754701	98.700207
2	EXP3	AgPd0	1.131760	98.697541
3	EXP4	AgPd0	2.261738	98.689468
4	EXP5	AgPd0	1.879079	97.346763
...	...	...	...	...
155	EXP73	AgPd3	1.007397	99.046351
156	EXP74	AgPd3	1.398936	98.709453
157	EXP75	AgPd3	2.070384	98.031919
158	EXP76	AgPd3	3.362043	95.984667
159	EXP77	AgPd3	0.612315	99.372770

160 rows × 4 columns

Aline experimental data

In [20]:
```
exp_results =[]

for i in range(0, len(df_all)):
    row = df_all.iloc[i]

    exp_results.append({
        "exp_numb": row["exp_numb"],
        "catalyst": row["catalyst"],
        "conv_exp": row["conv"],
        "selec_exp": row["selec"]
    })
exp_results_df = pd.DataFrame(exp_results)

display(exp_results_df)
```

	exp_numb	catalyst	conv_exp	selec_exp
0	EXP1	AgPd0	10.462575	99.716055
1	EXP2	AgPd0	20.991291	99.858475
2	EXP3	AgPd0	41.147780	98.759845
3	EXP4	AgPd0	89.976409	94.960579
4	EXP5	AgPd0	82.467170	93.278888
...	...	...	...	...
155	EXP73	AgPd3	91.372041	89.796865
156	EXP74	AgPd3	98.626359	79.081587
157	EXP75	AgPd3	99.552740	59.652920
158	EXP76	AgPd3	99.817177	9.624070
159	EXP77	AgPd3	69.231734	96.130660

160 rows × 4 columns

Merge both experimental and simulated data in tabel

In [21]:
```
comparison_df = pd.merge(sim_results_df,exp_results_df,on=["exp_numb","catalyst"],how="inner")

display(comparison_df)
```

	exp_numb	catalyst	conv_sim	selec_sim	conv_exp	selec_exp
0	EXP1	AgPd0	0.566098	98.701536	10.462575	99.716055

2	EXP3	AgPd0	1.131760	98.697541	41.147780	98.759845	
3	EXP4	AgPd0	2.261738	98.689468	89.976409	94.960579	
4	EXP5	AgPd0	1.879079	97.346763	82.467170	93.278888	
...	...	...	...	...	...	...	
155	EXP73	AgPd3	1.007397	99.046351	91.372041	89.795865	
156	EXP74	AgPd3	1.398936	98.709453	98.626359	79.081587	
157	EXP75	AgPd3	2.070384	98.031919	99.552740	59.652920	
158	EXP76	AgPd3	3.362043	95.984667	99.817177	9.624070	
159	EXP77	AgPd3	0.612315	99.372770	69.231734	96.130660	

160 rows × 6 columns

Error calculation

Calculate the RMSE for each experiment

$$RMSE = \sqrt{\frac{(conv_{sim} - conv_{exp})^2 + (selec_{sim} - selec_{exp})^2}{2}}$$

So first create the function

In [22]:
```
def calc_RMSE(conv_sim,conv_exp,selec_sim,selec_exp):
    RMSE = np.sqrt(((conv_sim-conv_exp)**2+(selec_sim-selec_exp)**2)/2)
    return RMSE
```

Add every RMSE to the final table

In [23]:
```
RMSE_results = []

for i in range(0,len(df_all)):
    row = comparison_df.iloc[i]

    RMSE = calc_RMSE(
        conv_sim = row["conv_sim"],
        conv_exp = row["conv_exp"],
        selec_sim = row["selec_sim"],
        selec_exp = row["selec_exp"]
        )

    RMSE_results.append(RMSE)

comparison_df["RMSE"] = RMSE_results

display(comparison_df)

# Overall RMSE:
print("Overall RMSE =", np.sum(RMSE_results)/len(df_all.iloc[0,:]))
```

	exp_numb	catalyst	conv_sim	selec_sim	conv_exp	selec_exp	RMSE
0	EXP1	AgPd0	0.566098	98.701536	10.462575	99.716055	7.034541
1	EXP2	AgPd0	0.754701	98.700207	20.991291	99.858475	14.332850
2	EXP3	AgPd0	1.131760	98.697541	41.147780	98.759845	28.295634
3	EXP4	AgPd0	2.261738	98.689468	89.976409	94.960579	62.079659
4	EXP5	AgPd0	1.879079	97.346763	82.467170	93.278888	57.056937
...	...	...	...	...	...	...	...
155	EXP73	AgPd3	1.007397	99.046351	91.372041	89.795865	64.231380
156	EXP74	AgPd3	1.398936	98.709453	98.626359	79.081587	70.137098
157	EXP75	AgPd3	2.070384	98.031919	99.552740	59.652920	74.080218
158	EXP76	AgPd3	3.362043	95.984667	99.817177	9.624070	91.547107
159	EXP77	AgPd3	0.612315	99.372770	69.231734	96.130660	48.575385

160 rows × 7 columns

```
Overall RMSE = 472.42114881328184
```

Results (check)

To also quickly see how the simulated and experimental data compare to each other we plot it per catalyst. The dotted line is the perfect prediction

Done through for-loop, currently only checking AgPd2 so not in use. But in the end can be used to quickly plot all the catalysts unique X and S plots.

(Then everything below can be deleted)

Conversion and selectivity:

In [24]:
```
for catalyst_name in comparison_df["catalyst"].unique():
    df_cat = comparison_df[comparison_df["catalyst"] == catalyst_name]

    plt.figure()
    plt.scatter(df_cat["conv_exp"], df_cat["conv_sim"])
    plt.plot([0, 100], [0, 100], linestyle="--")
    plt.xlabel("Experimental conversion (%)")
    plt.ylabel("Simulated conversion (%)")
    plt.title(f"Parity plot: C2H2 conversion - {catalyst_name}")
    plt.show()

    plt.figure()
    plt.scatter(df_cat["selec_exp"], df_cat["selec_sim"])
    plt.plot([0, 100], [0, 100], linestyle="--")
    plt.xlabel("Experimental selectivity (%)")
    plt.ylabel("Simulated selectivity (%)")
    plt.title(f"Parity plot: C2H4 selectivity - {catalyst_name}")
    plt.show()
```



Optimisation

Need to use: differential_evolution

- Not all. Each catalyst separately because each catalyst has its own kinetics. Done \ Code wise:
- k1,k2,K1,K2, Kinh now become an argument.

After differential_evolution, solve simulation again with optimised parameters.

Start fitting with the "best" data

In [25]:
```
# Only AgPd0
df_fit = df_all[df_all["catalyst"] == "AgPd0"].copy()

##All data
#df_fit = df_all.copy()

display(df_fit)
```

	exp_numb	flow (mL/min)	WHSV (cm^3/ h/ g cat.)	vol. % C2H2	vol. % C2H4	vol. % H2	vol. % He	vol. % CO	mass cat. (mg)	conv	...	y_C2H4	y_H2	y_He	y_CO	F_C2H2_in	F_C2H4_in	F_H2_in	F_He_in	F_CO_in	F_C2H6_in
0	EXP1	200	120000	1	20	20	59	0	100	10.462575	...	0.2	0.2	0.59	0.0	0.000001	0.000021	0.000021	0.000063	0.0	0.0
1	EXP2	150	90000	1	20	20	59	0	100	20.991291	...	0.2	0.2	0.59	0.0	0.000001	0.000016	0.000016	0.000048	0.0	0.0
2	EXP3	100	60000	1	20	20	59	0	100	41.14778	...	0.2	0.2	0.59	0.0	0.000001	0.000011	0.000011	0.000032	0.0	0.0
3	EXP4	50	30000	1	20	20	59	0	100	89.976409	...	0.2	0.2	0.59	0.0	0.0	0.000005	0.000005	0.000016	0.0	0.0
4	EXP5	100	60000	0.5	20	20	59.5	0	100	82.46717	...	0.2	0.2	0.595	0.0	0.0	0.000011	0.000011	0.000032	0.0	0.0
5	EXP6	100	60000	0.5	20	10	69.5	0	100	48.097891	...	0.2	0.1	0.695	0.0	0.0	0.000011	0.000005	0.000037	0.0	0.0
6	EXP7	100	60000	0.5	20	5	74.5	0	100	26.268002	...	0.2	0.05	0.745	0.0	0.0	0.000011	0.000003	0.00004	0.0	0.0
7	EXP8	100	60000	0.5	20	25	54.5	0	100	99.251859	...	0.2	0.25	0.545	0.0	0.0	0.000011	0.000013	0.000029	0.0	0.0
8	EXP9	100	60000	0.5	20	15	64.5	0	100	78.19268	...	0.2	0.15	0.645	0.0	0.0	0.000011	0.000008	0.000035	0.0	0.0
9	EXP10	50	30000	1	20	20	59	0	100	98.791597	...	0.2	0.2	0.59	0.0	0.0	0.000005	0.000005	0.000016	0.0	0.0
10	EXP11	50	30000	1	30	20	58	0	100	94.846143	...	0.3	0.2	0.58	0.0	0.0	0.000008	0.000005	0.000016	0.0	0.0
11	EXP12	50	30000	1	40	20	58	0	100	89.296816	...	0.4	0.2	0.58	0.0	0.0	0.000011	0.000005	0.000016	0.0	0.0
12	EXP13	50	30000	1	10	20	58	0	100	96.442351	...	0.1	0.2	0.58	0.0	0.0	0.000003	0.000005	0.000016	0.0	0.0
13	EXP14	100	60000	0.5	40	20	58	0	100	82.684527	...	0.4	0.2	0.58	0.0	0.0	0.000021	0.000011	0.000031	0.0	0.0
14	EXP15	100	60000	0.5	10	20	58	0	100	93.611079	...	0.1	0.2	0.58	0.0	0.0	0.000005	0.000011	0.000031	0.0	0.0
15	EXP16	100	60000	0.5	30	20	58	0	100	81.075988	...	0.3	0.2	0.58	0.0	0.0	0.000016	0.000011	0.000031	0.0	0.0
16	EXP17	100	60000	0.5	20	20	58	0	100	85.488353	...	0.2	0.2	0.58	0.0	0.0	0.000011	0.000011	0.000031	0.0	0.0
17	EXP18	100	60000	0.5	40	5	58	0	100	19.658322	...	0.4	0.05	0.58	0.0	0.0	0.000021	0.000003	0.000031	0.0	0.0
18	EXP19	100	60000	0.5	40	15	58	0	100	62.632562	...	0.4	0.15	0.58	0.0	0.0	0.000021	0.000008	0.000031	0.0	0.0
19	EXP20	100	60000	0.5	40	20	58	0	100	77.053764	...	0.4	0.2	0.58	0.0	0.0	0.000021	0.000011	0.000031	0.0	0.0
20	EXP21	100	60000	0.5	40	10	58	0	100	41.838696	...	0.4	0.1	0.58	0.0	0.0	0.000021	0.000005	0.000031	0.0	0.0
21	EXP22	100	60000	0.5	10	20	58	0	100	91.86307	...	0.1	0.2	0.58	0.0	0.0	0.000005	0.000011	0.000031	0.0	0.0
22	EXP23	100	60000	2	10	20	58	0	100	19.894769	...	0.1	0.2	0.58	0.0	0.000001	0.000005	0.000011	0.000031	0.0	0.0
23	EXP24	100	60000	1.5	10	20	58	0	100	28.921287	...	0.1	0.2	0.58	0.0	0.000001	0.000005	0.000011	0.000031	0.0	0.0
24	EXP25	100	60000	1	10	20	58	0	100	46.5194	...	0.1	0.2	0.58	0.0	0.000001	0.000005	0.000011	0.000031	0.0	0.0
25	EXP26	100	60000	0.5	20	20	58	0	100	90.692707	...	0.2	0.2	0.58	0.0	0.0	0.000011	0.000011	0.000031	0.0	0.0
26	EXP27	100	60000	2	20	20	58	0	100	21.782473	...	0.2	0.2	0.58	0.0	0.000001	0.000011	0.000011	0.000031	0.0	0.0
27	EXP28	100	60000	1.5	20	20	58	0	100	25.05544	...	0.2	0.2	0.58	0.0	0.000001	0.000011	0.000011	0.000031	0.0	0.0
28	EXP29	100	60000	1	20	20	58	0	100	68.35193	...	0.2	0.2	0.58	0.0	0.000001	0.000011	0.000011	0.000031	0.0	0.0
29	EXP30	100	60000	0.5	30	20	58	0	100	81.797062	...	0.3	0.2	0.58	0.0	0.0	0.000016	0.000011	0.000031	0.0	0.0
30	EXP31	50	30000	2	30	20	58	0	100	43.369806	...	0.3	0.2	0.58	0.0	0.000001	0.000008	0.000005	0.000016	0.0	0.0
31	EXP32	50	30000	1.5	30	20	58	0	100	61.208434	...	0.3	0.2	0.58	0.0	0.0	0.000008	0.000005	0.000016	0.0	0.0
32	EXP33	50	30000	1	30	20	58	0	100	92.784525	...	0.3	0.2	0.58	0.0	0.0	0.000008	0.000005	0.000016	0.0	0.0
33	EXP35	50	30000	2	20	10	68	0	100	30.538149	...	0.2	0.1	0.68	0.0	0.000001	0.000005	0.000003	0.000018	0.0	0.0
34	EXP36	50	30000	1.5	20	10	68.5	0	100	40.469403	...	0.2	0.1	0.685	0.0	0.0	0.000005	0.000003	0.000018	0.0	0.0
35	EXP37	50	30000	1	20	10	69	0	100	55.909774	...	0.2	0.1	0.69	0.0	0.0	0.000005	0.000003	0.000019	0.0	0.0
36	EXP38	50	30000	0.5	20	10	69.5	0	100	97.753323	...	0.2	0.1	0.695	0.0	0.0	0.000005	0.000003	0.000019	0.0	0.0
37	EXP39	50	30000	2	20	5	73	0	100	22.908086	...	0.2	0.05	0.73	0.0	0.000001	0.000005	0.000001	0.00002	0.0	0.0
38	EXP40	50	30000	1.5	20	5	73.5	0	100	13.471929	...	0.2	0.05	0.735	0.0	0.0	0.000005	0.000001	0.00002	0.0	0.0
39	EXP41	50	30000	1	20	5	74	0	100	30.786808	...	0.2	0.05	0.74	0.0	0.0	0.000005	0.000001	0.00002	0.0	0.0
40	EXP42	50	30000	0.5	20	5	74.5	0	100	63.78491	...	0.2	0.05	0.745	0.0	0.0	0.000005	0.000001	0.00002	0.0	0.0

41 rows × 26 columns

In [26]:
```
#For now, all the same. Can later be further improved
bounds = [
    (1e-10, 1),                 #k1
    (1e-10, 1),                 #k2
    (1e-10, 1000),              #K1
    (1e-10, 1000),              #K2
    (1e-10, 1000),              #Kinh
    (1e-10, 1000)               #KH
]
```

Function that ensures that for every k1, k2, K1 and K2 another simulation is done

In [27]:
```
def make_comparison_df(df, k1, k2, K1, K2, Kinh, KH):
    sim_results = []

    for i in range(0, len(df)):
        row = df.iloc[i]

        F0 = [
            row["F_C2H2_in"],
            row["F_C2H4_in"],
            row["F_C2H6_in"],
            row["F_H2_in"],
            row["F_He_in"],
            row["F_CO_in"]
        ]

        z, F = solve_pfr(
            F0=F0,
            L_total=L_total_AgPd0,
            LHHW=LHHW,
            T=T,
            P_total=P_total,
            dN_dz=dN_dz,
            k1=k1,
            k2=k2,
            K1=K1,
            K2=K2,
            Kinh=Kinh,
            KH=KH
        )

        F_out = F[:, -1]

        conversion_sim, selectivity_sim, C2H2_consumed, C2H6_formed = conv_selec_sim(
            F0=F0,
            F_out=F_out,
            F_C4=0
        )

        sim_results.append({
            "exp_numb": row["exp_numb"],
            "catalyst": row["catalyst"],
            "conv_sim": conversion_sim,
            "selec_sim": selectivity_sim
        })

    sim_results_df = pd.DataFrame(sim_results)

    exp_results = []

    for i in range(0, len(df)):
        row = df.iloc[i]

        exp_results.append({
            "exp_numb": row["exp_numb"],
            "catalyst": row["catalyst"],
            "conv_exp": row["conv"],
            "selec_exp": row["selec"]
        })

    exp_results_df = pd.DataFrame(exp_results)

    comparison_df = pd.merge(
        sim_results_df,
        exp_results_df,
        on=["exp_numb", "catalyst"],
        how="inner"
    )

    return comparison_df
```

Calculates total RMSE over all experiments

In [28]:
```
def calc_overall_RMSE(comparison_df):
    rmse_values = []

    for i in range(0, len(comparison_df)):
        row = comparison_df.iloc[i]

        rmse_i = calc_RMSE(
            conv_sim=row["conv_sim"],
            conv_exp=row["conv_exp"],
            selec_sim=row["selec_sim"],
            selec_exp=row["selec_exp"]
        )

        rmse_values.append(rmse_i)

    overall_RMSE = np.mean(rmse_values)

    return overall_RMSE
```

In [29]:
```
def objective_function(params, df):
```

```
    k1, k2, K1, K2, Kinh, KH = params

    comparison_df = make_comparison_df(df, k1, k2, K1, K2, Kinh, KH)

    rmse = calc_overall_RMSE(comparison_df)

    if not np.isfinite(rmse):
        return 1e10

    return rmse
```

Test RMSE

In [30]:
```
test_rmse = objective_function([1, 1, 1, 1, 1, 1], df_fit)

print("Test RMSE:", test_rmse)
```

```
Test RMSE: 2298.780351881826
```

Calculation of diff_evolution

In [31]:
```
result = differential_evolution(objective_function, bounds=bounds, args=(df_fit,),strategy="best1bin",maxiter=50,popsize=42,tol=1e-8,seed=42,polish=True,disp=True)

best_k1, best_k2, best_K1, best_K2, best_Kinh, best_KH = result.x

print("Best k1:", best_k1)
print("Best k2:", best_k2)
print("Best K1:", best_K1)
print("Best K2:", best_K2)
print("Best Kinh:", best_Kinh)
print("Best_KH:", best_KH)
print("Best RMSE:", result.fun)
```

```
differential_evolution step 1: f(x)= 688.314
differential_evolution step 2: f(x)= 231.12
differential_evolution step 3: f(x)= 231.12
differential_evolution step 4: f(x)= 231.12
differential_evolution step 5: f(x)= 231.12
differential_evolution step 6: f(x)= 231.12
differential_evolution step 7: f(x)= 201.635
differential_evolution step 8: f(x)= 77.5653
differential_evolution step 9: f(x)= 77.5653
differential_evolution step 10: f(x)= 77.5653
differential_evolution step 11: f(x)= 77.5653
differential_evolution step 12: f(x)= 36.9625
differential_evolution step 13: f(x)= 36.9625
differential_evolution step 14: f(x)= 36.9625
differential_evolution step 15: f(x)= 36.9625
differential_evolution step 16: f(x)= 32.6532
differential_evolution step 17: f(x)= 30.9133
differential_evolution step 18: f(x)= 29.7839
differential_evolution step 19: f(x)= 19.976
differential_evolution step 20: f(x)= 19.976
differential_evolution step 21: f(x)= 19.976
differential_evolution step 22: f(x)= 19.976
differential_evolution step 23: f(x)= 18.9423
differential_evolution step 24: f(x)= 18.9423
differential_evolution step 25: f(x)= 18.7673
differential_evolution step 26: f(x)= 18.7673
differential_evolution step 27: f(x)= 18.7548
differential_evolution step 28: f(x)= 18.5798
differential_evolution step 29: f(x)= 18.5798
differential_evolution step 30: f(x)= 18.5798
differential_evolution step 31: f(x)= 18.5798
differential_evolution step 32: f(x)= 18.5798
differential_evolution step 33: f(x)= 18.5514
differential_evolution step 34: f(x)= 18.5514
differential_evolution step 35: f(x)= 18.5514
differential_evolution step 36: f(x)= 18.415
differential_evolution step 37: f(x)= 18.415
differential_evolution step 38: f(x)= 18.3345
differential_evolution step 39: f(x)= 17.8773
differential_evolution step 40: f(x)= 17.8187
differential_evolution step 41: f(x)= 17.7188
differential_evolution step 42: f(x)= 17.7188
differential_evolution step 43: f(x)= 16.6017
differential_evolution step 44: f(x)= 15.9631
differential_evolution step 45: f(x)= 15.9631
differential_evolution step 46: f(x)= 11.8981
differential_evolution step 47: f(x)= 11.8981
differential_evolution step 48: f(x)= 11.8981
differential_evolution step 49: f(x)= 9.63176
differential_evolution step 50: f(x)= 9.63176
Polishing solution with 'L-BFGS-B'
Best k1: 7.185213653332459e-05
Best k2: 1.6491690134537864e-05
Best K1: 936.2138142510355
Best K2: 10.207782660587807
Best Kinh: 971.9975246336327
Best_KH: 3.3319351185636328
Best RMSE: 6.439535080722887
```

In [32]:
```
comparison_de_df = make_comparison_df(
    df=df_fit,
    k1=best_k1,
    k2=best_k2,
    K1=best_K1,
    K2=best_K2,
    Kinh=best_Kinh,
    KH=best_KH
)

display(comparison_de_df)
```

	exp_numb	catalyst	conv_sim	selec_sim	conv_exp	selec_exp
0	EXP1	AgPd0	25.588462	99.604854	10.462579	99.716055
1	EXP2	AgPd0	34.662012	99.552724	20.991291	99.858475
2	EXP3	AgPd0	53.531036	99.384633	41.147780	98.759845
3	EXP4	AgPd0	97.258010	95.039758	89.976409	94.960579
4	EXP5	AgPd0	93.820644	91.181972	82.467170	93.278888
5	EXP6	AgPd0	63.302732	97.382293	48.097891	99.018459
6	EXP7	AgPd0	33.659843	98.422807	26.268002	99.536018
7	EXP8	AgPd0	97.589334	86.266364	99.251859	87.648772
8	EXP9	AgPd0	83.829820	95.123030	78.192680	92.597644
9	EXP10	AgPd0	97.258010	95.039758	98.791597	90.843585
10	EXP11	AgPd0	84.468912	97.486847	94.846143	90.274211
11	EXP12	AgPd0	71.140083	97.849873	89.296816	89.835532
12	EXP13	AgPd0	99.995423	86.969765	96.442351	94.666728

13	EXP14	AgPd0	64.867739	93.911833	82.684527	84.948803
14	EXP15	AgPd0	99.904689	83.099671	93.611079	93.032813
15	EXP16	AgPd0	78.978778	93.638647	81.075988	88.064639
16	EXP17	AgPd0	93.987448	91.114613	86.488353	90.955986
17	EXP18	AgPd0	20.536155	97.245387	19.658322	97.871800
18	EXP19	AgPd0	53.196505	95.320607	62.632562	89.463707
19	EXP20	AgPd0	64.867739	93.911833	77.053764	85.348316
20	EXP21	AgPd0	38.395894	96.419664	41.838696	93.056003
21	EXP22	AgPd0	99.904689	83.099671	91.863070	93.217509
22	EXP23	AgPd0	19.620162	99.955598	19.894769	99.093028
23	EXP24	AgPd0	33.597584	99.906565	28.921287	98.090568
24	EXP25	AgPd0	69.957263	99.576627	46.519400	97.538916
25	EXP26	AgPd0	93.987448	91.114613	90.692707	89.875899
26	EXP27	AgPd0	17.428309	99.906452	21.782473	99.112238
27	EXP28	AgPd0	28.488276	99.810940	25.055440	98.164887
28	EXP29	AgPd0	53.641550	99.388620	68.351930	95.040856
29	EXP30	AgPd0	78.978778	93.638647	81.797062	86.166601
30	EXP31	AgPd0	32.609866	99.810936	43.369806	98.404244
31	EXP32	AgPd0	51.907760	99.548771	61.208434	96.725177
32	EXP33	AgPd0	84.468912	97.486847	92.784525	90.647609
33	EXP35	AgPd0	17.864374	99.904945	30.538149	98.660829
34	EXP36	AgPd0	29.471287	99.807457	40.469403	99.984229
35	EXP37	AgPd0	56.590588	99.344845	55.909774	98.006326
36	EXP38	AgPd0	96.695252	88.002025	97.752323	85.407464
37	EXP39	AgPd0	8.786395	99.914495	22.908086	99.599562
38	EXP40	AgPd0	14.345291	99.841055	13.471929	99.434254
39	EXP41	AgPd0	27.423373	99.595301	30.786808	97.504127
40	EXP42	AgPd0	65.320921	97.258708	63.784910	94.820135

Conversion and selectivity plots

In [33]:

```
cat_name = "AgPd0"

de_steps = result.nit
best_RMSE = result.fun

de_steps = result.nit
df_cat_2 = comparison_de_df[comparison_de_df["catalyst"] == cat_name]

fig, axes = plt.subplots(1, 2, figsize=(12, 5))

#error band
band = 0.10                                                                    #±10% relative deviation band
x_line = np.linspace(0, 100, 200)

#points within ±10% error band
conv_within_10pct = (
    (df_cat_2["conv_sim"] >= df_cat_2["conv_exp"] * (1 - band)) &
    (df_cat_2["conv_sim"] <= df_cat_2["conv_exp"] * (1 + band))
)

selec_within_10pct = (
    (df_cat_2["selec_sim"] >= df_cat_2["selec_exp"] * (1 - band)) &
    (df_cat_2["selec_sim"] <= df_cat_2["selec_exp"] * (1 + band))
)

n_conv_within = conv_within_10pct.sum()
n_selec_within = selec_within_10pct.sum()

total_cat_points = len(df_cat_2)

perc_conv_within = n_conv_within / total_cat_points * 100
perc_selec_within = n_selec_within / total_cat_points * 100

#calculate error per X or S
rmse_conv = np.sqrt(np.mean((df_cat_2["conv_sim"] - df_cat_2["conv_exp"])**2))
rmse_selec = np.sqrt(np.mean((df_cat_2["selec_sim"] - df_cat_2["selec_exp"])**2))

# Conversion parity plot
axes[0].scatter(df_cat_2["conv_exp"], df_cat_2["conv_sim"])
axes[0].plot([0, 100], [0, 100], linestyle="--", color="gray")                 #parity-line
axes[0].plot(x_line, x_line * (1 + band), linestyle="--", alpha=0.45, color='green', label="±10% band")   #error band
axes[0].plot(x_line, x_line * (1 - band), linestyle="--", alpha=0.45, color='green')                       #error band
axes[0].set_xlabel("Experimental (%)")
axes[0].set_ylabel("Simulated (%)")
axes[0].set_title("C2H2 conversion")
axes[0].text(
    0.05, 0.95,
    f"RMSE: {rmse_conv:.2f}\n"
    f"Within ±10% rel.: {n_conv_within}/{total_cat_points}\n({perc_conv_within:.1f}%)",
    transform=axes[0].transAxes,
    verticalalignment="top",
    bbox=dict(boxstyle="round", facecolor="white", alpha=0.8)
)
axes[0].grid(True, alpha = 0.1)

# Selectivity parity plot
axes[1].scatter(df_cat_2["selec_exp"], df_cat_2["selec_sim"])
axes[1].plot([0, 100], [0, 100], linestyle="--", color="gray")                 #parity-line
axes[1].plot(x_line, x_line * (1 + band), linestyle="--", alpha=0.45, color='green', label="±10% band")   #error band
axes[1].plot(x_line, x_line * (1 - band), linestyle="--", alpha=0.45, color='green')                       #error band
axes[1].set_xlabel("Experimental (%)")
axes[1].set_ylabel("Simulated (%)")
axes[1].set_title("C2H4 selectivity")
axes[1].text(
    0.05, 0.95,
    f"RMSE: {rmse_selec:.2f}\n"
    f"Within ±10% rel.: {n_selec_within}/{total_cat_points}\n({perc_selec_within:.1f}%)",
    transform=axes[1].transAxes,
    verticalalignment="top",
    bbox=dict(boxstyle="round", facecolor="white", alpha=0.8)
)
axes[1].grid(True, alpha = 0.1)

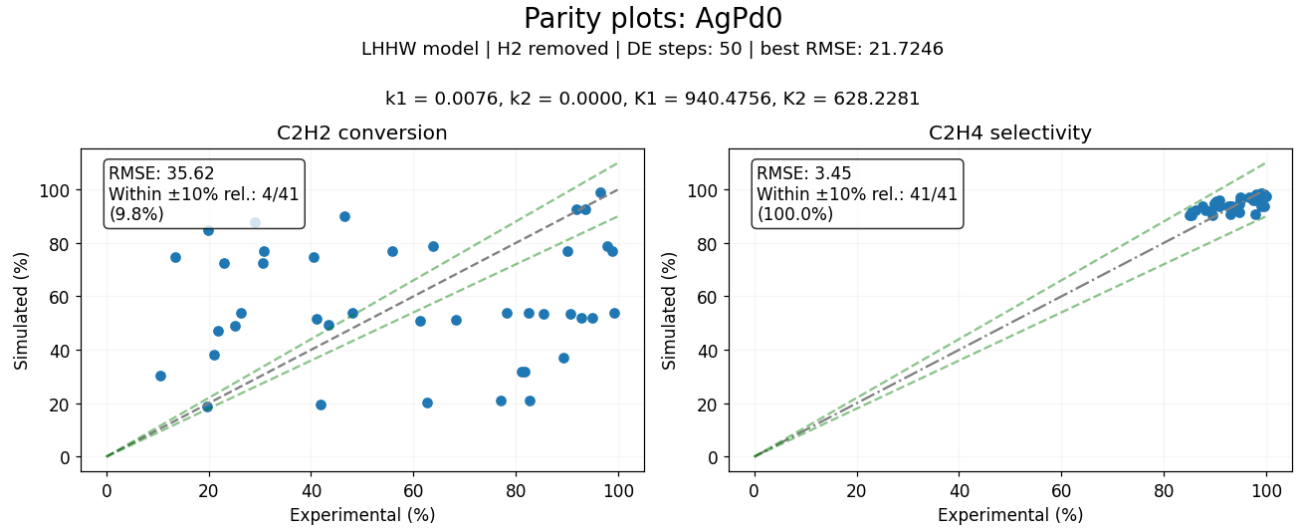
fig.suptitle("Parity plots: AgPd0",fontsize=18)
fig.text(0.5, 0.89,f"LHHW with all K's implemented and r2 inhibition | DE steps: {de_steps} | best RMSE: {best_RMSE:.4f}",ha="center",fontsize=12)
```

```
fig.text(0.5, 0.8,f"k1 = {best_k1:.4f}, k2 = {best_k2:.4f}, K1 = {best_K1:.4f}, K2 = {best_K2:.4f}, Kinh = {best_Kinh:.4f}, KH = {best_KH:.4f}", ha="center",fontsize=12
fig.text(0.5, 0.8,f"k1 = {best_k1:.4f}, k2 = {best_k2:.4f}, K1 = {best_K1:.4f}, K2 = {best_K2:.4f}, Kinh = {best_Kinh:.4f}, KH = {best_KH:.4f}", ha="center",fontsize=12

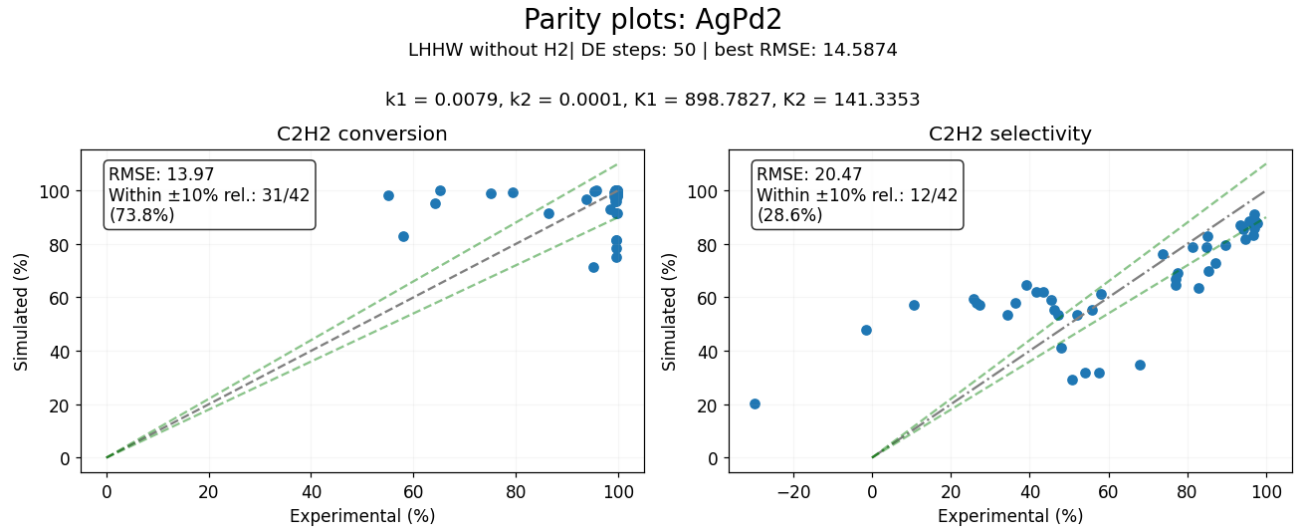
plt.tight_layout(rect=[0, 0, 1, 0.88])
plt.show()
```

G

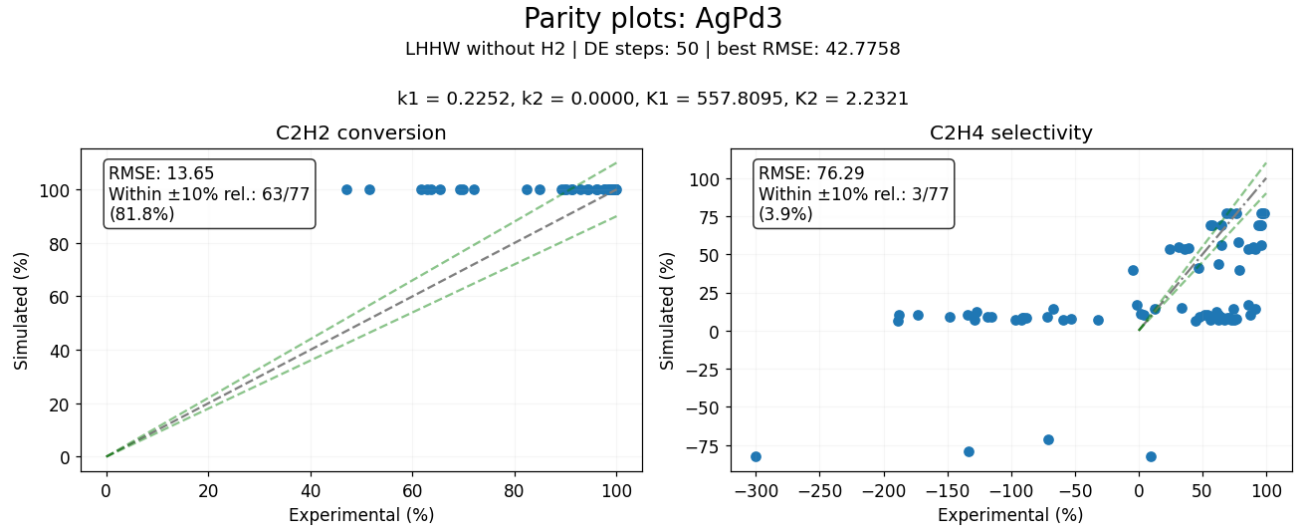
Plots simplified LHHW model, v4a



(a)

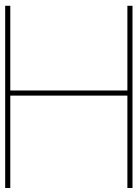


(b)

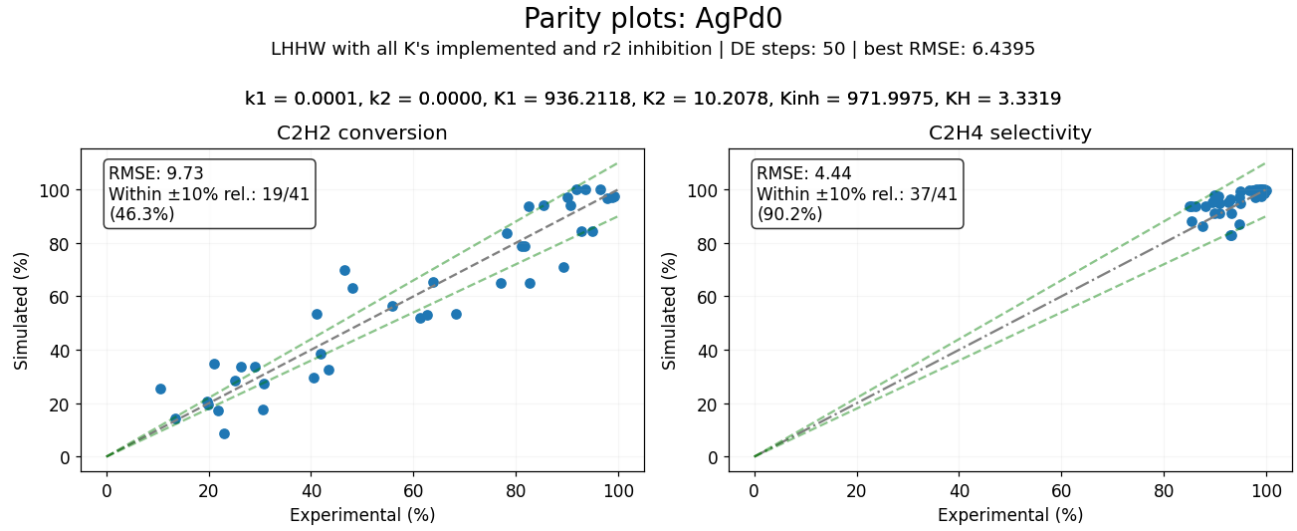


(c)

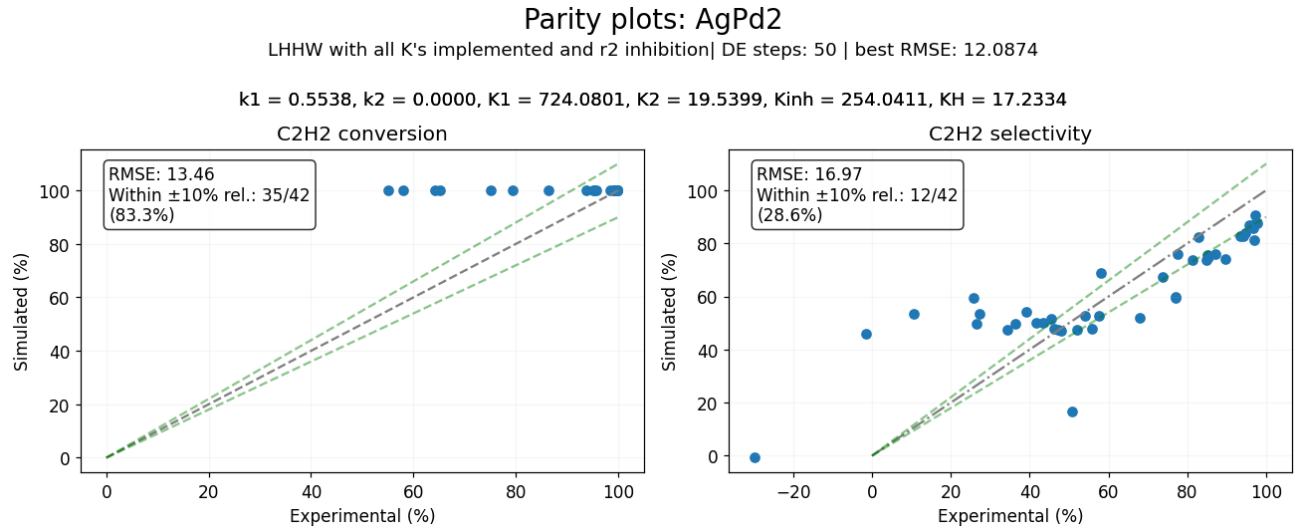
Figure G.1: Removal of hydrogen term in LHHW model: (a) Pd_0Ag, (b) Pd_2Ag, (c) Pd_3Ag



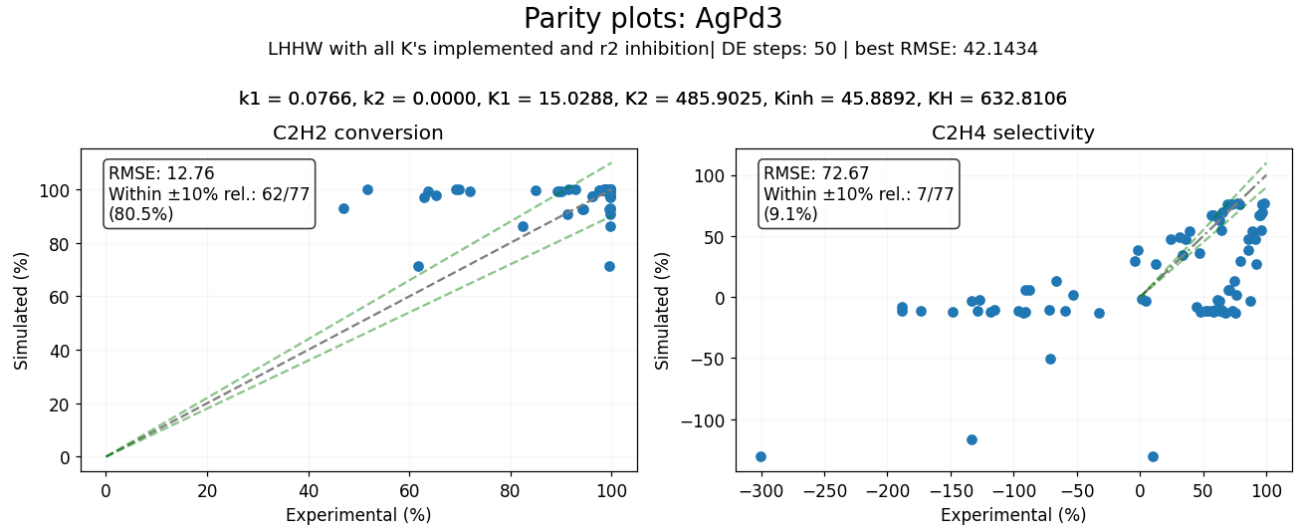
Plots with inhibition term LHHW model, v5



(a)



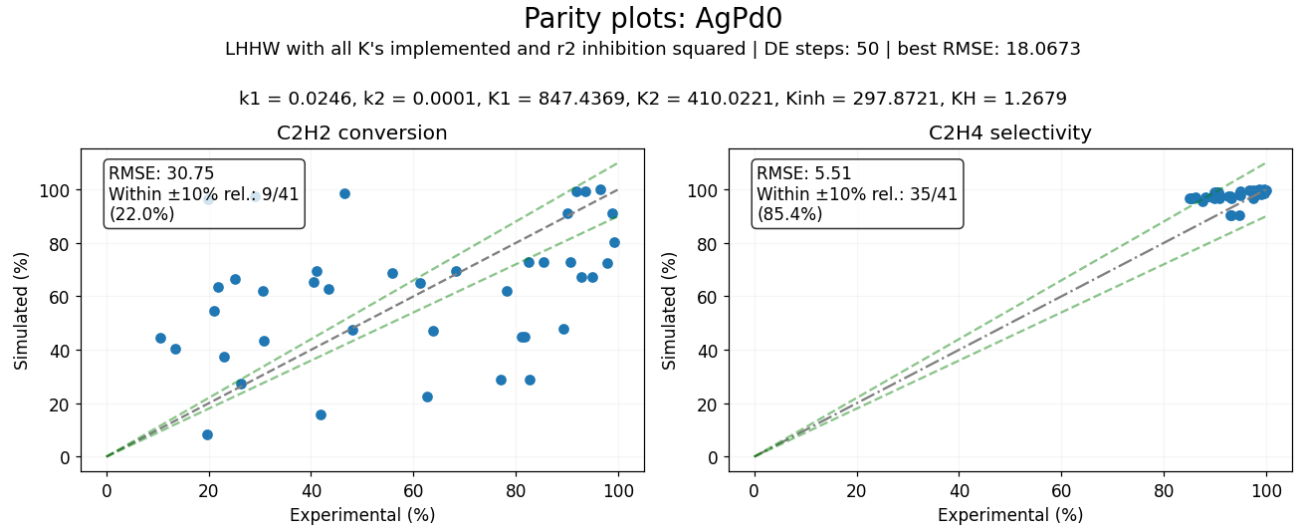
(b)



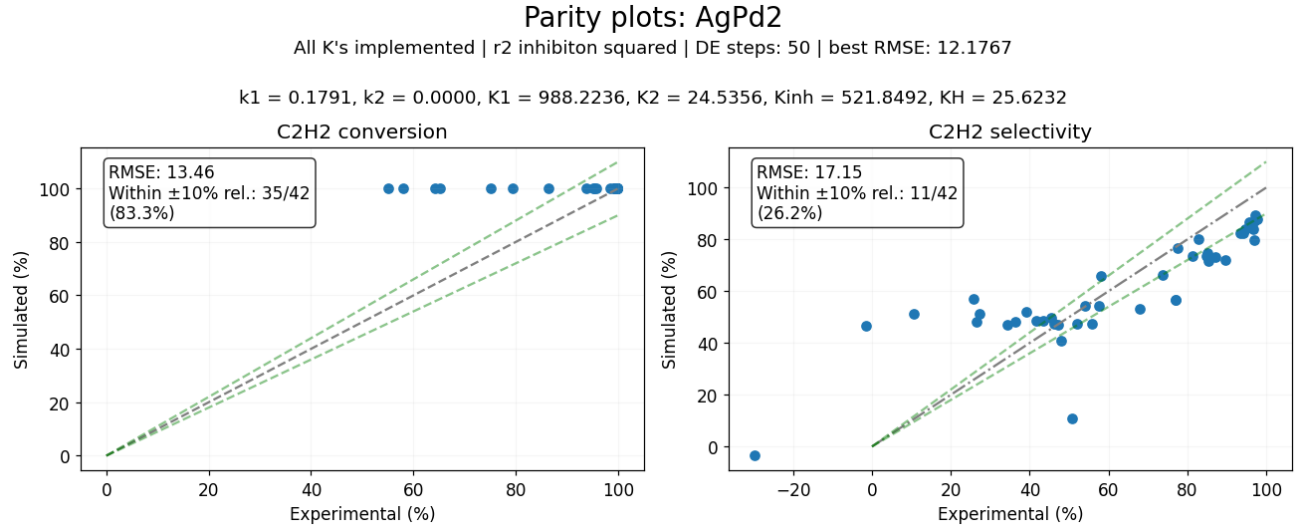
(c)

Figure H.1: Addition of acetylene inhibition term in LHHW model: (a) Pd_0Ag, (b) Pd_2Ag, (c) Pd_3Ag

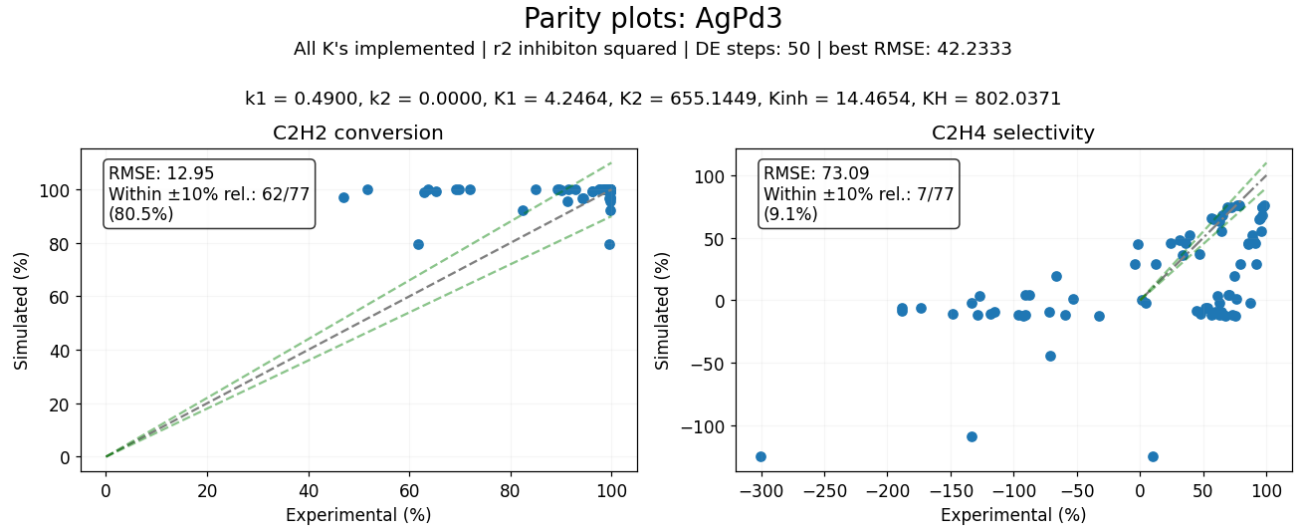
Result plots with inhibition term squared LHHW model, v5d



(a)



(b)



(c)

Figure I.1: Addition of squared acetylene inhibition term in LHHW model: (a) Pd_0Ag, (b) Pd_2Ag, (c) Pd_3Ag

J

Declaration of AI use

J.1. Responsible use

In this project, ChatGPT (GPT-5.5) was used as an AI assistant. The AI was used for tasks such as creating images, checking spelling, assisting with literature research, and helping to write Python code. Throughout the use of AI, no experimental data was uploaded. All calculations were performed by myself, although AI assisted in creating and debugging parts of the code. When AI was used for rewriting paragraphs in the results chapter, fake results were used.

ChatGPT assisted throughout this BEP with the following functions:

- Checking spelling, improving flow, and reviewing specific claims. Small paragraphs were uploaded to receive feedback on these points.
- Literature-search assistance. AI helped identify possible sources for specific claims. When AI suggested a source, the source was read and verified by me. AI was not cited as a source.
- Generating general images. AI assisted in creating specific figures, including the front page, Figure 3.1, and Figure 3.3.
- Writing and debugging Python code. AI assisted in finding mistakes in my own code and in improving code structure.

Overview

Table J.1: Overview of AI use in this project.

Tool	Section	Use
ChatGPT (GPT-5.5)	Introduction	Feedback on structure, spelling, and flow
ChatGPT (GPT-5.5)	Literature background	Feedback on structure, spelling, and flow
ChatGPT (GPT-5.5)	Literature background	Assistance in finding sources
ChatGPT (GPT-5.5)	Appendix C	Assistance with writing Python code
ChatGPT (GPT-5.5)	Literature background	Generation of images to support explanations
ChatGPT (GPT-5.5)	Appendix B	Generation of tables based on code-file names

Acknowledging limitations

Throughout the use of AI, its limitations became clear. Sometimes, when asked to review a paragraph, it suggested additions that were not relevant to the scope of the project. In addition, when writing code, the model did not always produce working code immediately. Therefore, the process remained a collaboration between my own understanding of the problem and the AI's suggestions.

Responsible use
Throughout the BEP, no specific previously unwritten claims were copied directly from AI output. Every new claim suggested by AI was checked by me and reviewed using my own understanding of the subject. The code written with AI assistance was checked by me, including unit checks on paper.

Reflection
The use of AI was helpful for improving scientific writing and sentence and paragraph structure. Every source suggested by AI was checked by me before it was added to the BEP. AI was also useful for creating specific images that could not be found online for the specific BEP case. Every AI-generated image was first sketched in Photoshop before being generated.

At some points, AI showed clear bias, mainly by strongly suggesting topics that were not within the scope of this research. These suggestions were filtered out before the remaining AI output was used.

For my next assignment, AI will probably be used again, mainly because of its assistance with writing. In specific cases where no clear images can be found, AI may also be used again to create images. Every image created with AI was first sketched in Photoshop before being put into the image generator.

J.2. Transparent use

Example 1
Prompt:

```
Update this code in such a way that per tested subject (GHSV, H2, C2H4 and C2H2) an extra p
axis.
```

ChatGPT answer:

```
Done -- I updated the notebook so every tested variable now gets an extra combined plot wi
- conversion (X_C2H2) and selectivity (S) on the same y-axis
- red line for conversion
- blue dashed line for selectivity
- log x-axis for GHSV, H2, and C2H2
- linear x-axis for C2H4
- multiple sets shown with different markers/labels where needed

You can download the updated notebook here:
[Download updated notebook](sandbox:/mnt/data/plot_data_AgPd0_with_combined_X_S_plots

I did not execute the notebook because it still loads the Excel file from your local path:
/Users/Victor/Desktop/Uni/MST 4e jaar/BEP/Acetylene_Data.xlsx

So just run it on your own laptop where that file exists.
```

Example 2
Prompt:

```
Is this a good source:
https://www.sciencedirect.com/science/article/pii/S0926337324003655?casa_token=0HcR_Q
kqLOyW7yfzHzYizHOSID3-JlzjekSFKZXvLrbExI88AlDPa54D_021sDVUc3irWZ

for defending this claim?
”These oxide supports prevent this phenomenon by separating the metal atoms and anchoring
```

ChatGPT answer:

```
Yes, but only partly.
```

The article is useful because it is about zeolite-encapsulated PdAg alloy catalysts, so it can

A safer sentence would be:
”Oxide and zeolitic supports can reduce sintering by dispersing the metal species and strength
dispersed metal particles under reaction conditions.”

That wording is better than ”prevent”, because supports usually reduce or inhibit sintering ra